

References

- [1] Amaro, R.; Pereira, R.; da Silva, M. M. (2023). Capabilities and practices in DevOps: A multivocal literature review. *IEEE Transactions on Software Engineering*, 49(3), 883–901. <https://doi.org/10.1109/TSE.2022.3166626>
- [2] Theunissen, T.; van Heesch, U.; Avgeriou, P. (2022). A mapping study on documentation in continuous software development. *Information and Software Technology*, 141, 106733. <https://doi.org/10.1016/j.infsof.2021.106733>
- [3] Laukkarinen, T.; Kuusinen, K.; Mikkonen, T. (2018). Regulated software meets DevOps. *Information and Software Technology*, 97, 176–178. <https://doi.org/10.1016/j.infsof.2018.01.011>
- [4] Marnewick, C.; Langerman, J. (2020). DevOps and organizational performance: The fallacy of chasing maturity. *IEEE Software*, 38(5), 48–55. <https://doi.org/10.1109/MS.2020.3023298>
- [5] Gall, M.; Pigni, F. (2022). Taking DevOps mainstream: A critical review and conceptual framework. *European Journal of Information Systems*, 31(5), 548–567. <https://doi.org/10.1080/0960085X.2021.1997100>
- [6] Liao, L. (2023). Addressing performance regressions in DevOps: Can we escape from system performance testing? In *45th IEEE/ACM International Conference on Software Engineering — Companion (ICSE Companion)*, 203–207. <https://doi.org/10.1109/ICSE-Companion58688.2023.00056>
- [7] Delgado, A.; García, F.; Ruiz, F. (2023). BizDevOps support for business process microservices-based applications. In *Service-Oriented Computing – ICSOC 2022 Workshops (LNCS 13821)*, 274–286. https://doi.org/10.1007/978-3-031-26507-5_22
- [8] Leite, L.; Rocha, C.; Kon, F.; Milojicic, D.; Meirelles, P. (2019). A survey of DevOps concepts and challenges. *ACM Computing Surveys*, 52(6), 1–35. <https://doi.org/10.1145/3359981>
- [9] Luz, W. P.; Pinto, G.; Bonifácio, R. (2019). Adopting DevOps in the real world: A theory, a model, and a case study. *Journal of Systems and Software*, 157, 110384. <https://doi.org/10.1016/j.jss.2019.07.083>
- [10] Pérez, J. E.; González-Prieto, Á.; Díaz, J.; López-Fernández, D.; García-Martín, J.; Yagüe, A. (2022). DevOps research-based teaching using qualitative research and inter-coder agreement. *IEEE Transactions on Software Engineering*, 48(9), 3378–3393. <https://doi.org/10.1109/TSE.2021.3092705>
- [11] Lwakatare, L. E.; Kilamo, T.; Karvonen, T.; Sauvola, T.; Heikkilä, V.; Itkonen, J.; Kuvaja, P.; Mikkonen, T.; Oivo, M.; Lassenius, C. (2019). DevOps in practice: A multiple case study of five companies. *Information and Software Technology*, 114, 217–230. <https://doi.org/10.1016/j.infsof.2019.06.010>
- [12] Myrbakken, H.; Colomo-Palacios, R. (2017). DevSecOps: A multivocal literature review. In *Software Process Improvement and Capability Determination (SPICE 2017)*, CCIS 770, 17–29. https://doi.org/10.1007/978-3-319-67383-7_2
- [13] Bogdanović, Z.; Žakić, M. D.; Barać, D.; Labus, A.; Radenković, M. (2023). The role of DevOps in sustainable enterprise development. In *International Series in Operations Research & Management Science*, 333, 217–237. https://doi.org/10.1007/978-3-031-16620-4_12
- [14] Leiter, A.; Hegyi, A.; Kispál, I.; Boósy, P.; Galambosi, N., & Tar, G. Z. (2023). GitOps and Kubernetes operator-based network function configuration. In *2023 IEEE/IFIP Network Operations and Management Symposium (NOMS)*. IEEE. <https://doi.org/10.1109/NOMS56928.2023.10154212>
- [15] Amaro, R.; Pereira, R.; da Silva, M. M. (2023). Capabilities and metrics in DevOps: A design science study. *Information & Management*, 60(5), 103809. <https://doi.org/10.1016/j.im.2023.103809>
- [16] Gupta, S., Bhatia, M., Memoria, M., & Manani, P. (2022). Prevalence of GitOps, DevOps in fast CI/CD cycles. In *2022 International Conference on Machine Learning, Big Data, Cloud and Parallel Computing (COM-IT-CON)*, Faridabad, India, 26–27 May 2022, pp. 589–596. <https://doi.org/10.1109/COM-IT-CON54601.2022.9850786>

- [17] Ganugapati, S. A.; Prabhu, S. (2023). Unifying governance, risk and controls framework using SDLC, CI/CD and DevOps. In 8th International Conference on Communication and Electronics Systems (ICCES). <https://doi.org/10.1109/ICCES57224.2023.10192730>.
- [18] Leite, L. A. F.; Lago, N.; Melo, C.; Kon, F.; Meirelles, P. (2023). A theory of organizational structures for development and infrastructure professionals. *IEEE Transactions on Software Engineering*, 49(4): 1898–1911. <https://doi.org/10.1109/TSE.2022.3199169>.
- [19] Arsecularatne, M., & Wickramarachchi, R. (2023). The adoptability of chaos engineering with DevOps to stimulate software delivery performance: A systematic literature review. 2023 IEEE 8th International Conference for Convergence in Technology (I2CT). <https://doi.org/10.1109/I2CT57861.2023.10126414>.
- [20] Wiedemann, A.; Wiese, M.; Gewald, H.; Krcmar, H. (2023). Integrating development and operations teams: A control approach for DevOps. *Information and Organization*, 33(3): 100474. <https://doi.org/10.1016/j.infoandorg.2023.100474>.
- [21] Thome, J.; Johnson, J.; Dawson, I.; Bolkensteyn, D.; Henriksen, M.; Art, M. (2023). SourceWarp: A scalable, SCM-driven testing and benchmarking approach to support data-driven and agile decision making for CI/CD tools and DevOps platforms. 2023 IEEE/ACM International Conference on Automation of Software Test (AST), pp. 68–78. <https://doi.org/10.1109/AST58925.2023.00011>.
- [22] Deekshetha, H. R., & Tyagi, A. K. (2023). Automated and intelligent systems for next-generation-based smart applications. In Tyagi, A. K. (Ed.), *Data Science for Genomics* (pp. 213–232). CRC Press. <https://doi.org/10.1201/9781003262130-13>.
- [23] Dewi, K. C., & Ayuni, N. W. D. (2020). Factors affecting acceptance of e-marketplace based on hybrid model of modified TAM-TRI. 2020 Fifth International Conference on Informatics and Computing (ICIC), Sulawesi, Indonesia, 3–4 Nov 2020, pp. 1–8. <https://doi.org/10.1109/ICIC50835.2020.9288626>.
- [24] Lohrasbinasab, I.; Acharya, P. B.; Palacios, R. C. (2020). BizDevOps: A multivocal literature review. In Gervasi, O. et al. (Eds.), *Computational Science and Its Applications – ICCSA 2020. Lecture Notes in Computer Science*, vol. 12249, pp. 305–320. Springer, Cham. https://doi.org/10.1007/978-3-030-58811-3_23.
- [25] Fisher, D. A. (2021). BubbleUp: Supporting DevOps with data visualization. *IEEE Computer Graphics and Applications*, 41(1): 99–105. <https://doi.org/10.1109/MCG.2020.3024039>.
- [26] Forbrig, P. (2018). BizDevOps and the role of S-BPM. In *S-BPM ONE 2018: Proceedings of the 10th International Conference on Subject-Oriented Business Process Management*, Linz, Austria, 5–6 April 2018, pp. 1–8. ACM. <https://doi.org/10.1145/3178248.3178250>.
- [27] Angara, J.; Prasad, S.; Sridevi, G. (2020). DevOps project management tools for sprint planning, estimation and execution maturity. *Cybernetics and Information Technologies*, 20(2): 79–92. <https://doi.org/10.2478/cait-2020-0018>.
- [28] Mudadi, A., & Lotriet, H. H. (2023). An analysis of DevOps' impact on information technology organisations: A case study. *South African Journal of Industrial Engineering*, 34(1): 155–167. <https://doi.org/10.7166/34-1-2859>.
- [29] Mathieson, J. T. J.; Mazzuchi, T.; Sarkani, S. (2020). The systems engineering DevOps lemniscate and model-based system operations. *IEEE Systems Journal*, 14(3): 3775–3786. <https://doi.org/10.1109/JSYST.2020.2969136>.
- [30] Cifo, M. A. S.; Bermejo, P.; Navarro, E. (2022). DevOps: Is there a gap between education and industry? *Journal of Software: Evolution and Process*, 34(12): e2495. <https://doi.org/10.1002/smr.2495>.
- [31] Dehghani, R., & Ramsin, R. (2022). A knowledge management-driven and DevOps-based method for situational method engineering. *Information Technology and Management*, 23(4): 227–247. <https://doi.org/10.1007/s10799-022-00364-4>.
- [32] Das, S.; Deb, N.; Chaki, N.; Cortesi, A. (2023). Minimizing conflicts among run-time non-functional requirements within DevOps. *Systems Engineering*, 26(4): 399–413. <https://doi.org/10.1002/sys.21688>.

- [33] Sanjurjo, E.; Pedreira, O.; García, F.; Piattini, M. (2020). Process reference model for BizDevOps. 2020 15th Iberian Conference on Information Systems and Technologies (CISTI), Sevilla, Spain, 24–27 June 2020, pp. 1–6. <https://doi.org/10.23919/CISTI49556.2020.9141079>.
- [34] Chen, T., & Suo, H. (2023). Architecture design of enterprise information system based on Docker and DevOps technology. Proceedings of the 7th International Conference on Management Engineering, Software Engineering and Service Sciences (ICMSS 2023), pp. 138–143. ACM. <https://doi.org/10.1145/3589610.3589637>.
- [35] Tanzil, M. H.; Sarker, M.; Uddin, G.; Iqbal, A. (2023). A mixed method study of DevOps challenges. Information and Software Technology, 161: 107244. <https://doi.org/10.1016/j.infsof.2023.107244>.
- [36] Zarour, M.; Alhammad, N.; Alenezi, M.; Alsarayrah, K. (2020). DevOps process model adoption in Saudi Arabia: An empirical study. Jordanian Journal of Computers and Information Technology (JJCIT), 6(3): 267–280. <https://doi.org/10.5455/jjcit.71-1588276534>.
- [37] Azad, N., & Hyrynsalmi, S. (2023). DevOps critical success factors: A systematic literature review. Information and Software Technology, 152: 107043. <https://doi.org/10.1016/j.infsof.2022.107043>.
- [38] Chouhan, U.; Tiwari, V.; Agrawal, K. K. (2024). Optimizing cloud-based e-learning platforms: A comparative analysis of server-based and serverless deployment strategies. Proceedings of the 2024 IEEE 9th International Conference for Convergence in Technology (I2CT), Pune, India, 5–7 April 2024, pp. 1–6. IEEE. <https://doi.org/10.1109/I2CT61223.2024.10543608>.
- [39] Khan, S. U.; Khan, A. W.; Lee, Y. (2023). Factors influencing vendor organizations in the selection of DevOps for global software development: An exploratory study using a systematic literature review. Cognition, Technology & Work, 25(4): 411–426. <https://doi.org/10.1007/s10111-023-00733-7>.
- [40] Kumar, R., & Goyal, R. (2020). Modeling continuous security: A conceptual model for automated DevSecOps using open-source software over cloud (ADOC). Computers & Security, 97: 101913. <https://doi.org/10.1016/j.cose.2020.101913>.
- [41] Solayman, H. E., & Qasga, R. P. (2023). Seamless integration of DevOps tools for provisioning automation of the IoT application on multi-infrastructures. Proceedings of the 3rd International Conference on Intelligent Communication and Computational Techniques (ICCT), Jaipur, India, 21–22 Jan 2023, pp. 348–354. IEEE. <https://doi.org/10.1109/ICCT57119.2023.10087369>.
- [42] Yang, L., & Rossi, D. (2021). Quality monitoring and assessment of deployed deep learning models for network AIOps. IEEE Network, 35(6): 207–213. <https://doi.org/10.1109/MNET.011.2100226>.
- [43] Dileepkumar, S. R., & Mathew, J. (2023). Enhancing DevOps and continuous integration in software engineering: A comprehensive approach. Proceedings of the 2nd International Conference on Electrical, Electronics, Information and Communication Technologies (ICEEICT 2023), pp. 1–6. IEEE. <https://doi.org/10.1109/ICEEICT58398.2023.10155620>.
- [44] Scekcic, M.; Gazivoda, M.; Scepanovic, S.; Nikolic, J. (2018). Application of DevOps approach in developing business intelligence system in bank. 2018 7th Mediterranean Conference on Embedded Computing (MECO), Budva, Montenegro, 10–14 June 2018, pp. 1–4. IEEE. <https://doi.org/10.1109/MECO.2018.8405994>.
- [45] Gloor, M.; Mazalin, M.; Zimmermann, A. (2021). Automated payment terminal testing: How to achieve continuous integration for systems that are almost impossible to virtualize. IEEE Software, 38(6): 19–24. <https://doi.org/10.1109/MS.2020.3019809>.
- [46] Sokolowski, D.; Spielmann, D.; Salvaneschi, G. (2024). Unleashing the giants: Enabling advanced testing for infrastructure as code. 2024 IEEE/ACM 46th International Conference on Software Engineering: Companion Proceedings (ICSE-Companion '24), Lisbon, Portugal, 14–20 Apr 2024, pp. 300–301. ACM. <https://doi.org/10.1145/3639478.3643078>.
- [47] Gopal, B., & Cooper, S. (2023). Process oriented guided inquiry-based learning-like pedagogy (POGIL-like) in online software testing and DevOps – A replication study. 2023 IEEE International Conference on Software Testing, Verification and Validation Workshops (ICSTW), Dublin, Ireland, 16–20 Apr 2023, pp. 137–144. IEEE. <https://doi.org/10.1109/ICSTW58534.2023.00030>.

- [48] Rajapakse, R. N.; Zahedi, M.; Babar, M. A.; Shen, H. (2022). Challenges and solutions when adopting DevSecOps: A systematic review. *Information and Software Technology*, 141: 106700. <https://doi.org/10.1016/j.infsof.2021.106700>.
- [49] Sub, J. G.; Swift, S.; Escott, E. (2021). Using DevOps toolchains in agile model-driven engineering. In Bruel, J.-M., Mazzara, M., Meyerovich, L. A., & Meyer, B. (Eds.), *Software Engineering Aspects of Continuous Development and New Paradigms of Software Production and Deployment (DEVOPS 2020)*. Lecture Notes in Computer Science, vol. 12577, pp. 131–147. Springer, Cham. https://doi.org/10.1007/978-3-030-67445-8_9.
- [50] Pastrana Pardo, M. A.; Ordoñez Erazo, H. A.; Cobos Lozada, C. A. (2021). Documenting and implementing DevOps good practices with test automation and continuous deployment tools through software refinement. *Periodicals of Engineering and Natural Sciences (PEN)*, 9(4): 854–863. <https://doi.org/10.21533/PEN.V9I4.2239>.
- [51] Alves, I.; Pérez, J.; Díaz, J.; López-Fernández, D.; Pais, M.; Kon, F.; Rocha, C. (2023). Harmonizing DevOps taxonomies: Theory operationalization and testing. arXiv preprint arXiv:2302.00033, Jan 2023. URL: arXiv:2302.00033.
- [52] Jayakody, J. A. V. M. K., & Wijayanayake, W. M. J. I. (2023). Process improvement framework for DevOps adoption in software development. 2023 International Research Conference on Smart Computing and Systems Engineering (SCSE), University of Kelaniya, Sri Lanka, 29 Jun 2023, pp. 1–6. <https://doi.org/10.1109/SCSE59836.2023.10214992>.
- [53] Sravan, S. S.; Ganesh, C. S.; Kiran, K. V. D.; Kandra, T. A.; Aparna, K.; Vignesh, T. (2023). Significant challenges to espouse DevOps culture in software organisations by AWS. 2023 9th International Conference on Advanced Computing and Communication Systems (ICACCS), Coimbatore, India, 17–18 Mar 2023, pp. 1462–1467. IEEE. <https://doi.org/10.1109/ICACCS57248.2023.10107054>.
- [54] Ramadoni, W.; Utami, E.; Fatta, H. A. (2021). Analysis on the use of declarative and pull-based deployment models on GitOps using Argo CD. 2021 4th International Conference on Information and Communications Technology (ICOIACT), Yogyakarta, Indonesia, 30–31 Aug 2021, pp. 84–89. IEEE. <https://doi.org/10.1109/ICOIACT53268.2021.9564037>.
- [55] Waseem, M.; Liang, P.; Shahin, M. (2020). A systematic mapping study on microservices architecture in DevOps. *Journal of Systems and Software*, 170: 110798. <https://doi.org/10.1016/j.jss.2020.110798>.
- [56] e Souza, I. S.; Franco, D. P.; Silva, J. P. S. G. (2023). Infrastructure as code as a foundational technique for increasing the DevOps maturity level: Two case studies. *IEEE Software*, 40(1): 63–68. <https://doi.org/10.1109/MS.2022.3213228>.
- [57] Charan, K.; Karthikeya, N.; Narasimharao, P. B.; Devi, S. A.; Abhilash, V.; Srinivas, P. V. V. S. (2023). Effective code testing strategies in DevOps: A comprehensive study of techniques and tools for ensuring code quality and reliability. 2023 4th International Conference on Electronics and Sustainable Communication Systems (ICESC), Coimbatore, India, 6–8 July 2023, pp. 302–309. IEEE. <https://doi.org/10.1109/ICESC57686.2023.10193275>.
- [58] Colantoni, A.; Horváth, B.; Horváth, Á.; Berardinelli, L.; Wimmer, M. (2021). Towards continuous consistency checking of DevOps artefacts. 2021 ACM/IEEE 24th International Conference on Model Driven Engineering Languages and Systems Companion (MODELS-C), 449–453. <https://doi.org/10.1109/MODELS-C53483.2021.00069>.
- [59] Narang, P., & Mittal, P. (2022). Hybrid model for software development: An integral comparison of DevOps automation tools. *Indonesian Journal of Electrical Engineering and Computer Science*, 27(1): 456–465. <https://doi.org/10.11591/ijeecs.v27.i1.pp456-465>.
- [60] Kumar, A.; Nadeem, M.; Shameem, M. (2023). Systematic literature review of metrics for measuring DevOps success. *AIP Conference Proceedings*, 2724(1): 030005. <https://doi.org/10.1063/5.0128883>.
- [61] Ebert, C., & Hochstein, L. (2023). DevOps in practice. *IEEE Software*, 40(1): 29–36. <https://doi.org/10.1109/MS.2022.3213285>.

- [62] Baertschi, S.; Guenthardt, L.; Sabani, R.; & Krey, M. (2023). A method for the adoption of DevOps in the banking industry. 2023 International Conference on Information Management (ICIM), Oxford, UK, 17–19 Mar 2023, pp. 31–36. <https://doi.org/10.1109/ICIM58774.2023.00012>.
- [63] Limoncelli, T. A. (2018). GitOps: A path to more self-service IT. *Communications of the ACM*, 61(9): 38–42. <https://doi.org/10.1145/3233241>.
- [64] Airaj, M. (2017). Enable cloud DevOps approach for industry and higher education. *Concurrency and Computation: Practice and Experience*, 29(5): e3937. <https://doi.org/10.1002/cpe.3937>.
- [65] Srivastava, S.; Allam, K.; Mustyala, A. (2023). Software automation enhancement through the implementation of DevOps. *International Journal of Research Publication and Reviews*, 4(6): 2050–2054. <https://doi.org/10.55248/gengpi.4.623.45947>.
- [66] Maheshwari, K.; Jain, K.; Patel, K.; Mittal, K. (2022). DevOps. *International Research Journal of Modernization in Engineering Technology and Science*, 4(10). <https://doi.org/10.56726/IRJMETS30906>.
- [67] Al Masud, S. M. R.; Masnun, M.; Sultana, M. A.; Sultana, A.; Ahmed, F.; Begum, N. (2022). DevOps enabled agile: Combining agile and DevOps methodologies for software development. *International Journal of Advanced Computer Science and Applications (IJACSA)*, 13(11): 278–283. <https://doi.org/10.14569/IJACSA.2022.0131131>.
- [68] Sen, A.; Falter, S.; Mayer, N. (2022). Using DevOps paradigm to deploy web applications. *Issues in Information Systems*, 23(4): 142–156. https://doi.org/10.48009/4_iis_2022_113.
- [69] Koche, A.; Masram, S.; Meshram, P.; Ingole, S. (2023). DevOps using AWS. *International Research Journal of Modernization in Engineering Technology and Science*, 5(6). <https://doi.org/10.56726/IRJMETS41357>.
- [70] Pardo, C. J.; Guerrero, J.; Suescún Monsalve, E. (2022). DevOps model in practice: Applying a novel reference model to support and encourage the adoption of DevOps in a software development company as case study. *Periodicals of Engineering and Natural Sciences (PEN)*, 10(3): 221–235. <https://doi.org/10.21533/pen.v10i3.3086>.
- [71] Spinellis, D. (2016). Being a DevOps developer. *IEEE Software*, 33(3): 4–5. <https://doi.org/10.1109/MS.2016.76>.
- [72] Kumar, N., & Gondkar, R. R. (2021). Role of ITOps in DevOps. *Proceedings of the International Conference on Innovative Computing & Communication (ICICC 2021)*. SSRN. <https://doi.org/10.2139/ssrn.3835090>.
- [73] Amrit, A.; Akhil, P. J.; Pranjal; Rakshith Raj, N.; Shylaja, B. S. (2022). Microservices evolving DevOps pipelines. *International Research Journal of Modernization in Engineering Technology and Science*, 4(9). <https://doi.org/10.56726/IRJMETS29808>.
- [74] Sen, A., & Skrobot, I. (2021). Implementation of DevOps paradigm to deployment and provisioning of microservices. *Issues in Information Systems*, 22(1): 136–148. https://doi.org/10.48009/1_iis_2021_136-148.
- [75] Dörnenburg, E. (2018). The path to DevOps. *IEEE Software*, 35(5): 71–75. <https://doi.org/10.1109/MS.2018.290110337>.
- [76] Prabhakar, S. (2020). DevOps phenomenon: A review on trends and challenges. *Journal of Xidian University*, 14(4): 688–695. <https://doi.org/10.37896/jxu14.4/085>.
- [77] Bento, A. C.; Cantú Delgado, D. A.; Camacho-León, S. (2023). Experimental survey results on Azure.DevOps application for management students' projects. In *2023 Future of Educational Innovation—Workshop Series: Data in Action (Digital Ecosystem and Emerging Tools for Education)*, Monterrey, Mexico, 16–18 Jan 2023. IEEE. Online ISBN: 978-1-6654-9416-8; Print-on-Demand ISBN: 978-1-6654-9417-5. <https://doi.org/10.1109/IEEECONF56852.2023.10104593>.
- [78] Yin, L., & Filkov, V. (2020). Team discussions and dynamics during DevOps tool adoptions in OSS projects. *35th IEEE/ACM International Conference on Automated Software Engineering (ASE '20)*,

September 21–25, 2020, Virtual Event, Australia. ACM, New York, NY, USA, 12 pages.
<https://doi.org/10.1145/3324884.3416640>.

[79] Sánchez-Gordón, M., & Colomo-Palacios, R. (2020). Security as culture: A systematic literature review of DevSecOps. 2020 IEEE/ACM 42nd International Conference on Software Engineering Workshops (ICSEW), 266–269. <https://doi.org/10.1145/3387940.3392233>.

[80] Morales, J. A.; Scanlon, T. P.; Volkmann, A.; Yankel, J.; Yasar, H. (2020). Security impacts of sub-optimal DevSecOps implementations in a highly regulated environment. Proceedings of the 15th International Conference on Availability, Reliability and Security (ARES '20), Article 63, 1–8. ACM. <https://doi.org/10.1145/3407023.3409186>.

[81] Barros, D. D. R.; Horita, F. E. A.; Fantinato, D. G. (2020). Data mining tool to discover DevOps trends from public repositories: Predicting Release Candidates with gthbmining.rc. Proceedings of the XXXIV Brazilian Symposium on Software Engineering (SBES '20), Natal, Brazil, 21–23 Oct 2020, pp. 658–663. ACM. <https://doi.org/10.1145/3422392.3422501>.

[82] Ashenden, D., & Ollis, G. (2020). Putting the sec in DevSecOps: Using social practice theory to improve secure software development. New Security Paradigms Workshop 2020 (NSPW '20) – Post-Proceedings, 34–44. ACM. <https://doi.org/10.1145/3442167.3442178>.

[83] Li, C.-J., & Shih, H.-J. (2021). The one-key seamless integrating platform for open-source DevOps tools, to solve interconnecting issues and boost CI/CD efficiency. ITCC '21: Proceedings of the 2021 3rd International Conference on Information Technology and Computer Communications, Guangzhou, China, 23–25 June 2021, pp. 80–83. ACM. <https://doi.org/10.1145/3473465.3473479>.

[84] Rajapakse, R. N.; Zahedi, M.; Babar, M. A. (2021). An empirical analysis of practitioners' perspectives on security tool integration into DevOps. ESEM '21: Proceedings of the 15th ACM/IEEE International Symposium on Empirical Software Engineering and Measurement, Article 6, 1–12. <https://doi.org/10.1145/3475716.3475776>.

[85] Nadgowda, S.; Luan, L. (2021). tapiseri: Blueprint to modernize DevSecOps for real world. WoC '21: Proceedings of the Seventh International Workshop on Container Technologies and Container Clouds, co-located with Middleware 2021, Virtual Event, Canada, 6 Dec 2021, pp. 13–18. <https://doi.org/10.1145/3493649.3493655>.

[86] Ramaj, X. (2022). A DevSecOps-enabled framework for risk management of critical infrastructures. 2022 IEEE/ACM 44th International Conference on Software Engineering: Companion Proceedings (ICSE-Companion '22), Pittsburgh, PA, USA, 22–24 May 2022, pp. 242–244. <https://doi.org/10.1145/3510454.3517053>.

[87] Scanlon, T. P.; Morales, J. A. (2022). Revelations from an agile and DevSecOps transformation in a large organization: An experiential case study. ICSSP '22: 16th International Conference on Software and System Process (co-located with ICGSE 2022), Virtual Event / Pittsburgh, PA, USA, 19–20 May 2022, pp. 77–81. <https://doi.org/10.1145/3529320.3529329>.

[88] Yasar, H.; Teplov, S. E. (2022). DevSecOps in embedded systems: An empirical study of past literature. Proceedings of the 17th International Conference on Availability, Reliability and Security (ARES '22), Vienna, Austria, 23–26 Aug 2022, Article 155, 1–6. ACM. <https://doi.org/10.1145/3538969.3544451>.

[89] Cankar, M.; Petrović, N.; Pita Costa, J.; Černivec, A.; Antić, J.; Martinčič, T.; Štepec, D. (2023). Security in DevSecOps: Applying tools and machine learning to verification and monitoring steps. ICPE '23 Companion: Companion of the 2023 ACM/SPEC International Conference on Performance Engineering, Coimbra, Portugal, 15–19 Apr 2023, pp. 201–205. <https://doi.org/10.1145/3578245.3584943>.

[90] Rahman, A.; Shamim, S. I.; Bose, D. B.; Pandita, R. (2023). Security misconfigurations in open source Kubernetes manifests: An empirical study. ACM Transactions on Software Engineering and Methodology, 32(4): 1–36. <https://doi.org/10.1145/3579639>.

[91] Matthies, C.; Heinrich, R.; Wohlrab, R. (2023). Investigating software engineering artifacts in DevOps through the lens of boundary objects. Proceedings of the 27th International Conference on Evaluation and

Assessment in Software Engineering (EASE '23), Oulu, Finland, 14–16 June 2023, pp. 12–21.
<https://doi.org/10.1145/3593434.3593441>.

[92] Pauzi, Z.; Thind, R.; Capiluppi, A. (2023). Artifact traceability in DevOps: An industrial experience report. Proceedings of the 27th International Conference on Evaluation and Assessment in Software Engineering (EASE '23), Oulu, Finland, 14–16 June 2023, pp. 180–183.
<https://doi.org/10.1145/3593434.3593451>.

[93] Trihinas, D.; Tryfonos, A.; Dikaiakos, M. D.; Pallis, G. (2018). DevOps as a service: Pushing the boundaries of microservice adoption. *IEEE Internet Computing*, 22(3): 65–71.
<https://doi.org/10.1109/MIC.2018.032501519>.

[94] Alnafessah, A.; Gias, A. U.; Wang, R.; Zhu, L.; Casale, G.; Filieri, A. (2021). Quality-aware DevOps research: Where do we stand? *IEEE Access*, 9: 44476–44489.
<https://doi.org/10.1109/ACCESS.2021.3064867>.

[95] Narang, P., & Mittal, P. (2022). Implementation of DevOps-based hybrid model for project management and deployment using Jenkins automation tool with plugins. *International Journal of Computer Science and Network Security*, 22(8): 249–259. <https://doi.org/10.22937/IJCSNS.2022.22.8.31>.

[96] Petrović, N. (2023). Chat GPT-based design-time DevSecOps. 2023 58th International Scientific Conference on Information, Communication and Energy Systems and Technologies (ICEST), Niš, Serbia, 29 Jun–1 Jul 2023, pp. 143–146. <https://doi.org/10.1109/ICEST58410.2023.10187247>.

[97] Gasparaitė, M.; Naudziūnaitė, K.; Ragaisis, S. (2020). Systematic literature review of DevOps models. In *Quality of Information and Communications Technology (QUATIC 2020): 13th International Conference*, Faro, Portugal, 9–11 Sept 2020, Proceedings. Communications in Computer and Information Science (CCIS), 1266, 184–198. Springer, Cham. https://doi.org/10.1007/978-3-030-58793-2_15.

[98] Sanjurjo, E.; Pedreira, O.; García, F.; Piattini, M. (2020). Measuring the maturity of BizDevOps. *Quality of Information and Communications Technology (QUATIC 2020): 13th International Conference*, Faro, Portugal, 9–11 Sept 2020, Proceedings. Communications in Computer and Information Science, 1266: 199–210. Springer, Cham. https://doi.org/10.1007/978-3-030-58793-2_16.

[99] Casola, V.; De Benedictis, A.; Rak, M.; Salzillo, G. (2020). A cloud SecDevOps methodology: From design to testing. *Quality of Information and Communications Technology (QUATIC 2020): 13th International Conference*, Faro, Portugal, 9–11 Sept 2020, Proceedings. Communications in Computer and Information Science, 1266: 317–331. Springer, Cham. https://doi.org/10.1007/978-3-030-58793-2_26.

[100] Kersten, M. (2018). A Cambrian explosion of DevOps tools. *IEEE Software*, 35(2): 14–17.
<https://doi.org/10.1109/MS.2018.1661330>.

[101] Anisetti, M.; Bena, N.; Berto, F.; Jeon, G. (2022). A DevSecOps-based assurance process for big data analytics. 2022 IEEE International Conference on Web Services (ICWS), Barcelona, Spain, 11–15 July 2022, pp. 1–10. <https://doi.org/10.1109/ICWS55610.2022.00017>.

[102] Zunnurhain, K.; Duclervil, S. R. (2019). A new project management tool based on DevSecOps. 2019 International Conference on Computational Science and Computational Intelligence (CSCI), 239–243.
<https://doi.org/10.1109/CSCI49370.2019.00049>.

[103] Shah, J.; Dubaria, D.; Widhalm, J. (2018). A survey of DevOps tools for networking. 2018 9th IEEE Annual Ubiquitous Computing, Electronics & Mobile Communication Conference (UEMCON), New York, NY, USA, pp. 185–188. <https://doi.org/10.1109/UEMCON.2018.8796814>.

[104] Aljohani, M. A., & Alqahtani, S. S. (2023). A unified framework for automating software security analysis in DevSecOps. 2023 International Conference on Smart Computing and Application (ICSCA), Ha'il, Saudi Arabia, pp. 1–6. IEEE. <https://doi.org/10.1109/ICSCA57840.2023.10087568>.

[105] Pérez-Sánchez, J.; Ros, J. N.; de Gea, J. M. C.; Fernández-Alemán, J. L. (2022). DevOps certifications for IT professionals. *Computer (IEEE Computer Society)*, 55(11): 62–73.
<https://doi.org/10.1109/MC.2022.3144068>.

- [106] Dabab, M.; Freiling, M.; Sagalowicz, D. (2021). Expanding horizons in business decision making. 2021 IEEE Technology & Engineering Management Conference – Europe (TEMSCON-EUR), Dubrovnik (virtual), 17–21 May 2021. <https://doi.org/10.1109/TEMSCON-EUR52034.2021.9488619>.
- [107] Poonam; Mittal, P. (2019). Performance measure of project management automation tool based on DevOps selection criteria for a general-purpose software system. *International Journal of Engineering and Advanced Technology*, 9(1): 6263–6267. <https://doi.org/10.35940/ijeat.A2051.109119>
- [108] Ramzan, S.; Khan, S. U. R.; Hussain, S.; Wang, W.-L.; Tang, M.-H. (2023). Identification of influential factors for successful adoption of DevOps and Cloud. *Proceedings of the 27th International Conference on Evaluation and Assessment in Software Engineering (EASE '23)*, Oulu, Finland, 14–16 June 2023, pp. 423–429. ACM. <https://doi.org/10.1145/3593434.3594239>.
- [109] Nuerk, J.; Dařena, F. (2023). Activating supply chain business models' value potentials through systems engineering. *Systems Engineering*, 26(5): 660–674. <https://doi.org/10.1002/sys.21676>.
- [110] Tomas, N.; Li, J.; Huang, H. (2019). An empirical study on culture, automation, measurement, and sharing of DevSecOps. 2019 International Conference on Cyber Security and Protection of Digital Services (Cyber Security), Oxford, UK, 3–4 June 2019, pp. 1–8. <https://doi.org/10.1109/CyberSecPODS.2019.8884935>.
- [111] Petrović, N.; Cankar, M.; Luzar, A. (2022). Automated approach to IaC code inspection using Python-based DevSecOps tool. 2022 30th Telecommunications Forum (TELFOR), Belgrade, Serbia, 16–17 Nov 2022, pp. 1–4. <https://doi.org/10.1109/TELFOR56187.2022.9983681>.
- [112] Wörndle, P., Terrill, S., & Dinsing, T. (2023). Automating telecom software deployment with GitOps. *Ericsson Technology Review*, 2023(1). Available at: <https://www.ericsson.com/en/ericsson-technology-review/articles/automating-telecom-software-deployment-with-gitops>
- [113] Beetz, F.; Harrer, S. (2022). GitOps: The evolution of DevOps? *IEEE Software*, 39(4): 70–75. <https://doi.org/10.1109/MS.2021.3119106>.
- [114] López-Viana, R.; Díaz, J.; Pérez-Martínez, J. E. (2022). Continuous deployment in IoT edge computing: A GitOps implementation. 2022 17th Iberian Conference on Information Systems and Technologies (CISTI), Madrid, Spain, 22–25 June 2022, pp. 1–6. <https://doi.org/10.23919/CISTI54924.2022.9820108>.
- [115] Bomström, H.; Kelanti, M.; Annanperä, E.; Liukkunen, K.; Kilamo, T.; Sievi-Korte, O.; Systä, K. (2023). Information needs and presentation in agile software development. *Information and Software Technology*, 162: 107265. <https://doi.org/10.1016/j.infsof.2023.107265>.
- [116] López-Fernández, D.; Díaz, J.; García-Martín, J.; Pérez, J.; González-Prieto, Á. (2022). DevOps team structures: Characterization and implications. *IEEE Transactions on Software Engineering*, 48(10): 3716–3736. <https://doi.org/10.1109/TSE.2021.3102982>.
- [117] Narendiran, A.; Abhishek, D.; Adithya, P.; Ray, D.; Auradkar, P. K.; Phalachandra, H. L. (2023). Integrated log-aware CI/CD pipeline with custom bot for monitoring. 2023 8th International Conference on Cloud Computing and Big Data Analytics (ICCCBDA), Chengdu, China, 26–28 Apr 2023, pp. 257–262. <https://doi.org/10.1109/ICCCBDA56900.2023.10154891>.
- [118] Ahmed, S.; Singh, M.; Doherty, B.; Ramlan, E. I.; Harkin, K.; Bucholc, M.; Coyle, D. (2023). Knowledge-based intelligent system for IT incident DevOps. 2023 IEEE/ACM International Workshop on Cloud Intelligence & AIOps (AIOps), Melbourne, Australia, 15 May 2023, pp. 1–7. <https://doi.org/10.1109/AIOps59134.2023.00005>.
- [119] Bertolino, A.; De Angelis, G.; Guerriero, A.; Miranda, B.; Pietrantuono, R.; Russo, S. (2023). DevOpRET: Continuous reliability testing in DevOps. *Journal of Software: Evolution and Process*, 35(3): e2298. <https://doi.org/10.1002/smr.2298>.
- [120] Roy, M.; Deb, N.; Cortesi, A.; Chaki, R.; Chaki, N. (2021). CARO: A conflict-aware requirement ordering tool for DevOps. 2021 IEEE 29th International Requirements Engineering Conference (RE), 442–443. <https://doi.org/10.1109/RE51729.2021.00061>.

- [121] Shahin, M.; Rezaei Nasab, A.; Babar, M. A. (2023). A qualitative study of architectural design issues in DevOps. *Journal of Software: Evolution and Process*, 35(5): e2379. <https://doi.org/10.1002/smr.2379>.
- [122] Andel, W. (2022). Continuous documentation: Automating document preparation with your DevSecOps pipeline. 2022 IEEE 29th Annual Software Technology Conference (STC), 156–165. <https://doi.org/10.1109/STC55697.2022.00029>.
- [123] Khan, M. S.; Khan, A. W.; Khan, F.; Khan, M. A.; Whangbo, T. K. (2022). Critical challenges to adopt DevOps culture in software organizations: A systematic review. *IEEE Access*, 10: 14339–14349. <https://doi.org/10.1109/ACCESS.2022.3145970>.
- [124] Díaz, J.; López-Fernández, D.; Pérez, J.; González-Prieto, Á. (2021). Why are many businesses instilling a DevOps culture into their organization? *Empirical Software Engineering*, 26(2): 25. <https://doi.org/10.1007/s10664-020-09919-3>.
- [125] Gür, I.; Möller, F.; Hupperz, M.; Uzun, D.; Otto, B. (2022). Requirements for DataOps to foster dynamic capabilities in organizations – A mixed methods approach. 2022 IEEE 24th Conference on Business Informatics (CBI), Amsterdam, Netherlands, 15–17 June 2022, pp. 166–175. <https://doi.org/10.1109/CBI54897.2022.00025>.
- [126] Yu, S.; Chen, T.; Han, L.; Demartini, G.; Sadiq, S. (2023). DataOps-4G: On supporting generalists in data quality discovery. *IEEE Transactions on Knowledge and Data Engineering*, 35(5): 4668–4681. <https://doi.org/10.1109/TKDE.2022.3151605>.
- [127] Sokolowski, D. (2021). Deployment coordination for cross-functional DevOps teams. *Proceedings of the 29th ACM Joint Meeting on European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE 2021)*, 1630–1634. <https://doi.org/10.1145/3468264.3473101>.
- [128] Sun, X.; Cheng, Y.; Qu, X.; Li, H. (2021). Design and implementation of security test pipeline based on DevSecOps. 2021 IEEE 4th Advanced Information Management, Communicates, Electronic and Automation Control Conference (IMCEC), Chongqing, China, 18–20 June 2021, pp. 532–535. <https://doi.org/10.1109/IMCEC51613.2021.9482270>.
- [129] Chen, T.; Suo, H. (2022). Design and practice of security architecture via DevSecOps technology. 2022 IEEE 13th International Conference on Software Engineering and Service Science (ICSESS), Beijing, China, 21–23 Oct 2022, pp. 310–313. <https://doi.org/10.1109/ICSESS54813.2022.9930212>.
- [130] Trigo, A.; Varajão, J.; Sousa, L. (2022). DevOps adoption: Insights from a large European Telco. *Cogent Engineering*, 9(1): 2083474. <https://doi.org/10.1080/23311916.2022.2083474>.
- [131] Ibrahim, A.; Yousef, A. H.; Medhat, W. (2022). DevSecOps: A security model for infrastructure as code over the cloud. 2022 2nd International Mobile, Intelligent, and Ubiquitous Computing Conference (MIUCC), Cairo, Egypt, 8–9 May 2022, pp. 284–288. <https://doi.org/10.1109/MIUCC55081.2022.9781709>.
- [132] Efendi, M.; Raharjo, T.; Suhanto, A. (2021). DevSecOps approach in software development case study: Public company logistic agency. 2021 International Conference on Informatics, Multimedia, Cyber and Information System (ICIMCIS), Virtual Conference, 28–29 Oct 2021, pp. 96–101. <https://doi.org/10.1109/ICIMCIS53775.2021.9699316>.
- [133] Underwood, M. (2023). Continuous metadata in continuous integration, stream processing and enterprise DataOps. *Data Intelligence*, 5(1): 275–288. https://doi.org/10.1162/dint_a_00193.
- [134] Ferino, S.; Fernandes, M.; Cirilo, E.; Agnez, L.; Batista, B.; Kulesza, U.; Aranha, E.; Treude, C. (2023). Overcoming challenges in DevOps education through teaching methods. 2023 IEEE/ACM 45th International Conference on Software Engineering: Software Engineering Education and Training (ICSE-SEET), pp. 166–178. <https://doi.org/10.1109/ICSE-SEET58685.2023.00022>.
- [135] Palma, S. D., Catolino, G., Di Nucci, D., Tamburri, D. A., & van den Heuvel, W. J. (2023). DevOps supports regression testing. In *Proceedings of the 3rd International Conference on Engineering, Technology and Innovative Researches (AIP Conference Proceedings, Vol. 2482, Article 140003-1-140003-9)*. AIP Publishing. <https://doi.org/10.1063/5.0110556>.

- [136] Dalla Palma, S.; Catolino, G.; Di Nucci, D.; Tamburri, D. A.; van den Heuvel, W.-J. (2022). Go serverless with RADON! A practical DevOps experience report. *IEEE Software*, 39(5): 82–90. IEEE. <https://doi.org/10.1109/MS.2022.3170153>.
- [137] Fluri, J.; Fornari, F.; Pustulka, E. (2023). Measuring the benefits of CI/CD practices for database application development. 2023 IEEE/ACM International Conference on Software and System Processes (ICSSP), 46–57. <https://doi.org/10.1109/ICSSP59042.2023.00015>.
- [138] Zeller, M. (2023). DevCertOps: Strategies to realize continuous delivery of safe software in regulated domain. 2023 IEEE/ACM 45th International Conference on Software Engineering: Companion Proceedings (ICSE-Companion '23), 334–335. <https://doi.org/10.1109/ICSE-Companion58688.2023.00094>.
- [139] Carter, K. (2017). Francois Raynaud on DevSecOps. *IEEE Software*, 34(5): 93–96. <https://doi.org/10.1109/MS.2017.3571578>.
- [140] Anjaria, D.; Kulkarni, M. (2022). Effective DevSecOps implementation: A systematic literature review. *Cardiometry*, 24: 410–417. <https://doi.org/10.18137/cardiometry.2022.24.410417>.
- [141] Ramaj, X.; Sánchez-Gordón, M.; Gkioulos, V.; Chockalingam, S.; Colomo-Palacios, R. (2022). Holding on to compliance while adopting DevSecOps: An SLR. *Electronics*, 11(22): 3707. <https://doi.org/10.3390/electronics11223707>.
- [142] Alonso, J.; Piliszek, R.; Cankar, M. (2023). Embracing IaC through the DevSecOps philosophy: Concepts, challenges, and a reference framework. *IEEE Software*, 40(1): 56–62. <https://doi.org/10.1109/MS.2022.3212194>.
- [143] Narang, P., & Mittal, P. (2022). Performance assessment of traditional software development methodologies and DevOps automation culture. *Engineering, Technology & Applied Science Research*, 12(6): 9726–9731. <https://doi.org/10.48084/etasr.5315>.
- [144] Rodríguez, C. A.; Molinari, L.; Tinetti, F. G. (2017). Evolution of handling web applications up to the current DevOps tools. 2017 International Conference on Computational Science and Computational Intelligence (CSCI), 987–992. <https://doi.org/10.1109/CSCI.2017.170>.
- [145] Alawneh, M.; Abbadi, I. M. (2022). Expanding DevSecOps practices and clarifying the concepts within Kubernetes ecosystem. 2022 9th International Conference on Software Defined Systems (SDS), Paris, France, 12–15 Dec 2022, pp. 1–7. <https://doi.org/10.1109/SDS57574.2022.10062874>.
- [146] Arseneault, E.; Boudreau, D.; Lien, J.; Young, G. (2022). Experience-based guidelines for effective planning & management of software integration & test activities in the Agile/DevSecOps environment. 2022 IEEE 29th Annual Software Technology Conference (STC). <https://doi.org/10.1109/STC55697.2022.00028>.
- [147] Narasimhulu, M.; Mounika, D. V.; Varshini, P.; Amarendra, K.; Rao, T. K. Rama Krishna. (2023). Investigating the impact of containerization on the deployment process in DevOps. 2023 2nd International Conference on Edge Computing and Applications (ICECAA), Namakkal, India, 19–21 July 2023, pp. 679–685. <https://doi.org/10.1109/ICECAA58104.2023.10212240>.
- [148] Petrović, N. (2023). Machine learning-based run-time DevSecOps: ChatGPT against traditional approach. 2023 10th International Conference on Electrical, Electronic and Computing Engineering (IcETRAN), East Sarajevo, Bosnia and Herzegovina, June 2023, pp. 1–5. <https://doi.org/10.1109/IcETRAN59631.2023.10192161>.
- [149] Sravani, D.; Reddy, J. R.; Viswas, P. S.; Jyothi, N. M.; Chandukiran, P. (2023). Python security in DevOps: Best practices for secure coding, configuration management, and continuous testing and monitoring. 2023 4th International Conference on Electronics and Sustainable Communication Systems (ICESC), Coimbatore, India, 6–8 July 2023, pp. 514–520. IEEE. <https://doi.org/10.1109/ICESC57686.2023.10193128>.
- [150] Zhang, J.; Wang, Y.; Liu, X. (2022). Cloud-native CI/CD platform. Proceedings of the International Conference on Networks, Communications and Information Technology (NCIT 2022). <https://doi.org/10.1109/NCIT56646.2022.10153748>.

- [151] Hrusto, A.; Engström, E.; Runeson, P. (2023). Towards optimization of anomaly detection in DevOps. *Information and Software Technology*, 160: 107241. <https://doi.org/10.1016/j.infsof.2023.107241>.
- [152] Zhu, L.; Bass, L.; Champlin-Scharff, G. (2016). DevOps and its practices. *IEEE Software*, 33(3): 32–34. <https://doi.org/10.1109/MS.2016.81>.
- [153] Balalaie, A.; Heydarnoori, A.; Jamshidi, P. (2016). Microservices architecture enables DevOps: Migration to a cloud-native architecture. *IEEE Software*, 33(3): 42–52. <https://doi.org/10.1109/MS.2016.64>.
- [154] Callanan, M., & Spillane, A. (2016). DevOps: Making it easy to do the right thing. *IEEE Software*, 33(3): 53–59. <https://doi.org/10.1109/MS.2016.66>.
- [155] Ebert, C.; Gallardo, G.; Hernantes, J.; Serrano, N. (2016). DevOps. *IEEE Software*, 33(3): 94–100. <https://doi.org/10.1109/MS.2016.68>.
- [156] Zhou, X.; Mao, R.; Zhang, H.; Dai, Q.; Huang, H.; Shen, H.; Li, J.; Rong, G. (2023). Revisit security in the era of DevOps: An evidence-based inquiry into DevSecOps industry. *IET Software*, 17(4): 435–454. <https://doi.org/10.1049/sfw2.12132>.
- [157] Putra, A. M.; Kabetta, H. (2022). Implementation of DevSecOps by integrating static and dynamic security testing in CI/CD pipelines. 2022 IEEE International Conference of Computer Science and Information Technology (ICOSNIKOM), Toba, Indonesia, 19–21 Oct 2022. <https://doi.org/10.1109/ICOSNIKOM56551.2022.10034883>.
- [158] Popentiu-Vladicescu, F.; Albeanu, G. (2022). Increasing SoS dependability by DevSecOps. 2022 International Conference on Emerging Technologies in Electronics, Computing and Communication (ICETECC), Jamshoro, Pakistan, 7–9 Dec 2022, pp. 271–276. <https://doi.org/10.1109/ICETECC56662.2022.10069468>.
- [159] Dupont, S.; Ginis, G.; Malacario, M.; Porretti, C.; Maunero, N.; Ponsard, C.; Massonet, P. (2021). Incremental Common Criteria certification processes using DevSecOps practices. 2021 IEEE European Symposium on Security and Privacy Workshops (EuroS&PW), Vienna (virtual), 6–10 Sept 2021, pp. 12–23. <https://doi.org/10.1109/EuroSPW54576.2021.00009>.
- [160] Bahaa, A.; Abdelaziz, A.; Sayed, A.; Elfangary, L.; Fahmy, H. (2021). Monitoring real-time security attacks for IoT systems using DevSecOps: A systematic literature review. *Information*, 12(4): 154. <https://doi.org/10.3390/info12040154>.
- [161] Ahmed, Z.; Francis, S. C. (2019). Integrating security with DevSecOps: Techniques and challenges. 2019 International Conference on Digitization (ICD), Sharjah, UAE, 18–19 Nov 2019, pp. 178–182. <https://doi.org/10.1109/ICD47981.2019.9105789>.
- [162] Jäpel, L. M. (2019). BizDevOps: A systematic literature review. Seminar IT-Management in the Digital Age (Winter 2019), FH Wedel, Germany. Available at: https://www.fh-wedel.de/fileadmin/Mitarbeiter/Records/Jaepel_2019_-_BizDevOps_-_A_Systematic_Literature_Review.pdf
- [163] Rubasinghe, I.; Meedeniya, D.; Perera, I. (2021). SAT-Analyser traceability management tool support for DevOps. *Journal of Information Processing Systems*, 17(5): 972–988. <https://doi.org/10.3745/JIPS.04.0225>.
- [164] Chasioti, K. (2019). BizDevOps: A process model for the alignment of DevOps with business goals. Master's thesis, Utrecht University. Available at: [https://studenttheses.uu.nl/bitstream/handle/20.500.12932/33534/K.Chasioti\(Final_Thesis\).pdf?sequence=1](https://studenttheses.uu.nl/bitstream/handle/20.500.12932/33534/K.Chasioti(Final_Thesis).pdf?sequence=1)
- [165] Costea, L.; Economakis, S. (2022). *Argo CD in practice: The GitOps way of managing cloud-native applications*. Packt Publishing, Birmingham—Mumbai. ISBN (print): 978-1-80323-332-1; ISBN (eBook): 978-1-80323-282-9. (Pub. date: Nov 18, 2022).

- [166] Miñón, R.; Díaz-de-Arcaya, J.; Torre-Bastida, A. I.; Hartlieb, P. (2022). Pangea: An MLOps tool for automatically generating infrastructure and deploying analytic pipelines in edge, fog and cloud layers. *Sensors*, 22(12): 4425. <https://doi.org/10.3390/s22124425>.
- [167] McZara, J.; Kafle, S.; Shin, D. (2020). Modeling and analysis of dependencies between microservices in DevSecOps. 2020 IEEE International Conference on Smart Cloud (SmartCloud), Washington, DC, USA, 6–8 Nov 2020, pp. 140–147. <https://doi.org/10.1109/SmartCloud49737.2020.00034>.
- [168] Sojan, A.; Rajan, R.; Kuvaja, P. (2021). Monitoring solution for cloud-native DevSecOps. 2021 IEEE 6th International Conference on Smart Cloud (SmartCloud), Newark, NJ, USA, 6–8 Nov 2021, pp. 125–131. <https://doi.org/10.1109/SmartCloud52277.2021.00029>.
- [169] Biswas, D. (2022). MLOps for Compositional AI. In *Challenges in Deploying and Monitoring Machine Learning Systems (DMML) (Workshop at NeurIPS 2022)*, Virtual, 9 December 2022. Available at: <https://neurips.cc/virtual/2022/62156>.
- [170] Nguyen, T. T. P. (2021). DataOps for product information management: A study of adoption readiness. Master's thesis, Arcada UAS. Available at: https://www.theseus.fi/bitstream/handle/10024/746750/Nguyen_Thi_Thanh_Phuong.pdf?sequence=3
<https://ceur-ws.org/Vol-3407/paper1.pdf>
- [171] Morais, G. (2023). Enabling Systematic Microservices Reuse Through DevOps. CAiSE 2023 Doctoral Consortium, Proceedings, CEUR Workshop Proceedings, Vol. 3407, pp. 1–10. Available at: <https://ceur-ws.org/Vol-3407/paper1.pdf>.
- [172] Dalla Palma, S.; Di Nucci, D.; Tamburri, D. A. (2020). AnsibleMetrics: A Python library for measuring Infrastructure-as-Code blueprints in Ansible. *SoftwareX*, 12: 100633. <https://doi.org/10.1016/j.softx.2020.100633>.
- [173] Mao, R.; Zhang, H.; Dai, Q.; Huang, H.; Rong, G.; Shen, H.; Chen, L.; Lu, K. (2020). Preliminary findings about DevSecOps from grey literature. 2020 IEEE 20th International Conference on Software Quality, Reliability and Security (QRS), Macau, China, 11–14 Dec 2020, pp. 450–457. <https://doi.org/10.1109/QRS51102.2020.00064>.
- [174] Rafi, S.; Yu, W.; Akbar, M. A.; Alsanad, A.; Gumaei, A. (2020). Prioritization based taxonomy of DevOps security challenges using PROMETHEE. *IEEE Access*, 8: 105426–105446. <https://doi.org/10.1109/ACCESS.2020.2998819>.
- [175] Samir, M.; Wassif, K. T.; Makady, S. H. (2023). Proactive auto-scaling approach of production applications using an ensemble model. *IEEE Access*, 11: 25008–25019. <https://doi.org/10.1109/ACCESS.2023.3256302>.
- [176] Wang, Z.; Guo, G.; Liu, C.; Zhu, W. (2022). Research on railway DevSecOps system construction based on “people–process–technology”. 2022 2nd International Signal Processing, Communications and Engineering Management Conference (ISPCEM), Montréal, Canada, 25–27 Nov 2022, pp. 19–23. <https://doi.org/10.1109/ISPCEM57418.2022.00010>.
- [177] Kendall, R. P.; Hariharan, N. S.; Sears, D. R.; Post, D. E. (2023). Successful adoption of DevOps practices in software development in DoD acquisition programs: The CREATE example. *Defense Acquisition Research Journal*, 30(2): 124–145. <https://doi.org/10.22594/dau.22-896.30.02>.
- [178] Fernandez-Gauna, B.; Rojo, N.; Graña, M. (2023). Automatic feedback and assessment of team-coding assignments in a DevOps context. *International Journal of Educational Technology in Higher Education*, 20: 17. <https://doi.org/10.1186/s41239-023-00386-6>.
- [179] Noorani, N. M.; Zamani, A. T.; Alenezi, M.; Shameem, M.; Singh, P. (2022). Factor prioritization for effectively implementing DevOps in software development organizations: A SWOT-AHP approach. *Axioms*, 11(10): 498. <https://doi.org/10.3390/axioms11100498>.
- [180] Lévy, L.-N.; Bosom, J.; Guerard, G.; Ben Amor, S.; Bui, M.; Tran, H. (2022). DevOps model approach for monitoring smart energy systems. *Energies*, 15(15): 5516. <https://doi.org/10.3390/en15155516>.

- [181] Singh, G., Chaudhary, J., & Laddhani, L. (2022). Taxonomic analysis of DevOps tools. *Journal of Algebraic Statistics*, 13(3), 2725–2731. Available at: <https://publishoa.com/index.php/journal/article/view/938>.
- [182] Miller, A. W.; Giachetti, R. E.; Van Bossuyt, D. L. (2022). Challenges of adopting DevOps for the combat systems development environment. *Defense Acquisition Research Journal*, 29(1): 22–48. <https://doi.org/10.22594/dau.21-870.29.01>.
- [183] Rafi, S.; Akbar, M. A.; AlSanad, A. A.; AlSuwaidan, L.; Abdulaziz Al-Alshaikh, H.; AlSagri, H. S. (2022). Decision-making taxonomy of DevOps success factors using Preference Ranking Organization Method for Enrichment Evaluation (PROMETHEE). *Mathematical Problems in Engineering*, 2022, Article ID 2600160, 15 pages. <https://doi.org/10.1155/2022/2600160>.
- [184] Ebert, E. (2019). DevOps and storage: APIs and flexibility key. *ComputerWeekly.com*, 10 October 2019. Available at: <https://www.computerweekly.com/feature/Storage-and-DevOps-What-is-required-for-agile-development>.
- [185] Angarita, L. B.; Fernández Del Carpio, A.; Osorio Londoño, A. A. (2023). A bibliometric analysis of DevOps metrics. *DESIDOC Journal of Library & Information Technology*, 42(6): 387–396. <https://doi.org/10.14429/djlit.42.6.18365>.
- [186] Marin, I.; Marin, M.-P.; Nicolae, C. G. (2020). Farm supply chain management improvement through the use of Lean, Agile and DevOps methodologies. *Scientific Papers. Series D. Animal Science*, 63(2), 329–332. Available at: <https://www.researchgate.net/publication/344553685>.
- [187] Skenderi, M., Luma-Osmari, S., & Imeri, F. (2020). Ethics in DevOps: The attitude of programmers towards it. *Journal of Natural Sciences and Mathematics of UT*, 5(9–10), 69–85. University of Tetova. <https://eprints.unite.edu.mk/605/1/7.pdf>.
- [188] Lacek, J.-M. (2019). Changing the DevOps culture one security scan at a time. *ISSA Journal*, 17(11), 32–37. ISSA. Available online: <https://issala.org/wp-content/uploads/2019/11/November19.pdf>.
- [189] Rice, T. (2019). Secure DevOps before DevSecOps. *ISSA Journal*, 17(11), 16–19. ISSA. Available: <https://issala.org/wp-content/uploads/2019/11/November19.pdf>.
- [190] Stoyanova, M. (2019). Smart concept for project management – transition to DevOps. *Knowledge – International Journal*, 34(1), 93–97. Available at: <https://www.researchgate.net/publication/337076816>.
- [191] Galup, S.; Dattero, R.; Quan, J. (2020). What do agile, lean, and ITIL mean to DevOps? *Communications of the ACM*, 63(10): 48–53. <https://doi.org/10.1145/3372114>.
- [192] Narang, P., & Mittal, P. (2022). Performance analysis of DevOps-based hybrid models integrated with different automation tool chains for quality software development. *International Journal on Information Technologies & Security*, 14(4): 25–30. <https://ijits-bg.com/2022.v14.i4.03>.
- [193] Liu, C.-H., & Chen, S.-L. (2022). An operation-injection approach to detect runtime permission crashes of Android apps. *Journal of Information Science and Engineering*, 38(6), 1149–1169. [https://doi.org/10.6688/JISE.202211_38\(6\).0004](https://doi.org/10.6688/JISE.202211_38(6).0004).
- [194] Wiedemann, A.; Forsgren, N.; Wiese, M.; Gewald, H.; Krcmar, H. (2019). Research for Practice: The DevOps phenomenon. *Communications of the ACM*, 62(8): 44–49. <https://doi.org/10.1145/3331138>.
- [195] Limoncelli, T. A. (2019). SQL is no excuse to avoid DevOps. *Communications of the ACM*, 62(1): 46–49. <https://doi.org/10.1145/3287299>.
- [196] Seremet, Z.; Rakic, K. (2022). Platform engineering and site reliability engineering: The path to DevOps success. In B. Katalinic (Ed.), *DAAAM International Scientific Book 2022* (Chap. 13, pp. 155–162). DAAAM International, Vienna. <https://doi.org/10.2507/daaam.scibook.2022.13>.
- [197] Metzger, S.; Durden, D.; Sturtevant, C.; Luo, H.; Pingintha-Durden, N.; Sachs, T.; Serafimovich, A.; Hartmann, J.; Li, J.; Xu, K.; Desai, A. R. (2017). eddy4R 0.2.0: A DevOps model for community-extensible processing and analysis of eddy-covariance data based on R, Git, Docker, and HDF5. *Geoscientific Model Development*, 10: 3189–3206. <https://doi.org/10.5194/gmd-10-3189-2017>.

- [198] Šeremet, Z.; Rakić, K. (2021). Best approach to security in Azure DevOps. In B. Katalinić (Ed.), DAAAM International Scientific Book 2021 (Chap. 18, pp. 223–230). DAAAM International, Vienna. <https://doi.org/10.2507/daaam.scibook.2021.18>.
- [199] Rafi, S.; Yu, W.; Akbar, M. A.; Mahmood, S.; Alsanad, A.; Gumaei, A. (2021). Readiness model for DevOps implementation in software organizations. *Journal of Software: Evolution and Process*, 33(4): e2323. <https://doi.org/10.1002/smr.2323>.
- [200] Forsgren, N.; Kersten, M. (2018). DevOps metrics. *Communications of the ACM*, 61(4): 44–48. <https://doi.org/10.1145/3159169>.
- [201] Schlossnagle, T. (2018). Monitoring in a DevOps world. *Communications of the ACM*, 61(3): 58–61. <https://doi.org/10.1145/3168505>.
- [202] Hart, M., & Burke, J. (2020). An exploratory study on the DevOps IT alignment model. *Interdisciplinary Journal of Information, Knowledge, and Management*, 15: 127–154. <https://doi.org/10.28945/4595>.
- [203] Remillard, J. (2019). DevOps and Infosec. *ISSA Journal*, November 2019. ISSA. Available online: <https://issala.org/wp-content/uploads/2019/11/November19.pdf>.
- [204] Henkel, J.; Bird, C.; Lahiri, S. K.; Reps, T. (2020). Learning from, understanding, and supporting DevOps artifacts for Docker. *Proceedings of the 42nd International Conference on Software Engineering (ICSE '20)*, Seoul, Republic of Korea, 23–29 May 2020, pp. 38–49. ACM. <https://doi.org/10.1145/3377811.3380406>.
- [205] Hémon, A.; Lyonnet, B.; Rowe, F.; Fitzgerald, B. (2020). From agile to DevOps: Smart skills and collaborations. *Information Systems Frontiers*, 22(4): 927–945. <https://doi.org/10.1007/s10796-019-09905-1>.
- [206] Ali, N.; Daneth, H.; Hong, J.-E. (2020). A hybrid DevOps process supporting software reuse: A pilot project. *Journal of Software: Evolution and Process*, 32(7): e2248. <https://doi.org/10.1002/smr.2248>.
- [207] Khan, A. A., & Shameem, M. (2020). Multicriteria decision-making taxonomy for DevOps challenging factors using analytical hierarchy process. *Journal of Software: Evolution and Process*, 32(10): e2263. <https://doi.org/10.1002/smr.2263>.
- [208] Rafi, S.; Akbar, M. A.; Mahmood, S.; Alsanad, A.; Alothaim, A. (2022). Selection of DevOps best test practices: A hybrid approach using ISM and fuzzy TOPSIS analysis. *Journal of Software: Evolution and Process*, 34(5): e2448. <https://doi.org/10.1002/smr.2448>.
- [209] Erich, F. M. A.; Amrit, C.; Daneva, M. (2017). A qualitative study of DevOps usage in practice. *Journal of Software: Evolution and Process*, 29(6): e1885. <https://doi.org/10.1002/smr.1885>.
<https://www.dbta.com/Magazine/Database-Trends-and-Applications-Magazine-February-March-2023-Issue-12538.pdf>.
- [210] Sprauve, C. (2023). DevOps TDM for hybrid and multi-cloud acceleration. *Database Trends and Applications*, Feb–Mar 2023 issue. <https://www.dbta.com/Magazine/Database-Trends-and-Applications-Magazine-February-March-2023-Issue-12538.pdf>.
- [211] Azeta, A. A.; Iboroma, D.-O. A.; Oyeyemi, I. O.; Abimbola, F. O.; Ogunde, B. (2018). A DevOps software architecture for recommender systems in digital library. In E. Ivala (Ed.), *Proceedings of the 13th International Conference on e-Learning (ICEL 2018)*, Cape Town, South Africa, 5–6 July 2018. Academic Conferences and Publishing International. <https://dl.acm.org/doi/pdf/10.1145/3529372.3533284>.
- [212] Rahman, A. (2018). Characteristics of defective infrastructure as code scripts in DevOps. *Proceedings of the 40th International Conference on Software Engineering: Companion Proceedings (ICSE-Companion '18)*, Gothenburg, Sweden, 27 May–3 June 2018, pp. 476–479. ACM. <https://doi.org/10.1145/3183440.3183452>.
- [213] Ben Mesmia, W.; Escheikh, M.; Barkaoui, K. (2021). DevOps workflow verification and duration prediction using non-Markovian stochastic Petri nets. *Journal of Software: Evolution and Process*, 33(3): e2329. <https://doi.org/10.1002/smr.2329>.

- [214] Domoney, C. (2018). DevSecOps: What skills are needed to thrive in today's app economy? *Software World*, 48(6).
- [215] Adato, L.; Geak, H. (2018). 2018 IT predictions: IoT computing, DevSecOps, security, data breaches and more. *Database and Network Journal*, 47(6).
- [216] Rowell, W., Oimoen, S., Truett, L., & Shelton, T. (2021). Applying scientific test and analysis techniques in the Agile/DevSecOps T&E environment. *The ITEA Journal of Test and Evaluation*, 42(2), June. Available at: <https://itea.org/the-itea-journal/2021-itea-journal-issue-abstracts>.
- [217] Ekoramaradhya, M., & Thorpe, C. (2022). A novel DevSecOps model for robust security in an MQTT Internet of Things. *Proceedings of the 17th International Conference on Cyber Warfare and Security (ICCWS 2022)*, Albany, NY, USA, 17–18 Mar 2022, pp. 63–71. <https://doi.org/10.34190/iccws.17.1.31>.
- [218] Martin, L. (2022). Wherefor DevSecOps? *ISSA Journal*, June 2022.
- [219] Naidoo, R.; Möller, N. (2022). Building software applications securely with DevSecOps: A socio-technical perspective. *Proceedings of the 21st European Conference on Cyber Warfare and Security (ECCWS 2022)*, Chester, UK, 16–17 June 2022, pp. 198–205. <https://doi.org/10.34190/eccws.21.1.295>.
- [220] Martin, L. (2022). DevSecOps at the U.S. DoD. *ISSA Journal*, October 2022.
- [221] Jeganathan, S. (2019). DevSecOps: A Systemic Approach for Secure Software Development. *ISSA Journal*, 17(11), 20–27. Available at: <https://issa.mydigitalpublication.com/articles/devsecops-a-systemic-approach-for-secure-software-development>.
- [222] Martin, L. (2023). DevSecOps at NIST. *ISSA Journal*, March 2023.
- [223] Buker, J. (2023). Privacy by design and default in DevSecOps. *ISSA Journal*, May 2023.
- [224] Sriraman, G.; Raghunathan, S. (2023). A systems thinking approach to improve sustainability in software engineering—A grounded capability maturity framework. *Sustainability*, 15(11): 8766. <https://doi.org/10.3390/su15118766>.
- [225] Ambade, P.; Solanki, D.; Deb, N. (2021). RV-SLC: A tool for regression validation of safety and liveness constraints on goal models in DevOps environment. *2021 IEEE 29th International Requirements Engineering Conference (RE '21)*, Notre Dame, IN, USA, 20–24 Sept 2021, pp. 452–453. <https://doi.org/10.1109/RE51729.2021.00066>.
- [226] Hernández, R.; Moros, B.; Nicolás, J. (2023). Requirements management in DevOps environments: A multivocal mapping study. *Requirements Engineering*, 28(3): 317–346. <https://doi.org/10.1007/s00766-023-00396-w>.
- [227] Kumara, I.; Mundt, P.; Tokmakov, K.; Radolović, D.; Maslennikov, A.; Sosa González, R.; Fernández-Fabeiro, J.; Quattrocchi, G.; Meth, K. Z.; Di Nitto, E.; Tamburri, D. A.; van den Heuvel, W.-J.; Meditskos, G. (2021). SODALITE@RT: Orchestrating applications on cloud-edge infrastructures. *Journal of Grid Computing*, 19(3): 29. <https://doi.org/10.1007/s10723-021-09572-0>.
- [228] Lombardi, F.; Fanton, A. (2023). From DevOps to DevSecOps is not enough: CyberDevOps—an extreme shifting-left architecture to bring cybersecurity within the software security lifecycle pipeline. *Software Quality Journal*, 31(2): 619–654. <https://doi.org/10.1007/s11219-023-09619-3>.
- [229] Mohan, V.; Ben Othmane, L. (2016). SecDevOps: Is it a marketing buzzword?—Mapping research on security in DevOps. *2016 11th International Conference on Availability, Reliability and Security (ARES)*, Salzburg, Austria, 31 Aug–2 Sep 2016, pp. 542–547. IEEE. <https://doi.org/10.1109/ARES.2016.92>.
- [230] Díaz, J.; Pérez, J. E.; López-Peña, M. A.; Mena, G. A.; Yagüe, A. (2019). Self-service cybersecurity monitoring as enabler for DevSecOps. *IEEE Access*, 7: 100283–100295. <https://doi.org/10.1109/ACCESS.2019.2930000>.

- [231] Martini, A.; Stray, V.; Besker, T.; Moe, N. B.; Bosch, J. (2023). Process debt: Definition, risks and management. SSRN Electronic Journal. <https://doi.org/10.2139/ssrn.4328073>.
- [232] Lee, J. S. (2018). The DevSecOps and agency theory. 2018 IEEE International Symposium on Software Reliability Engineering Workshops (ISSREW), 243–244. <https://doi.org/10.1109/ISSREW.2018.00013>.
- [233] Palihawadana, S.; Wijeweera, C. H.; Sanjitha, M. G. T. N.; Liyanage, V. K.; Perera, I.; Meedeniya, D. A. (2017). Tool support for traceability management of software artefacts with DevOps practices. 2017 Moratuwa Engineering Research Conference (MERCon), Moratuwa, Sri Lanka, 29–31 May 2017, pp. 129–134. IEEE. <https://doi.org/10.1109/MERCon.2017.7980469>.
- [234] Akbar, M. A.; Rafi, S.; Alsanad, A. A.; Qadri, S. F.; Alsanad, A.; Alothaim, A. (2022). Toward successful DevOps: A decision-making framework. IEEE Access, 10: 51343–51362. <https://doi.org/10.1109/ACCESS.2022.3174094>.
- [235] Kosińska, J.; Baliś, B.; Konieczny, M.; Malawski, M.; Zieliński, S. (2023). Toward the observability of cloud-native applications: The overview of the state-of-the-art. IEEE Access, 11: 73036–73052. <https://doi.org/10.1109/ACCESS.2023.3281860>.
- [236] Burkard, E. C. (2020). Usability testing within a DevSecOps environment. 2020 Integrated Communications, Navigation and Surveillance Conference (ICNS), pp. 1C1-1–1C1-7. <https://doi.org/10.1109/ICNS50378.2020.9222919>.
- [237] Verderame, L.; Caviglione, L.; Carbone, R.; Merlo, A. (2023). SecCo: Automated services to secure containers in the DevOps paradigm. Proceedings of the International Conference on Research in Adaptive and Convergent Systems (RACS '23), Gdańsk, Poland, 6–10 Aug 2023, 10:1–10:6. ACM. <https://doi.org/10.1145/3599957.3606222>.
- [238] Enoiu, E. P.; Truscan, D.; Sadovykh, A.; Mallouli, W. (2023). VeriDevOps software methodology: Security verification and validation for DevOps practices. Proceedings of the 18th International Conference on Availability, Reliability and Security (ARES '23), Benevento, Italy, Aug 29–Sept 1, 2023. ACM. <https://doi.org/10.1145/3600160.3605054>.
- [239] Winkler, F.; Westner, M. (2023). A systematic literature review of DevOps success factors and adoption models. Proceedings of the 12th International Symposium on Information and Communication Technology (SOICT '23), Ho Chi Minh City, Vietnam, 7–8 Dec 2023, pp. 525–532. ACM. <https://doi.org/10.1145/3628797.3628883>.
- [240] Andrade, Á. J. da C.; Veloso, E.; Santos, G. (2023). What we know about software dependability in DevOps – A tertiary study. SBQS '23: Proceedings of the XXII Brazilian Symposium on Software Quality, Brasília, Brazil, 7–10 Nov 2023, pp. 178–187. ACM. <https://doi.org/10.1145/3629479.3629502>.
- [241] Dunbar, W. (2024). Vendor-locked DevOps strategies: Using vendor-locked solutions and getting away with it. Communications of the ACM, 67(5): 43–44. <https://doi.org/10.1145/3633776>.
- [242] Schulz, M.; Ahmed, H.; Deng, X.; Echavarria, J.; Gammelmark, M.; Karlsson, S.; Kaya, E.; Reznak, M.; Schulz, L. B.; Tovey, M. (2024). From the Physics Lab to the Computer Lab: Towards Flexible and Comprehensive DevOps for Quantum Computing. CF '24 Companion: Proceedings of the 21st ACM International Conference on Computing Frontiers—Workshops and Special Sessions, Ischia, Italy, 7–9 May 2024, pp. 139–143. ACM. <https://doi.org/10.1145/3637543.3653432>.
- [243] Sarmiento-Calisaya, E.; Mamani-Aliaga, A.; Leite, J. C. (2024). Introducing Computer Science undergraduate students to DevOps technologies from Software Engineering fundamentals. ICSE-SEET '24: 46th IEEE/ACM International Conference on Software Engineering — Software Engineering Education and Training, Lisbon, Portugal, 14–20 Apr 2024. ACM. <https://doi.org/10.1145/3639474.3640071>.
- [244] Voggenreiter, M.; Angermeir, F.; Moyón, F.; Schöpp, U.; Bonvin, P. (2024). Automated security findings management: A case study in industrial DevOps. ICSE-SEIP '24: Proceedings of the 46th International Conference on Software Engineering—Software Engineering in Practice, Lisbon, Portugal, 14–20 Apr 2024, pp. 312–322. ACM. <https://doi.org/10.1145/3639477.3639744>.

- [245] Barrientos, A.; Duran Huanca, J. G.; Mayta Segovia, H. A. (2023). Implementation of a software engineering model with DevOps on Microsoft Azure. ICISE 2023: 8th International Conference on Information Systems Engineering, Bangkok, Thailand, 16–18 Dec 2023, pp. 1–6. ACM. <https://doi.org/10.1145/3641032.3641037>.
- [246] Kola-Olawuyi, A.; Weeraddana, N. R.; Nagappan, M. (2024). The impact of code ownership of DevOps artefacts on the outcome of DevOps CI builds. MSR '24: 21st International Conference on Mining Software Repositories, Lisbon, Portugal, 15–16 Apr 2024, pp. 543–555. ACM. <https://doi.org/10.1145/3643991.3644924>.
- [247] McIntosh, S. (2024). Mining our way back to incremental builds for DevOps pipelines. MSR '24: 21st International Conference on Mining Software Repositories — Tutorials, Lisbon, Portugal, 15–16 Apr 2024, pp. 48–49. ACM. <https://doi.org/10.1145/3643991.3649106>.
- [248] Amaro, R.; Pereira, R.; Mira da Silva, M. (2024). DevOps metrics and KPIs: A multivocal literature review. ACM Computing Surveys, 56(9): Article 231, 41 pages. <https://doi.org/10.1145/3652508>.
- [249] Azad, N., & Hyrnsalmi, S. (2024). Multivocal literature review on DevOps critical success factors. Proceedings of the 28th International Conference on Evaluation and Assessment in Software Engineering (EASE '24), Salerno, Italy, 18–21 June 2024, pp. 520–527. ACM. <https://doi.org/10.1145/3661167.3661236>.
- [250] Zohaib, M.; Altuwaijri, F. S.; Awais, M. (2024). Towards the successful execution of DevOps outsource software development process. Proceedings of the 28th International Conference on Evaluation and Assessment in Software Engineering (EASE '24), Salerno, Italy, 18–21 June 2024, pp. 534–537. ACM. <https://doi.org/10.1145/3661167.3661237>.
- [251] Rafi, S.; Akbar, M. A.; Mahmood, S. (2024). Aligning academic with industrial needs: Investigating DevOps teaching practices. Proceedings of the 28th International Conference on Evaluation and Assessment in Software Engineering (EASE '24), Salerno, Italy, 18–21 June 2024, pp. 516–519. ACM. <https://doi.org/10.1145/3661167.3661239>.
- [252] Nayanajith, A.; Wickramarachchi, R. (2024). Challenges affecting the successful adoption of DevOps practices: A systematic literature review. 2024 4th International Conference on Advanced Research in Computing (ICARC), Belihuloya, Sri Lanka, 21–24 Feb 2024. <https://doi.org/10.1109/ICARC61713.2024.10499735>.
- [253] Karunarathne, M. A. W.; Wijayanayake, W. M. J. I.; Prasadika, A. P. K. J. (2024). DevOps adoption in software development organizations: A systematic literature review. 2024 4th International Conference on Advanced Research in Computing (ICARC), Belihuloya, Sri Lanka, 21–24 Feb 2024, pp. 282–287. IEEE. <https://doi.org/10.1109/ICARC61713.2024.10499789>.
- [254] Jayakody, J. A. V. M. K.; Wijayanayake, W. M. J. I. (2024). DevOps maturity: A systematic literature review. 2024 International Research Conference on Smart Computing and Systems Engineering (SCSE), Colombo, Sri Lanka, 4 Apr 2024, pp. 437–442. <https://doi.org/10.1109/SCSE61872.2024.10550493>.
- [255] Babu, M. R.; Priya, K. R.; Harshitha, M. N.; Krishna, M. A. S.; Fayaz, S. (2024). DevOps transformation for enhanced airline booking system. 2024 3rd International Conference on Applied Artificial Intelligence and Computing (ICAAIC), Salem, India, 6–7 Jun 2024, pp. 1152–1155. IEEE. <https://doi.org/10.1109/ICAAIC60222.2024.10575224>.
- [256] Nayanajith, A.; Wickramarachchi, R. (2024). Exploring the challenges affecting the successful adoption of DevOps practices in Sri Lanka. 2024 International Research Conference on Smart Computing and Systems Engineering (SCSE), Colombo, Sri Lanka, 4 Apr 2024, p. 386. <https://doi.org/10.1109/SCSE61872.2024.10550479>.
- [257] Koren, I.; Rinker, F.; Meixner, K.; Kröger, M.; Zeng, M. (2023). Implementing DevOps practices in CPPS using microservices and GitOps. 2023 IEEE 28th International Conference on Emerging Technologies and Factory Automation (ETFA), Sinaia, Romania, 12–15 Sept 2023, pp. 1–4. <https://doi.org/10.1109/ETFA54631.2023.10275433>.

- [258] Pandiyavathi, T.; Sivakumar, B. (2024). Implementing various systems with DevOps to make successful decisions based on intelligent learning strategy. 2024 International Conference on Intelligent and Innovative Technologies in Computing, Electrical and Electronics (IITCEE), Bengaluru, India, 24–25 Jan 2024. IEEE. <https://doi.org/10.1109/IITCEE59897.2024.10468033>.
- [259] Supit, C. A.; Pangeran, A. A.; Laban, K. O. C.; Gutandjala, I. I.; Ramadhan, A. (2023). Incorporating cloud-native architecture and DevOps culture to improve company agility. 2023 3rd International Conference on Intelligent Cybernetics Technology & Applications (ICICyTA), Denpasar, Bali, Indonesia, 13–15 Dec 2023, pp. 290–294. IEEE. <https://doi.org/10.1109/ICICyTA60173.2023.10428951>.
- [260] Ali, M. S.; Puri, D. (2024). Optimizing DevOps methodologies with the integration of artificial intelligence. 2024 3rd International Conference for Innovation in Technology (INOCON), Bangalore, India, 1–3 Mar 2024, pp. 1028–1032. <https://doi.org/10.1109/INOCON60754.2024.10511490>.
- [261] Dornala, R. R.; Ponnappalli, S.; Sai, K. T.; Krishna Reddi, S. R.; Koteru, R. R.; Koteru, B. (2024). Ensemble resource allocation using optimized particle swarm optimization (PSO) in cloud computing. 2024 3rd International Conference on Sentiment Analysis and Deep Learning (ICSADL), pp. 342–348. IEEE. <https://doi.org/10.1109/ICSADL61749.2024.00062>.
- [262] Kluzek, E. (2024). Implementing CI/CD Philosophy for CTSM: The tale of b4b-dev. CESM Workshop 2024 — Software Engineering Working Group (SEWG), National Center for Atmospheric Research (<https://www.cesm.ucar.edu/sites/default/files/2024-06/2024cesmsewglkluzek.pdf>), Boulder, CO, USA, 11 June 2024.
- [263] Beckmann, N.; Lucia, B.; Gobieski, G.; Nowatzki, T.; Jackson, T. C.; Lallement, G.; Zhang, K.; Nagi, A.; Sathe, A.; Desai, H. (2024). Monza: An energy-minimal, general-purpose dataflow system-on-chip for the Internet of Things. IEEE Micro, 44(6): 52–62. <https://doi.org/10.1109/MM.2024.3426611>.
- [264] Jin, X.; Feng, Y.; Wang, C.; Liu, Y.; Hu, Y.; Gao, Y.; Xia, K.; Guo, L. (2024). PIPELINEASCODE: A CI/CD workflow management system through configuration files at ByteDance. 2024 IEEE International Conference on Software Analysis, Evolution and Reengineering (SANER), Rovaniemi, Finland, 12–15 Mar 2024, pp. 1011–1022. <https://doi.org/10.1109/SANER60148.2024.00109>.
- [265] Shubham; Saini, L. M. (2024). The impact of DevOps on software quality. 2024 Third International Conference on Smart Technologies and Systems for Next Generation Computing (ICSTSN), Villupuram, India, 18–19 Jul 2024, pp. 528–533. IEEE. <https://doi.org/10.1109/ICSTSN61422.2024.10670849>.
- [266] Gong, L. (2024). Research on lightweight ensemble algorithm for anomaly detection. 2024 IEEE 7th Advanced Information Technology, Electronic and Automation Control Conference (IAEAC), Chongqing, China, 15–17 Mar 2024, pp. 1424–1430. IEEE. <https://doi.org/10.1109/IAEAC59436.2024.10503755>.
- [267] Chung, J. (2024). DevSecOps metrics, benefits, and improvements. Doctoral dissertation, National University, School of Technology and Innovation Management, February 2024. <https://search.proquest.com/openview/f690cb1ab6a683b5744405ea8f0a754a/1>.
- [268] Kadaskar, H. R. (2024). Unleashing the power of DevOps in software development. International Journal of Scientific Research in Modern Science and Technology (IJSRMST), 3(3). <https://doi.org/10.59828/ijrmst.v3i3.185>.
- [269] Neupane, R. L.; Calyam, P.; Wang, S.; Neupane, K.; Pandey, A.; Cheng, X.; Gafurov, D.; Yeddulapalli, H. S.; Glaser, N.; Singh, K. P.; Gu, Y.; Li, S.; Srinivas, S. (2024). Online self-service learning platform for application-inspired cloud development and operations (DevOps) curriculum. IEEE Transactions on Learning Technologies, 17: 1906–1920. <https://doi.org/10.1109/TLT.2024.3428842>.
- [270] Lares, J. (2024). Near-zero downtime deployment of uWSGI applications. Master's thesis, University of Helsinki, Faculty of Science, Master's Programme in Computer Science, 18 December 2024. <https://helda.helsinki.fi/bitstreams/fe173cc2-64f6-4f08-94aa-9df67ef79ebf/download>.
- [271] Bryant, L.; Gardner, R. W.; Hu, F.; Jordan, D.; Taylor, R. P. (2024). Kubernetes deployment options for on-prem clusters. arXiv preprint arXiv:2407.01620, 28 June 2024.

- [272] Rajab, R.; Alnoukari, M. (2025). DevOps integration with Capability Model Maturity Integration: A systematic mapping review. *IEEE Access*, 13: 31829–31841. <https://doi.org/10.1109/ACCESS.2025.3542630>.
- [273] Patel, H.; Ramanan, B. A.; Khan, M. A.; Williams, T.; Friedman, B.; Drabeck, L. (2024). Automating code adaptation for MLOps—A benchmarking study on LLMs. *arXiv preprint arXiv:2405.06835*, 10 May 2024.
- [274] Bhatt, N. (2024). TESTynamo: An automated end-to-end CI/CD integrated GAN-powered unit test generator. Graduate project report, Concordia Institute for Information Systems Engineering (CIISE), Concordia University, Montreal, Canada.
- [275] Penagos, D. T.; Agudelo, N. (2024). Agile testing using user language automation with artificial intelligence in Enjisst. 2024 IEEE Latin American Conference on Computational Intelligence (LA-CCI). <https://doi.org/10.1109/LA-CCI62337.2024.10814749>.
- [276] Belouettar, S.; Amor, M. B.; Patzák, B.; Hu, H. (2025). DeeMa-Hub: Cloud-enabled semantic platform for data-driven multiscale co-design and co-simulation of composite materials and structures. *Composite Structures*, 360: 118980. <https://doi.org/10.1016/j.compstruct.2025.118980>.
- [277] Nieto, T. L.; González-Bañales, D. L. (2024). Good practices in ITSM learning: A business simulation-game-based approach for engineering students. 2024 IEEE Global Engineering Education Conference (EDUCON). <https://doi.org/10.1109/EDUCON60312.2024.10578827>.
- [278] Maulana, I.; Umar, R.; Yudhana, A. (2024). Implementasi deployment layanan website menggunakan Kubernetes dengan CI/CD Jenkins. *Jurnal SAINTIKOM (Jurnal Sains Manajemen Informatika dan Komputer)*, 23(2): 290–296. <https://doi.org/10.53513/jis.v23i2.9992>.
- [279] Gilbertson, C.; Milewicz, R.; Berquist, E.; Brundage, A.; Engelmann, J.; et al. (2024). Towards long-term scientific model sustainment at Sandia National Laboratories. *Proceedings of the 39th IEEE/ACM International Conference on Automated Software Engineering (ASE '24)*, Sacramento, CA, USA, 27 Oct–1 Nov 2024. <https://doi.org/10.1145/3691620.3695276>.
- [280] Tabbassum, A.; Malik, V.; Singh, J.; Surendranath, N. (2024). Integrating site reliability engineering principles with DevSecOps for enhanced security posture. 2024 International Conference on Intelligent Systems and Advanced Applications (ICISAA), Pune, India, Oct 2024. IEEE. <https://doi.org/10.1109/ICISAA62385.2024.10828869>.
- [281] Naresh, N. E.; Murthy, S. V. N.; Merikapudi, S.; Sreenivasa, N.; Rakhi, R. K. C. (2024). Continuous integration, testing, deployment and delivery in DevOps. 2024 International Conference on Knowledge Engineering and Communication Systems (ICKECS). IEEE. <https://doi.org/10.1109/ICKECS61492.2024.10616918>.
- [282] Begoug, M.; Chouchen, M.; Ouni, A. (2024). TerraMetrics: An open-source tool for infrastructure-as-code (IaC) quality metrics in Terraform. 2024 IEEE/ACM 32nd International Conference on Program Comprehension (ICPC), Lisbon, Portugal. <https://doi.org/10.1145/3643916.3644439>.
- [283] Vennapureddy, S. (2024). Poster: Vault Clinical Operations Management System (CTMS) used in research organizations for fast medicine & therapies available to the public. 2024 IEEE/ACM Conference on Connected Health: Applications, Systems and Engineering Technologies (CHASE). <https://doi.org/10.1109/CHASE60773.2024.00036>.
- [284] Nadeem, K.; Aslam, S. (2024). Cloud-native DevOps strategies: Redefining enterprise architecture with artificial intelligence. <https://www.researchgate.net/publication/337076816> preprint, Nov 2024. <https://doi.org/10.13140/RG.2.2.13823.83369>.
- [285] Sartaj, H.; Ali, S.; Gjøby, J. M. (2024). REST API testing in DevOps: A study on an evolving healthcare IoT application. *arXiv preprint arXiv:2410.12547v2*.
- [286] Wadhams, Z.; Reinhold, A. M.; Izurieta, C. (2024). Automating static code analysis through CI/CD pipeline integration. In 2024 IEEE International Conference on Software Analysis, Evolution and Reengineering – Companion (SANER-C). <https://doi.org/10.1109/SANER-C62648.2024.00021>.

- [287] Zheng, L.; Chen, Z.; Wang, D.; Deng, C.; Matsuoka, R.; Chen, H. (2024). LEMMA-RCA: A large multi-modal multi-domain dataset for root cause analysis. arXiv preprint arXiv:2406.05375.
- [288] Laitinen, A. K. (2024). Ensuring development efficiency with DevSecOps: A case study on streamlining dependency vulnerability management with Dependabot (Master's thesis, University of Jyväskylä, Faculty of Information Technology). https://thesis.jyu.fi/Record/jyx.123456789_95861.
- [289] Bonati, L.; Polese, M.; D'Oro, S.; del Prever, P. B.; Melodia, T. (2025). 5G-CT: Automated deployment and over-the-air testing of end-to-end open radio access networks. *IEEE Communications Magazine*, 63(1): 155–161. <https://doi.org/10.1109/MCOM.001.2300675>.
- [290] Carrillo-de-Gea, J. M.; Ebert, C.; Hosni, M.; Vizcaíno, A.; Nicolás, J.; Fernández-Alemán, J. L. (2024). Tools for requirements engineering. *IEEE Software*, 41(4): 30–33. <https://doi.org/10.1109/MS.2024.3385466>.
- [291] Sánchez Ruiz, J.-M., Domínguez Mayo, F. J., Oriol, X., Crespo, J. F., Benavides, D., & Teniente, E. (2024). A performance evaluation model for DevOps practices on open source software projects. SSRN Preprint. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4987820.
- [292] Khan, A. A.; Khan, J. A.; Akbar, M. A.; Zhou, P.; Fahmideh, M. (2024). Insights into software development approaches: Mining Q&A repositories. *Empirical Software Engineering*, 29: Article 8. <https://doi.org/10.1007/s10664-023-10417-5>.
- [293] Abbas, S. I.; Garg, A. (2024). AIOps in DevOps: Leveraging artificial intelligence for operations and monitoring. 2024 3rd International Conference on Sentiment Analysis and Deep Learning (ICSADL). IEEE. <https://doi.org/10.1109/ICSADL61749.2024.00016>.
- [294] Demchenko, Y. (2024). The importance of system engineering competences and knowledge in large scale digital research infrastructure projects. 2024 IEEE Global Engineering Education Conference (EDUCON). <https://doi.org/10.1109/EDUCON60312.2024.10578607>.
- [295] Vu, D. T. (2024). CI/CD automation's impact on microservices project management (Bachelor's thesis, Turku University of Applied Sciences, Degree Programme in Information and Communications Technology). https://www.theseus.fi/bitstream/10024/876023/2/Vu_Dieu%20Thu.pdf.
- [296] Feio, C.; Santos, N.; Escravana, N.; Pacheco, B. (2024). An empirical study of DevSecOps focused on continuous security testing. 2024 IEEE European Symposium on Security and Privacy Workshops (EuroS&PW). IEEE. <https://doi.org/10.1109/EuroSPW61312.2024.00074>.
- [297] Ágh, J.; Makarova, I.; Dakić, P. (2024). Enhancing SEO compliance of JavaScript web frameworks through automated testing inside CI/CD pipelines. In *SQAMIA 2024: Workshop on Software Quality, Analysis, Monitoring, Improvement, and Applications*, <https://ceur-ws.org/Vol-3407/paper1.pdf> Proceedings, Vol. 3845, paper 9 (12 pp).
- [298] Duong, T. N. B.; Meng, C. Y. (2024). Automatic grading of short answers using large language models in software engineering courses. 2024 IEEE Global Engineering Education Conference (EDUCON). <https://doi.org/10.1109/EDUCON60312.2024.10578839>.
- [299] Sokolowski, D. (2024). Reliable infrastructure as code for decentralized organizations (Doctoral dissertation, University of St. Gallen), Diss. No. 5462. <https://www.alexandria.unisg.ch/entities/publication/4ce8fbcf-2b17-46c8-920e-d60a864b63d5/full>.
- [300] Irfan, K.; Daniel, M. (2024). AI-augmented DevOps: A new paradigm in enterprise architecture and cloud management. <https://www.researchgate.net/publication/337076816> preprint, Nov 2024. <https://doi.org/10.13140/RG.2.2.24729.02406>.
- [301] Subramaniam, R., Palakkeel, P., Arunmozhi, M., Uthayakumar, U., & Sridharan, M. (2024). Sustainable data analysis in engineering: 6-year review emphasizing open-source tools, eco-friendly practices, and emerging trends for smart computing. In 2024 10th International Conference on Smart Computing and Communication (ICSCC). IEEE. <https://doi.org/10.1109/ICSCC62041.2024.10690466>.

- [302] Bělka, P. (2024). Cloud-native and microservice application development (Bachelor's thesis, Czech Technical University in Prague, Faculty of Electrical Engineering, Department of Computer Science). <https://dspace.cvut.cz/handle/10467/115489>.
- [303] Saurabh, S. K., & Kumar, D. (2024). Model to reduce DevOps pipeline execution time using SAST. *International Journal of System Assurance Engineering and Management*, 15(5), 1999–2009. <https://doi.org/10.1007/s13198-024-02262-6>.
- [304] Nayak, K., Route, S., Sundararajan, M., Jain, A., & S. R. (2024). Sustainable continuous testing in DevOps pipeline. In 2024 First IEEE International Conference on Communications and Computer Science (INCCCS). IEEE. <https://doi.org/10.1109/INCCCS60947.2024.10593566>.
- [305] Zhang, Y., Wu, Y., Chen, T., Wang, T., Liu, H., & Wang, H. (2024). How do developers talk about GitHub Actions? Evidence from online software development community. In Proceedings of the 2024 IEEE/ACM 46th International Conference on Software Engineering (ICSE '24). ACM. <https://doi.org/10.1145/3597503.3623327>.
- [306] Sirisha, N., Kiran, A., Arshad, M., & Mounika, M. (2024). Automating ML models using MLOps. In 2024 International Conference on Advancements in Smart, Secure and Intelligent Computing (ASSIC). IEEE. <https://doi.org/10.1109/ASSIC60049.2024.10507923>.
- [307] Kumar, O., & Narang, A. (2025). Securing microservices: Challenges and solutions. *International Journal of Innovative Research in Computer Science and Technology (IJIRCST)*, 13(1), 58–61. <https://doi.org/10.55524/ijircst.2025.13.1.8>.
- [308] Saeed, H., & Daniel, M. (2024). Smart enterprise architecture: Leveraging AI, cloud, and agile DevOps practices. <https://www.researchgate.net/publication/337076816>. <https://doi.org/10.13140/RG.2.2.18018.13761>.
- [309] Saleh, S. M., Madhavji, N., & Steinbacher, J. (2024). A systematic literature review on continuous integration and deployment (CI/CD) for secure cloud computing. In Proceedings of the 20th International Conference on Web Information Systems and Technologies (WEBIST 2024), 331–341. <https://doi.org/10.5220/0013018500003825>.
- [310] Bernardino, N. A., Sequeira, B., Piza, E., Henriques, F., Neves, F., & Reis, C. I. (2024). Enhancing DevSecOps: Three custom tools for continuous security. In 2024 IEEE 11th International Conference on Cyber Security and Cloud Computing (CSCloud). IEEE. <https://doi.org/10.1109/CSCloud62866.2024.00017>.
- [311] Fannouch, A., Gharib, J., & Gahi, Y. (2025). Enhancing DataOps practices through innovative collaborative models: A systematic review. *International Journal of Information Management Data Insights*, 5, 100321. <https://doi.org/10.1016/j.jjime.2025.100321>.
- [312] Bruno, F. (2024). Automation and provisioning of Kubernetes on bare-metal telco edge infrastructures (Master's thesis, Politecnico di Torino). <https://webthesis.biblio.polito.it/31755/1/tesi.pdf>.
- [313] Abdulrahman, M., Alqahtani, R., & Bashir, A. (2024). Review of opportunities, challenges and solutions for FaaS performance and security management. In Proceedings of the 7th International Conference on Inventive Computation Technologies (ICICT 2024). IEEE. <https://doi.org/10.1109/ICICT60155.2024.10544765>.
- [314] Dakić, P. (2024). Software compliance in various industries using CI/CD, dynamic microservices, and containers. *Open Computer Science*, 14, 20240013. <https://doi.org/10.1515/comp-2024-0013>.

- [315] Zarour, M., Akour, M., & Alenezi, M. (2024). Enhancing DevOps engineering education through system-based learning approach. *Open Education Studies*, 6, 20240012. <https://doi.org/10.1515/edu-2024-0012>.
- [316] Kağızmandere, Ö., & Arslan, H. (2024). Vulnerability analysis based on software bill of materials (SBOM): A model proposal for automated vulnerability scanning for CI/CD pipelines. *International Journal of Information Security Science*, 13(2), 33–42. <https://doi.org/10.55859/ijiss.1455039>.
- [317] Schubaur, P., Knauer, P., & Merli, D. (2025). Threats to the IoT device production processes – a blind spot in the product security lifecycle. In *IFIP AICT 738: IFIP International Conference on IoT (IFIPIoT 2024)*, 87–103. Springer. https://doi.org/10.1007/978-3-031-82065-6_7.
- [318] Silva-Martinez, J., & Desharnais, A. (2024). NASA Agile Community of Practice: 2023–2024 report. NASA/TM–20240010785. National Aeronautics and Space Administration. <https://ntrs.nasa.gov/api/citations/20240010785/downloads/20240010785.pdf>.
- [319] Avritzer, A., Cusick, J. J., Janes, A., Camilli, M., Russo, B., Trubiani, C., & van Hoorn, A. (2024). Dependability modeling in an industrial environment. In *2024 IEEE 35th International Symposium on Software Reliability Engineering Workshops (ISSREW)*. IEEE. <https://doi.org/10.1109/ISSREW63542.2024.00045>.
- [320] Haider, Z., & Yang, J. (2024). Revolutionizing enterprise architecture: Harnessing AI and cloud synergy with DevOps integration. <https://www.researchgate.net/publication/337076816>. <https://doi.org/10.13140/RG.2.2.33117.63200>.
- [321] Bhardwaj, A. K., Dutta, P. K., & Chintale, P. (2024). Securing container images through automated vulnerability detection in shift-left CI/CD pipelines. *Babylonian Journal of Networking*, 2024, 162–170. <https://doi.org/10.58496/BJN/2024/016>.
- [322] Seow, W. H., Lim, C. Y., & Ang, S. L. (2025). Random forest model for software build time prediction on CI/CD pipeline. *Pertanika Journal of Science & Technology*, 33(2), 1031–1048. <https://doi.org/10.47836/pjst.33.2.22>.
- [323] Krasnov, A., & Maiti, R. (2024). Overview of DevSecOps frameworks for software development lifecycle and its current limitations. *Journal of Software Engineering Practice*, 5(1).
- [324] Rahman, M. M., Barek, M. A., Akter, M. S., Riad, A. K. I., Rahman, M. A., Shahriar, H., Rahman, A., & Wu, F. (2024). Authentic learning on DevOps security with Labware: Git hooks to facilitate automated security static analysis. In *2024 IEEE 48th Annual Computers, Software, and Applications Conference (COMPSAC)*. IEEE. <https://doi.org/10.1109/COMPSAC61105.2024.00388>.
- [325] Cardoen, G. (2024). Towards an empirical analysis of code cloning and code reuse in CI/CD ecosystems. In *Proc. 23rd Belgium–Netherlands Software Evolution Workshop (BENEVOL 2024)*, CEUR-WS Vol-3941, 63–75. https://ceur-ws.org/Vol-3941/BENEVOL2024_TECH_paper8.pdf.
- [326] Linder, S., Lisetska, P., & Stutz, R. (2024). Network configuration automation with Infracore and Normir (Semester project, Department of Computer Science, OST – Eastern Switzerland University of Applied Sciences). <https://eprints.ost.ch/id/eprint/1272/1/HS%202024%202025-SA-EP-Stutz-Linder-Lisetska-Infrastructure%20as%20Code%20%E2%80%94%20Open%20Slot.pdf>.
- [327] Dubovský, D. (2024). User interface for data and metadata management in the EnviLab platform (Master's thesis, Faculty of Informatics, Masaryk University, Brno). <https://is.muni.cz/th/p3oom>.
- [328] Ho-Dac, H., & Vo, V. L. (2024). An approach to enhance CI/CD pipeline with open-source security tools. *European Modern Studies Journal*, 8(3), 408–411. [https://doi.org/10.59573/emsj.8\(3\).2024.30](https://doi.org/10.59573/emsj.8(3).2024.30).

- [329] Mack, S. D. (2024). *The DevSecOps playbook: Deliver continuous security at speed*. John Wiley & Sons. <https://doi.org/10.1002/9781394189281>.
- [330] Ratuszniak, O., & Cabrero-Daniel, B. (2024). Designing digital twins for enhanced reusability. In 2024 IEEE/ACM 46th International Conference on Software Engineering: Companion Proceedings (ICSE-Companion '24). ACM. <https://doi.org/10.1145/3639478.3643102>.
- [331] Manchana, R. (2024). The power of convergence: Platform Ops as the unifying force for DevOps, DataOps, and MLOps. *International Journal of Science and Research (IJSR)*, 13(9). <https://doi.org/10.21275/SR24831222641>.
- [332] Philippe, J., Omond, A., Coullon, H., Prud'Homme, C., & Räis, I. (2024). Fast choreography of cross-DevOps reconfiguration with Ballet: A multi-site OpenStack case study. In 2024 IEEE International Conference on Software Analysis, Evolution and Reengineering (SANER). IEEE. <https://doi.org/10.1109/SANER60148.2024.00007>.
- [333] Llosa, S., Forrester, C., Weihe, G., Collaco, O., & Gifford, K. (2025). Improved radio astronomy interference characterization using DevOps. In 2025 USNC-URSI National Radio Science Meeting (NRSM), Boulder, CO, USA. https://www.researchgate.net/publication/389657281_Improved_Radio_Astronomy_Interference_Characterization_Using_DevOps.
- [334] Kersbergen, B., Sprangers, O., Kootte, F., Guha, S., de Rijke, M., & Schelter, S. (2024). ETUDE – Evaluating the inference latency of session-based recommendation models at scale. In 2024 IEEE 40th International Conference on Data Engineering (ICDE). IEEE. <https://doi.org/10.1109/ICDE60146.2024.00389>.
- [335] Ali, M. I., Di Modica, G., Rondinini, G., Sabbioni, A., & Corradi, A. (2024). Function profiling to support smart serverless service deployment in the cloud continuum. In 2024 IEEE Symposium on Computers and Communications (ISCC). IEEE. <https://doi.org/10.1109/ISCC61673.2024.10733710>.
- [336] Rajendra, A., Reddy, S. P., Vignesh, B. S. P., & Rao, T. S. M. (2024). Setting up a CI/CD pipeline in the cloud for a web application. In 2024 International Conference on Expert Clouds and Applications (ICOECA). IEEE. <https://doi.org/10.1109/ICOECA62351.2024.00048>.
- [337] Sanagana, D. P. R., & Tummalachervu, C. K. (2024). Securing cloud computing environment via optimal deep learning-based intrusion detection systems. In 2024 Second International Conference on Data Science and Information System (ICDSIS). IEEE. <https://doi.org/10.1109/ICDSIS61070.2024.10594404>.
- [338] Chiş, A., Bălan, T.-C., Cotfas, P.-A., & Cotfas, D.-T. (2024). Deployment of a security solution for cloud. In 2024 15th International Conference on Communications (COMM). IEEE. <https://doi.org/10.1109/COMM62355.2024.10741412>.
- [339] Wäspi, R. (2024). Quantitative maturity assessment of DevSecOps practices using metrics (Independent study, University of Zurich, Communication Systems Group). <https://files.ifl.uzh.ch/CSG/staff/vonderassen/extern/theses/is-waespi.pdf>.
- [340] Stefani, T., Christensen, J. M., Hoemann, E., Giriya, A. A., Köster, F., Krueger, T., & Hallerbach, S. (2024). Applying model-based system engineering and DevOps on the implementation of an AI-based collision avoidance system. In 33rd Congress of the International Council of the Aeronautical Sciences (ICAS 2024). https://www.icas.org/icas_archive/icas2024/data/papers/icas2024_0869_paper.pdf.
- [341] Kahlhofer, M., & Rass, S. (2024). Application layer cyber deception without developer interaction. arXiv preprint arXiv:2405.12852.

- [342] Prates, L., & Pereira, R. (2025). DevSecOps practices and tools. *International Journal of Information Security*, 24, 11. <https://doi.org/10.1007/s10207-024-00914-z>.
- [343] Samir, A., Dagenborg, H., & EL Ioini, N. (2025). Misconfiguration of cluster and IoT systems recovery: Extended experiments. Conference paper, The Arctic University of Norway and University of Nottingham Malaysia. https://doi.org/10.1007/978-3-031-68165-3_8.
- [344] Chaabi, Y., Alwesabi, K., Lekdioui, K., Haider, H., Al-Ashmoery, Y., Al-Fuhaidi, B., & Zaid, A. (2024). Impact of integrating Lean, Agile, and DevOps with ITIL4 framework for modern IT service management. In *2024 1st International Conference on Emerging Technologies for Dependable Internet of Things (ICETI)*. IEEE. <https://doi.org/10.1109/ICETI63946.2024.10777262>.
- [345] Dacheppally, R. (2025). DevSecOps: Shifting security left with automated scanning tools. *International Journal of Advances in Developmental Research (IJAIDR)*, 16(1). [345] DOI: <https://doi.org/10.5281/zenodo.14993296>.
- [346] Badampudi, D., Usman, M., & Chen, X. (2025). Large scale reuse of microservices using CI/CD and InnerSource practices – a case study. *Empirical Software Engineering*, 30, 41. <https://doi.org/10.1007/s10664-024-10595-w>.
- [347] Chai, S., & Ostrowski, S. (2024). Optimized and secured AI for the tactical edge. *Proceedings of SPIE*, 13054, 130540P. <https://doi.org/10.1117/12.3014167>.
- [348] Mascheroni, M., Dost, J. M., Bockelman, B. P., et al. (2025). Exploiting Kubernetes to simplify the deployment and management of the multi-purpose CMS pilot job factory. CMS Conference Report CMS CR-2025/045, presented at CHEP 2024, CERN. <https://indico.cern.ch/event/1338689/contributions/6011013>.
- [349] Chowdhary, N., Dutta, T., Chattopadhyay, S., & Chakraborty, S. (2025). AutoPAC: Exploring LLMs for automating policy-to-code conversion in business organizations. Conference paper, IIT Kharagpur & IDRBT Hyderabad. DOI: <https://doi.org/10.1109/COMSNETS63942.2025.10885751>.
- [350] Shantyr, A. (2024). Methodology for quantitative assessment of quality characteristics in distributed systems and microservices. In *2024 IEEE 5th KhPI Week on Advanced Technology (KhPIWeek)*. IEEE. <https://doi.org/10.1109/KHPIWEEK61434.2024.10878097>.
- [351] Puppala, R., Goutham, P., Rohan, S. A., Sainadh, J. T. K., & David, T. J. (2025). Serverless computing and DevOps: A synergistic approach to modern software development. In *EAI International Conference on Computational Intelligence and Generative AI (EAI/Springer Innovations in Communication and Computing)*, pp. 123–137. Springer. https://doi.org/10.1007/978-3-031-76610-7_9.
- [352] Chen, M., Liang, B., & Lu, X. (2024). The practice and application of a novel DevSecOps platform on security. In *2024 5th International Seminar on Artificial Intelligence, Networking and Information Technology (AINIT)*. IEEE. <https://doi.org/10.1109/AINIT61980.2024.10581700>.
- [353] Farao, A. (2024). Advanced cyber security solutions for critical infrastructure and applications: Smart grid and cyber insurance (PhD thesis, Department of Digital Systems, University of Piraeus). <https://dione.lib.unipi.gr/xmlui/handle/unipi/16272>.
- [354] Sravan, N. K., Theeda, A., Vuyyuru, V. A., Peddineni, L., & Gottapu, P. (2024). Agile management tools: Technological evaluations and future archetype. In *2024 4th International Conference on Pervasive Computing and Social Networking (ICPCSN)*, pp. 914–918. IEEE. <https://doi.org/10.1109/ICPCSN62568.2024.00154>.
- [355] Saxena, A., Yadav, N. L., Singh, S., Yang, T., Prakash, S., Rathore, R. S., & Singh, S. (2024). Design and implementation of Flutter based multi-platform Docker controller app. In *2024 International Conference on Decision Aid Sciences and Applications (DASA)*. IEEE. <https://doi.org/10.1109/DASA63652.2024.10836516>.

- [356] Martinez, D. R., & Kifle, B. M. (2024). Artificial intelligence: A systems approach from architecture principles to deployment. MIT Press (Lincoln Laboratory Series). <https://doi.org/10.7551/mitpress/14806.001.0001>.
- [357] Wilkes, B., Milani, A. M. P., & Storey, M.-A. (2023). A framework for automating the measurement of DevOps Research and Assessment (DORA) metrics. In Proceedings of the 2023 IEEE International Conference on Software Maintenance and Evolution (ICSME). IEEE. DOI: 10.1109/ICSME58846.2023.00018.
- [358] Jonášová, A. (2024). Web application for visual analysis of geospatial information of natural ecosystems (Master's thesis, Faculty of Informatics, Masaryk University, Brno). <https://is.muni.cz/th/i0k61/>
- [359] Esther, D. (2024). The role of CI/CD in accelerating software deployment cycles. <https://www.researchgate.net/publication/337076816>.
- [360] Herati, D. A., Aderne, M. C., & Kon, F. (2025). The road to sustainable DevOps. arXiv preprint (arXiv:2503.08845). <https://doi.org/10.48550/arXiv.2503.08845>.
- [361] Mahimalur, R. K. (2024). The ephemeral DevOps pipeline: Building for self-destruction (a ChaosSecOps approach). Technical White Paper. Confluence. DOI: <https://doi.org/10.2139/ssrn.5182397>.
- [362] Peng, Y. (2024). Development and design of an intelligent financial asset management system based on big data analysis and Kubernetes. *Procedia Computer Science*, 243, 482–489. Elsevier. <https://doi.org/10.1016/j.procs.2024.09.059>.
- [363] Liu, Y.-C., & Kuo, C.-L. (2024). Constructing spatio-temporal disaster knowledge graph from social media. *AGILE: GIScience Series*, 5, 37. <https://doi.org/10.5194/agile-giss-5-37-2024>.
- [364] AlSelek, M., Alcaraz-Calero, J. M., & Wang, Q. (2024). Agile AI and firmware management in IoT: DevOps for low-power microcontroller-based platforms. In 2024. <https://doi.org/10.1109/KHPIWEEK61434.2024.10878101>.
- [365] Tatineni, S., & Allam, K. (2024). DevOps security: Integrating security into the DevOps workflow. *EPH – International Journal of Humanities and Social Science*, 10(1). <https://doi.org/10.53555/ephijse.v10i1.233>.
- [366] Khan, A. (2024). Comparison of public cloud platforms using automated CI/CD pipelines (Master's thesis, ICT Innovation Programme, Major in Cloud and Network Infrastructures, Knowit Connectivity AB, Sweden, KTH Royal Institute of Technology and Aalto University). <https://www.diva-portal.org/smash/get/diva2%3A1900413/FULLTEXT01.pdf>.
- [367] Tua, A. (2018). The influence of DevOps on software development life cycle acceleration. *International Journal of Advanced and Innovative Research*, 8(1), 812–815. (No DOI). Available: https://www.researchgate.net/publication/383425400_The_Influence_of_DevOps_on_Software_Development_life_Cycle_Acceleration
- [368] Hamza, U., Syed-Mohamad, S. M., & Abdullah, N. L. (2024). Exploring the benefits, challenges and guidelines of DevOps adoption: A systematic literature review and an empirical study. *Journal of Mathematical Sciences and Informatics*, 4(2), 89–105. <https://doi.org/10.46754/jmsi.2024.10.007>.
- [369] Lumpp, F., Braga, D., Fummi, F., & Bombieri, N. (2024). Automating FinOps in cloud computing: An integrated solution for efficient data collection with dynamic scraper generation. In 2024 IEEE International Conference on Cloud Computing Technology and Science (CloudCom). IEEE. <https://doi.org/10.1109/CloudCom62794.2024.00025>.
- [370] Khan, H. H., Zubair, S., Nasim, F., Akhter, S., Ghazanfar, M. N., & Azeem, S. (2024). Role of Kubernetes in DevOps technology for the effective software product management. *Journal of Computing & Biomedical Informatics*, 7(1). <https://doi.org/10.56979/701/2024>.
- [371] Vadde, B. C., & Munagandla, V. B. (2024). DevOps in the age of machine learning: Bridging the gap between development and data science. *International Journal of Machine Learning Research in Cybersecurity and Artificial Intelligence*, 15(1).

https://www.researchgate.net/publication/385720844_DevOps_in_the_Age_of_Machine_Learning_Bridging_the_Gap_Between_Development_and_Data_Science.

[372] Ning, A. (2024). A container-based continuous integration development and operations platform. In 2024 IEEE 4th International Conference on Power, Electronics and Computer Applications (ICPECA). IEEE. <https://doi.org/10.1109/ICPECA60615.2024.10470998>.

[373] Tonesh, K., Vamsi, M., Nikitha, B., Tagore, S., & Rao, G. R. K. (2024). Transforming software delivery: A comprehensive exploration of DevOps principles, practices, and implications. *Journal of Data Acquisition and Processing*, 39(1). <https://doi.org/10.5281/zenodo.754687>.

[374] Nordström, A. (2024). Exploring GitOps for smaller-scale applications (Master's thesis, Faculty of Science and Engineering, Åbo Akademi University). <https://www.doria.fi/bitstream/handle/10024/188668>.

[375] Kovalev, D. (2024). Redefining data centers: Hetzner's cost-effective cloud solution (Bachelor's thesis, Metropolia University of Applied Sciences, Degree Programme in Information Technology). https://www.theseus.fi/bitstream/handle/10024/866835/Kovalev_Dmitry.pdf

[376] Rinker, F., Meixner, K., Vysoká, D., & Biffl, S. (2024). The MDM-CPPS framework: GitOps-enabled multi-domain modeling in cyber-physical production systems engineering. Technical Report No. CDL-SQI 2024-01. TU Wien, Austria. <https://doi.org/10.34726/5475>.

[377] Agbozo, E., & Gawali, M. B. (2025). A conceptual framework for improving effectiveness of DevOps teams – An organizational systems perspective. *AIP Conference Proceedings*, 3175, 020086. <https://doi.org/10.1063/5.0254098>.

[378] Ankit, A., Nimala, K., & Jadhav, M. (2024). Creation of continuous integration continuous deployment pipeline using cloud. In 2024 5th International Conference on Intelligent Communication Technologies and Virtual Mobile Networks (ICICV). IEEE. <https://doi.org/10.1109/ICICV62344.2024.00058>.

[379] Lykousas, N., Argyropoulos, V., & Casino, F. (2024). The potential of LLM-generated reports in DevSecOps. *arXiv preprint, arXiv:2410.01899*. <https://doi.org/10.48550/arXiv.2410.01899>.

[380] Faqih, A. R., Taufiqurrahman, A., Husen, J. H., & Sabariah, M. K. (2024). Empirical analysis of CI/CD tools usage in GitHub Actions workflows. *Journal of Informatics and Web Engineering*, 3(2). <https://doi.org/10.33093/jiwe.2024.3.2.18>.

[381] Sadovykh, A., & Ivanov, V. (2024). Enhancing DevSecOps with continuous security requirements analysis and testing. *Computer Research and Modeling*, 16(7), 1687–1702. <https://doi.org/10.20537/2076-7633-2024-16-7-1687-1702>.

[382] Setyoko, A. D., & Zahra, A. (2024). Efficiency comparison of multi-environment CI/CD processes through parallel and sequential implementations. *MALCOM: Indonesian Journal of Machine Learning and Computer Science*, 4(3), 911–925. <https://doi.org/10.57152/malcom.v4i3.1334>.

[383] Zulfikar, A., & Akbar, Y. (2025). Automation of Mikrotik router setting configuration backup using Ansible with network DevOps method. *MALCOM: Indonesian Journal of Machine Learning and Computer Science*, 5(1), 57–66. <https://doi.org/10.57152/malcom.v5i1.1591>.

[384] Kuncara, A. B., Kusumo, D. S., & Adrian, M. (2024). Comparison of Jenkins and GitLab CI/CD to improve delivery time of Basu Dairy Farm admin website. *Jurnal Teknik Informatika (JUTIF)*, 5(3), 747–756. <https://doi.org/10.52436/1.jutif.2024.5.3.1836>.

[385] Mahimalur, R. K. (2024). Immutable secrets management: A zero-trust approach to sensitive data in containers. Technical White Paper, Confluence. Available: <https://ssrn.com/abstract=5164225>.

[386] Patel, A. (2019). Research the use of machine learning models to predict and prevent failures in CI/CD pipelines and infrastructure. *International Journal of Artificial Intelligence and Machine Learning in Engineering*, SSRN Electronic Journal. Available: <https://www.researchgate.net/publication/384695412>.

[387] Salvati, M. (2024). DevOps for movie and cartoon production. In *SA Courses '24* (December 03–06, Tokyo, Japan). ACM. <https://doi.org/10.1145/3680532.3689583>.

- [388] Thati, B., Shyam, K. M., Sindhura, S., Dedeepya, P., & Chowdary, N. S. (2024). Continuous deployment in action: Developing a cloud-based image matching game. *International Journal of Innovative Technology and Interdisciplinary Sciences*, 7(2), 68–79. <https://doi.org/10.15157/ijitis.2024.7.2.68-79>.
- [389] Kothapalli, S., Nizamuddin, M., Talla, R. R., & Gummadi, J. C. S. (2024). "DevOps and software architecture: Bridging the gap between development and operations." *Journal of Computing and Digital Technologies*, 2(1), 51–64. <https://www.researchgate.net/publication/387722480>.
- [390] Lone, O., Stasiak, T., & Doran, H. D. (2024). Automated and orchestrated CI/CD pipelines in industrial protocol certification testing. In 2024 IEEE 20th International Conference on Factory Communication Systems (WFCS). IEEE. <https://doi.org/10.1109/WFCS60972.2024.10540983>.
- [391] Korrapati, R. (2024). Automating compliance in CI/CD pipelines: A modern software development framework. White Paper. SSRN. Available at <https://ssrn.com/abstract=5064454>.
- [392] Boddapati, V. N., Bauskar, S. R., Madhavaram, C. R., Galla, E. P., Sunkara, J. R., & Gollangi, H. K. (2024). Optimizing production efficiency in manufacturing using big data and AI/ML. Preprint. SSRN. Available at <https://ssrn.com/abstract=5080585>.
- [393] Feijó, D. N., de Almeida, C. D. A., & Rocha, L. S. (2024). Studying the impact of CI/CD adoption on atoms of confusion distribution and prevalence in open-source projects. *Journal of Software Engineering Research and Development*, 12, Article 17. <https://doi.org/10.5753/jserd.2024.4118>.
- [394] Rangineni, S., & Bhardwaj, A. K. (2024). Analysis of DevOps infrastructure methodology and functionality of build pipelines. *EAI Endorsed Transactions on Scalable Information Systems*, 11(4). <https://doi.org/10.4108/eetis.4977>.
- [395] Puolitaival, V. (2024). Enhancing development efficiency with DevSecOps template: A design science approach for IaC and CI/CD implementation for AWS (Master's thesis, Lappeenranta–Lahti University of Technology LUT). <https://lutpub.lut.fi/handle/10024/168498>.
- [396] Fitsilis, P., Damasiotis, V., Kyriatzis, V., & Tsoutsas, P. (2024). DOLLmC: DevOps for large language model customization. *International Journal of Software Engineering and Knowledge Engineering*, 34(1), 1–15. <https://doi.org/10.1142/S0218194024500015>.
- [397] Ramesh, R. S. (2024). DevOps and agile computing. In 2024 1st International Conference on Communications and Computer Science (INCCCS). IEEE. <https://doi.org/10.1109/INCCCS60947.2024.10593332>.
- [398] Malviya, R. K., Mandala, V., & Reddy, M. S. (2024). Reinforcement learning for collaborative decision-making in multi-agent systems: Applications in supply chain optimization. Preprint. SSRN. Available at <https://ssrn.com/abstract=5065476>.
- [399] Shesharam, M., Renganathan, S., & Allamuddin, S. (2024). Centralized software deployment control system: Unified solution for deployment oversight and risk mitigation. Technical Report, December 12, 2024. <https://doi.org/10.2139/ssrn.5073950>.
- [400] Saxena, M. C., Tamrakar, A., & Arranz, U. (2024). Automated testing and deployment strategies for quantum algorithms. In 2024 7th International Conference on Contemporary Computing and Informatics (IC3I). IEEE. <https://doi.org/10.1109/IC3I61595.2024.10829232>.
- [401] Jampani, S., Gudavalli, S., Ravi, V. K., Goel, P., Chhapola, A., & Shrivastav, A. (2024). Kubernetes and containerization for SAP applications. *Journal of Quantum Science and Technology (JQST)*, 1(4), 305–307. (ISSN 3048-6351). DOI: <https://ssrn.com/abstract=5068839>.
- [402] Barros, J. D., Jesus, L. C., Oliveira, J., & Padilha, J. (2024). Exploring the impact of DevOps and agile practices from the perspective of source code analysis. University Única / Federal University of Lavras (UFLA/ICTIN), Brazil. Available: <https://www.researchgate.net/publication/383185473>.
- [403] Erdenebat, B., & Kozsik, T. (2024). A comparative analysis of performance and security aspects in container build methods. Preprint submitted to Information and Software Technology. SSRN. Available at <https://ssrn.com/abstract=5060272>.

- [404] Grilo, A., Lourenço, L., Figueiras, P., Khodamoradi, A., Rêga, B., Costa, R., & Jardim-Gonçalves, R. (2024). Data analytics environment: Combining visual programming and MLOps for AI workflow creation. In 2024 IEEE International Conference on Engineering, Technology, and Innovation (ICE/ITMC). IEEE. <https://doi.org/10.1109/ICE/ITMC61926.2024.10794244>.
- [405] Eramo, R., Scoccia, G. L., Nolletti, M., Celi, A., & Autili, M. (2024). An empirical study on the role of DevOps in the development of mobile applications. Preprint submitted to Elsevier. SSRN. Available at <https://ssrn.com/abstract=5031770>.
- [406] Chaudhary, D., Vadlamani, S. L., Thomas, D., Nejati, S., & Sabetzadeh, M. (2024). Developing a Llama-based chatbot for CI/CD question answering: A case study at Ericsson. arXiv preprint, arXiv:2408.09277. <https://doi.org/10.48550/arXiv.2408.09277>.
- [407] Ticu-Jianu, R. (2024). Continuous resilience: DevSecOps strategies for cloud and quantum platforms. *Informatica Economică*, 28(4). <https://doi.org/10.24818/issn14531305/28.4.2024.05>.
- [408] Nadagouda, V. R. (2025). The evolution of modern CI/CD pipelines: A technical deep dive. *International Journal on Science and Technology (IJSAT)*, 16(1). <https://doi.org/10.71097/IJSAT.v16.i1.2364>.
- [409] Orosz, M., Hempen, J. M., Turner, J. M., Duffy, B., & Charlton, C. (2024). Managing resources in a DoD space-based Agile/DevSecOps program. In Proceedings of the Twenty-First Annual Acquisition Research Symposium, Naval Postgraduate School, Monterey, CA. Available: <https://dair.nps.edu/handle/123456789/5071>.
- https://www.researchgate.net/publication/384564681_AI-Driven_Phishing_Email_Detection_Leveraging_Big_Data_Analytics_for_Enhanced_Cybersecurity
- [410] Bauskar, S. R., Madhavaram, C. R., Galla, E. P., Sunkara, J. R., & Gollangi, H. K. (2024). AI-driven phishing email detection: Leveraging big data analytics for enhanced cybersecurity. *Library Progress International*, 44(3), 7211-7224. <https://doi.org/10.2139/ssrn.4980647>.
- [411] Berg, B. L. (2024). Service-oriented architecture over UDSocAN. In 2024 IEEE 100th Vehicular Technology Conference (VTC2024-Fall). IEEE. <https://doi.org/10.1109/VTC2024-Fall63153.2024.10757791>.
- [412] Maleh, Y. (2024). Understanding web application security. In *Web Application PenTesting: A Comprehensive Guide for Professionals* (pp. 39–69). River Publishers / Taylor & Francis. <https://doi.org/10.1201/9788770046985-2>.
- [413] Petrovics, A.-C. (2024). Defining and measuring efficiency gains through Infrastructure as Code adoption: A multivocal literature review and a case study on a company (Master's thesis, M.Sc. in Computer Science). Aalborg University, Denmark. Available at: https://vbn.aau.dk/ws/files/719468231/AlexPetrovics_Thesis.pdf.
- [414] Amgothu, S. (2024). An end-to-end CI/CD pipeline solution using Jenkins and Kubernetes. *International Journal of Science and Research (IJSR)*, 13(8). <https://doi.org/10.21275/SR24826231120>.
- [415] Kumar, S., Bala, N., Raj, Y., & Singh, A. P. (2024). Cloud-native continuous integration/continuous deployment (CI/CD) pipeline. In 2024 Second International Conference on Advanced Computing & Communication Technologies (ICACCTech). IEEE. <https://doi.org/10.1109/ICACCTech65084.2024.00054>.
- [416] Chapter 5: Web applications pentesting basics. (2024). In *Mastering Offensive Security: A Practical Guide to Modern Penetration Testing* (pp. 147–168). DOI: <https://doi.org/10.1201/9788770046985-5>.
- [417] Thota, R. C. (2020). CI/CD pipeline optimization: Enhancing deployment speed and reliability with AI and GitHub Actions. *International Journal of Innovative Research in Engineering & Multidisciplinary Physical Sciences*, 8(2). <https://doi.org/10.37082/IJIRMP.v8.i2.232185>.
- [418] Alfaro, S., Bergel, A., & Simmonds, J. (2024). "Detecting CI/CD workflow errors through visual inspection of logs." *Wiley Journal Preprint*. <https://doi.org/10.22541/au.172448303.32914989>.

- [419] Loisel, F., Zeqo, G., Morichetta, A., Lackinger, A., & Dustdar, S. (2024). RainCloud: Decentralized coordination and communication in heterogeneous IoT swarms. In 2024 International Symposium on Parallel Computing and Distributed Systems (PCDS). IEEE.
<https://doi.org/10.1109/PCDS61776.2024.10743766>.
- [420] Testi, M. (2024). Machine learning operations (MLOps) in healthcare (PhD thesis, Università Campus Bio-Medico di Roma). https://www.iris.unicampus.it/retrieve/c0a845e6-b568-4972-abd1-39246b3264f4/Matteo_Testi_Tesi_Machine%20Learning%20Operations%20in%20Healthcare_28_Nov_24.pdf.
- [421] Flores, A. (2024). A two-level model-driven engineering approach for reengineering CI/CD pipelines (Master's thesis, Faculdade de Engenharia da Universidade do Porto).
<https://web.fe.up.pt/~jacome/downloads/thesis/andreflores.pdf>.
- [422] Harve, B. M., Bidkar, D. M., Krishnappa, M. S., Pandey, G., Jayaram, V., Veerapaneni, P. K., & Mehta, G. (2024). The cloud-native revolution: Microservices in a cloud-driven world. In Proceedings of the 2024 International Conference on Intelligent Cybernetics Technology & Applications (ICICyTA) (pp. 1043–1052). IEEE. <https://doi.org/10.1109/ICICyTA.2024.00000>.
- [423] Gohari, H. E., Kadhe, S. R., Shah, S. Y., Adam, C., Adebayo, A., Adusumilli, P., Ahmed, F., Angel, N. B., et al. (2025). GneissWeb: Preparing high quality data for LLMs at scale. arXiv preprint, arXiv:2502.14907. <https://doi.org/10.48550/arXiv.2502.14907>.
- [424] Chatterjee, P. S., & Mittal, H. K. (2024). Enhancing operational efficiency through the integration of CI/CD and DevOps in software deployment. In 2024 Sixth International Conference on Computational Intelligence and Communication Technologies (CCICT). IEEE.
<https://doi.org/10.1109/CCICT62777.2024.00038>.
- [425] Madhavi, M. V. D. N. S., Rajani, D., Sairam, P. V. S., & Ponnappalli, S. (2024). Early prediction of rainfall using automated integrated prediction approach (AIPA). In Proceedings of the Second International Conference on Intelligent Cyber Physical Systems and Internet of Things (ICoICI 2024). IEEE.
<https://doi.org/10.1109/ICoICI62503.2024.10696408>.
- [426] Deshpande, G., Cheekati, S., Patel, S., Raj, P., Singh, M., Pindur, M., Al Soghyar, N., Zhao, B., Babolhavaej, P., Taher, M., Nathan, K., Spaeth, W., & Roozbahani, M. M. (2024). Transforming CS education with DevOps: Streamlined assignment validation and delivery @ scale. In Proceedings of the Eleventh ACM Conference on Learning @ Scale (L@S '24). ACM.
<https://doi.org/10.1145/3657604.3664676>.
- [427] Wu, T., Yao, H., Mai, T., Wang, Z., Wang, F., & Guizani, M. (2025). Using full-dimensional programmability to power self-driving 6G networks. IEEE Network, 39(1), 270–277.
<https://doi.org/10.1109/MNET.2024.3448288>.
- [428] Azad, N., Hyrynsalmi, S., & Smolander, K. (2025). Understanding DevOps critical success factors: A thematic analysis. In Proceedings of the International Conference on Digital Project Management (ICDPM 2024). Lecture Notes in Business Information Processing (LNBIP, 528) (pp. 28–43). Springer.
https://doi.org/10.1007/978-3-031-71515-0_3.
- [429] De La Cruz, E., Gonaygunta, H., Webb, J., Meduri, S. S., Nadeela, G. S., & Podicheti, S. (2024). Exploring TAM and DOI constructs as predictors of cybersecurity data analytics system success. Preprint. DOI: <https://doi.org/10.1109/ICoABCD63526.2024.10704426>.
- [430] Sharma, A., Cadar, C., & Metzman, J. (2024). Effective fuzzing within CI/CD pipelines (registered report). In Proceedings of the 3rd ACM International Fuzzing Workshop (FUZZING '24). ACM.
<https://doi.org/10.1145/3678722.3685534>.
- [431] Seco, J. C., & Aldrich, J. (2024). The Meerkat vision: Language support for live, scalable, reactive web apps. In Proceedings of the ACM SIGPLAN International Symposium on New Ideas, New Paradigms, and Reflections on Programming and Software (Onward! '24) (pp. 1–14). ACM.
<https://doi.org/10.1145/3689492.3690048>.

- [432] Lone, O., Stasiak, T., Meisterhans, J., & Doran, H. D. (2024). The future of industrial protocol certification testing: CI/CD pipelines, cloud-based, automated and orchestrated. In *Embedded World Conference 2024*. Institute of Embedded Systems ZHAW, Zurich, Switzerland. Available: <https://digitalcollection.zhaw.ch/bitstreams/f600873e-c914-471b-9ac1-d5b3d5cf2aad/download>.
- [433] Tenbergen, B., & Mead, N. R. (2024). Technology acquisition plans to foster supply chain risk management learning outcomes in project-based software development courses. In *36th International Conference on Software Engineering Education and Training (CSEET)*. IEEE. <https://doi.org/10.1109/CSEET62301.2024.10663048>.
- [434] Langerman, J., & Joseph, N. (2025). Operationalising information systems resilience within the financial services sector. In *Intelligent Sustainable Systems. Lecture Notes in Networks and Systems*, 1179. Springer. https://doi.org/10.1007/978-981-97-9327-3_24.
- [435] Giorgi, L., Puliafito, C., Virdis, A., & Mingozzi, E. (2024). Service continuity in edge computing through edge proxies and HTTP alternative services. *IEEE Open Journal of the Communications Society*, 5, 7057–7079. <https://doi.org/10.1109/OJCOMS.2024.3492695>.
- [436] Moura, F., Moura, J., de Assunção, M. D., & Macedo, M. (2024). Quantum computing as a service. In M. Felizardo, R. Braga, & R. Figueiredo (Eds.), *Quantum software engineering* (pp. 31–69). Springer. https://doi.org/10.1007/978-3-031-45499-6_3.
- [437] Johansson, A., & Paulsson, S. (2024). Microservice integration testing with hardware-in-the-loop in CI/CD pipelines (Master's thesis, Department of Mechanics and Maritime Sciences, Chalmers University of Technology, Gothenburg, Sweden). <http://hdl.handle.net/20.500.12380/308696>.
- [438] Uppala, M. (2024). Finding dependency vulnerabilities in software applications with CI/CD integration. *IEEE Software* (preprint version on TechRxiv). DOI: <https://doi.org/10.36227/techrxiv.172114411.13851660/v1>.
- [439] Nawaz, M. K. (2024). Exploring end-to-end data engineering: A GCP case study (Master's thesis, Centria University of Applied Sciences). <https://www.theseus.fi/handle/10024/858907>.
- [440] Gowda, P. G., Stanley, N. S. A., & Joyce, J. E. (2024). DevOps dynamics: Tools driving continuous integration and deployment. In *2024 IEEE International Conference on Information Technology, Electronics and Intelligent Communication Systems (ICITEICS)*. IEEE. <https://doi.org/10.1109/ICITEICS61368.2024.10624986>.
- [441] Gogineni, A. (2020). Automated deployment and rollback strategies for docker containers in continuous integration/continuous deployment (CI/CD) pipelines. *International Journal of Multidisciplinary Research and Growth Evaluation*, 1(5), 125–130. <https://doi.org/10.54660/IJMRGE.2020.1.5.125-130>.
- [442] Borra, P. (2024). Maximizing efficiency and collaboration with Microsoft Azure DevOps. *International Journal of Advanced Research in Science, Communication and Technology (IJARSCT)*, 4(2), 556–561. <https://doi.org/10.48175/IJARSCT-18864>.
- [443] El Aouni, F., Moumane, K., Idri, A., Najib, M., & Jan, S. U. (2024). A systematic literature review on agile, cloud, and DevOps integration: Challenges, benefits. *Information and Software Technology*, 171, 107529. Elsevier. <https://doi.org/10.1016/j.infsof.2024.107529>.
- [444] Zhou, X., Xu, J., Li, X., Cao, L., Li, L., Wang, Y., Li, S., & Liu, H. (2024). A blockchain-based and microservices-architected software composition analysis system. *Journal of Software: Evolution and Process*, 36(10), e2675. <https://doi.org/10.1002/smr.2675>.
- [445] Kunchenapalli, V. (2024). A case study: How DevOps enabled easy monolithic to microservice migration. *International Journal of Computer Engineering and Technology (IJCET)*, 15(5), 105–116. <https://doi.org/10.5281/zenodo.13737601>.
- [446] Srikanth, S. (2024). Using Azure DevOps for building end-to-end MLOps solutions. In *Proceedings of the 4th International Conference on Computational Intelligence and Data Engineering (ICCIDE 2024)* (pp. 309–318). Springer. https://doi.org/10.1007/978-981-97-1866-5_25.

- [447] Guo, Y., Bettaieb, S., & Casino, F. (2024). A comprehensive analysis on software vulnerability detection datasets: Trends, challenges, and road ahead. *International Journal of Information Security*, 23, 3311–3327. <https://doi.org/10.1007/s10207-024-00888-y>.
- [448] Shah, P. S. U., Ahmad, N., & Beg, M. O. (2024). Towards MLOps: A DevOps tools recommender system for machine learning systems. In *Proceedings of the 2024 International Conference on Emerging Software Systems and Applications (ICESSA)*. IEEE. <https://doi.org/10.1109/ICESSA.2024.00000>.
- [449] Muñoz, M., & Negrete Rodríguez, M. (2024). A guidance to implement or reinforce a DevOps approach in organizations: A case study. *Journal of Software: Evolution and Process*, 36(3), e2342. <https://doi.org/10.1002/smr.2342>.
- [450] Bedekar, M. M., & Mussbacher, G. (2024). A multi-platform specification language and dataset for the analysis of DevOps pipelines. In *Proceedings of the ACM/IEEE 27th International Conference on Model Driven Engineering Languages and Systems (MODELS Companion '24)*, September 2024, Linz, Austria. ACM. <https://doi.org/10.1145/3652620.3686247>.
- [451] Yanagawa, T., Agarwal, V., Watanabe, Y., Degenaro, L., & Sailer, A. (2024). A secure framework for continuous compliance across heterogeneous policy validation points. In *2024 IEEE 17th International Conference on Cloud Computing (CLOUD)*. IEEE. <https://doi.org/10.1109/CLOUD62652.2024.00029>.
- [452] Sharma, P., & Kulkarni, M. S. (2024). A study on unlocking the potential of different AI in continuous integration and continuous delivery (CI/CD). In *2024 4th International Conference on Innovative Practices in Technology and Management (ICIPTM)*. IEEE. <https://doi.org/10.1109/ICIPTM59628.2024.10563618>.
- [453] Tiwari, A., Vanshika, V., Aditi, A., & Kumar, A. (2024). Strategies for pull request validation in CI/CD practices. *SSRN Preprint*. <https://doi.org/10.2139/ssrn.4749815>.
- [454] Kumar, A., Nadeem, M., & Shameem, M. (2025). A systematic literature review for investigating DevOps metrics to implement in software development organizations. *Journal of Software: Evolution and Process*, 37, e2733. <https://doi.org/10.1002/smr.2733>.
- [455] Dunbar, W. (2024). Vendor-locked DevOps strategies: Using vendor-locked DevOps solutions and getting away with it. *Communications of the ACM*, 67(5), 43–47. <https://doi.org/10.1145/3633776>.
- [456] James, C. (2024). Test automation in CI/CD: Enhancing deployment workflows using TDD/BDD methodologies. https://www.researchgate.net/publication/386381578_Test_Automation_in_CICD_Enhancing_Deployment_Workflows_Using_TDD_BDD_Methodologies.
- [457] Pérez-Sánchez, J., Rafi, S., Carrillo de Gea, J. M., Ros, J. N., & Fernández Alemán, J. L. (2025). A theory on human factors in DevOps adoption. *Computer Standards & Interfaces*, 92, 103907. <https://doi.org/10.1016/j.csi.2024.103907>.
- [458] Sternhell, A. (2024). A trusted global data supply chain. In S. McClellan (Ed.), *Data, Security, and Trust in Smart Cities (Signals and Communication Technology)*. Springer. https://doi.org/10.1007/978-3-031-61117-9_1.
- [459] Iyer, G. N., Yisheng, A. G., Metilda, C. H. E., Choong, W. X., & Koh, S. W. (2024). A web-based IDE for DevOps learning in software engineering higher education. In *2024 IEEE International Conference on Teaching, Assessment and Learning for Engineering (TALE)*. IEEE. <https://doi.org/10.1109/TALE62452.2024.10834361>.
- [460] Josh, H. (2024). Self-healing infrastructure: AI-powered automation for fault-tolerant DevOps environments. <https://www.researchgate.net/publication/337076816> Article. <https://www.researchgate.net/publication/388636013>.
- [461] Flynn, C. (2024). AI-powered CI/CD pipelines: Enhancing automation in software development. https://www.researchgate.net/publication/388636013_AI-Powered_CICD_Pipelines_Enhancing_Automation_in_Software_Development.

- [462] Mallardi, G., Calefato, F., Quaranta, L., & Lanubile, F. (2024). An MLOps approach for deploying machine learning models in healthcare systems. In 2024 IEEE International Conference on Bioinformatics and Biomedicine (BIBM), 6832–6837. IEEE. <https://doi.org/10.1109/BIBM62325.2024.10822603>.
- [463] Caniglia, A., Dentamaro, V., Galantucci, S., & Impedovo, D. (2025). FOBICS: Assessing project security level through a metrics framework that evaluates DevSecOps performance. *Information and Software Technology*, 178, 107605. <https://doi.org/10.1016/j.infsof.2024.107605>.
- [464] Nandurkar, S. (2025). Navigating the DevOps landscape: Insights and implementation strategies. *Journal of Systems and Software*, 223, 112331. <https://doi.org/10.1016/j.jss.2024.112331>.
- [465] Oliva, A. (2024). Multi-tenancy in Kubernetes clusters (Master's thesis, Politecnico di Torino). <https://webthesis.biblio.polito.it/33340/1/tesi.pdf>.
- [466] Tsagkaropoulos, A., Verginadis, Y., & Mentzas, G. (2024). A multi-context severity-based SLO violation detector. In 2024 IEEE Symposium on Computers and Communications (ISCC). IEEE. <https://doi.org/10.1109/ISCC61673.2024.10733630>.
- [467] Cusick, J. J. (2024). Jumpstarting and evolving a software quality management initiative: From pre-Web to DevOps. In 2024 16th IIAI International Congress on Advanced Applied Informatics (IIAI-AAI). IEEE. <https://doi.org/10.1109/IIAI-AAI63651.2024.00112>.
- [468] Maier, R., Schlattl, A., Guess, T., & Mottok, J. (2024). CausalOps — Towards an industrial lifecycle for causal probabilistic graphical models. *Information and Software Technology*, 174, 107520. <https://doi.org/10.1016/j.infsof.2024.107520>.
- [469] Rinker, F., Meixner, K., Vysoká, D., & Biffel, S. (2024). Multi-domain modeling for change management in cyber-physical production systems engineering. In 2024 IEEE 29th International Conference on Emerging Technologies and Factory Automation (ETFA). IEEE. <https://doi.org/10.1109/ETFA61755.2024.10710656>.
- [470] Bodnar, L., Bodnar, M., Shulakova, K., Vasylenko, O., Tsarov, R., & Siemens, E. (2024). Practical experience in DevOps implementation. In Proceedings of the 12th International Conference on Applied Innovations in IT (ICAII 2024). <https://doi.org/10.25673/115639>.
- [471] Iyer, G. N., Sen, L. B., Chester, W. Z., & Chua, K. (2024). TROFOS – Agile project management platform for software engineering education. In 2024 IEEE International Conference on Teaching, Assessment and Learning for Engineering (TALE). IEEE. <https://doi.org/10.1109/TALE62452.2024.10834343>.
- [472] Riquet, N.; Devroey, X.; Vanderose, B. (2024). Debt stories: Capturing social and technical debt in the industry. In Proceedings of the 7th ACM/IEEE International Conference on Technical Debt (TechDebt '24), Lisbon, Portugal, April 14–15, 2024. ACM. <https://doi.org/10.1145/3644384.3644473>.
- [473] Drane, L., McDonnell, M., Petras, R., Stiner, C., Ruckman, A. J., Wiggins, G. M., Cage, G., Smith, R., Hitefield, S., McGaha, J., Ayres, A., Brim, M., Archibald, R., & Malviya-Thakur, A. (2025). Integrating scientific single-page applications with DevSecOps. *Future Generation Computer Systems*, 166, 107695. Elsevier. <https://doi.org/10.1016/j.future.2024.107695>.
- [474] Capocchi, L., Santucci, J.-F., Tigli, J.-Y., Gonnin, T., Lavirotte, S., & Rocher, G. (2025). Actuation conflict management in Internet of Things systems DevOps: A discrete event modeling and simulation approach. In IFIP International Conference on Internet of Things (IFIPIoT 2024), Advances in Information and Communication Technology, 737, 189–206. Springer. https://doi.org/10.1007/978-3-031-81900-1_12.
- [475] Luntovskyy, A., & Winkler, U. (2024). Advanced software technology paradigms and AI deployment. In Intelligent Systems and Applications (IntelliSys 2024), Lecture Notes in Networks and Systems, 1065, 590–605. Springer. https://doi.org/10.1007/978-3-031-66329-1_38.
- [476] Lin, Y. C., Chiu, P. K., & Suryajava, B. (2024). Advanced security approach for building secure containerized manufacturing digital twins. In IET International Conference on Engineering Technologies and Applications (ICETA 2024), Taipei, Taiwan, Oct. 25–27, 2024. The Institution of Engineering & Technology. <https://doi.org/10.1049/icp.2024.4151>.

- [477] Antara, F., Renuka, A., & Gopalakrishna Pandian, P. K. (2024). Advanced cloud automation workflows for CI/CD pipelines: Tools and techniques. *International Journal of Novel Trends and Innovation (IJNTI)*, 2(5), a296–a310. ISSN: 2984-908X. <https://ijpn.org/ijnti/viewpaperforall.php?paper=IJNTI2405031>.
- [478] Sakinala, K. (2025). Advancements in DevOps: The role of GitOps in modern infrastructure management. *International Journal of Information Technology and Management Information Systems (IJITMIS)*, 16(1), 632–646. https://doi.org/10.34218/IJITMIS_16_01_045.
- [479] Jagarlamudi, S. R. (2024). Advancements in software engineering: From performance optimization to quantum computing and user experience. *International Journal of Innovative Research in Science Engineering and Technology (IJIRSET)*, 13(8). <https://doi.org/10.15680/IJIRSET.2024.1308116>.
- [480] Arya, S., Chauhan, D., Tanishq, Anand, S., & Sharma, O. (2024). Beyond monoliths: An in-depth analysis of microservices adoption in the era of Kubernetes. In *2024 1st International Conference on Advanced Computing and Emerging Technologies (ACET)*. IEEE. <https://doi.org/10.1109/ACET61898.2024.10730456>.
- [481] Friman, O. (2023). Agile and DevSecOps oriented vulnerability detection and mitigation on public cloud (Master's thesis, Aalto University, School of Science, Espoo, Finland). https://aaltodoc.aalto.fi/items/06e4d727-a041-4f37-8169-438fd7c78903/master_Friman_Otso_2024.pdf.
- [482] Fu, M., Pasukmit, J., & Tantithamthavorn, C. (2025). AI for DevSecOps: A landscape and future opportunities. *ACM Transactions on Software Engineering and Methodology*. ACM. <https://doi.org/10.1145/3712190>.
- [483] Rayaprolu, R. (2024). AI enhanced cloud DevOps and automation. *Journal of Artificial Intelligence General Science (JAIGS)*, 4(1), 363–364. <https://doi.org/10.60087>.
- [484] Sharma, M., Aswathy, C., Ben, M., & Mehrotra, A. (2025). AI-driven DevOps: A tool selection guide. In *Intelligent Solutions for Smart Adaptation in Digital Era, Lecture Notes in Electrical Engineering*, 1278. Springer. https://doi.org/10.1007/978-981-97-8193-5_22.
- [485] Cui, J. (2024). The role of DevOps in enhancing enterprise software delivery success through R&D efficiency and source code management. *IEEE Conference Paper, Solbridge International School of Business*. Available: <https://www.researchgate.net/publication/388635365>.
- [486] Bishop, S., Adewale, T., & Ajayi, J. O. (2024). AI for continuous compliance and policy enforcement in DevOps security frameworks. https://www.researchgate.net/publication/388638315_AI_for_Continuous_Compliance_and_Policy_Enforcement_in_DevOps_Security_Frameworks.
- [487] Chittala, S. (2024). AIOps and DevOps: Catalysts of digital transformation in the age of automated operations. *International Journal of Scientific Research in Computer Science, Engineering and Information Technology*. <https://doi.org/10.32628/CSEIT24106163>.
- [488] Ajouaoui, L., Fritzsche, A., Soeldner, G., & Soeldner, J.-H. (2023). Platform engineering for cloud-native organizations. In *Proceedings of the 5th International Conference Business Meets Technology (BMT2023)*, Universitat Politècnica de València, Spain, 77–82. <https://doi.org/10.4995/BMT2023.2023.16741>.
- [489] Willman, A. (2024). Adoptions and effects of combining agile software development and DevOps practices – A literature review (Master's thesis, University of Helsinki). <https://helda.helsinki.fi/bitstreams/fe173cc2-64f6-4f08-94aa-9df67ef79ebf/download>.
- [490] Ahmad, A., Li, P., Piechocki, R., & Inacio, R. (2024). Anomaly detection in offshore open radio access network using long short-term memory models on a novel artificial intelligence-driven cloud-native data platform. Preprint submitted to Elsevier; preprint: arXiv:2409.02849. Available: <https://arxiv.org/abs/2409.02849>.

- [491] Paul, A., Manoj, R., & Udhayakumar, S. (2024). Amazon Web Services cloud compliance automation with Open Policy Agent. In 2024 International Conference on Expert Clouds and Applications (ICOECA). <https://doi.org/10.1109/ICOECA62351.2024.00063>.
- [492] Amiri, A., Steindl, G., & Kastner, W. (2024). Model-based systems engineering for the agile life-cycle management of IIoT based on DevOps. In 2024 IEEE International Conference on Software Services Engineering (SOSE), pp. 1–10. <https://doi.org/10.1109/SOSE60119.2024.00013>.
- [493] Amrit, C., & Narayanappa, A. K. (2025). An analysis of the challenges in the adoption of MLOps. *Journal of Innovation & Knowledge*, 10, 100637. <https://doi.org/10.1016/j.jik.2024.100637>.
- [494] Eramo, R., Said, B., Oriol, M., Bruneliere, H., & Morales, S. (2024). An architecture for model-based and intelligent automation in DevOps. *Journal of Systems and Software*, 217, 112180. <https://doi.org/10.1016/j.jss.2024.112180>.
- [495] Arbesser-Rastburg, G., Camhy, D., & Fuchs-Hanusch, D. (2022). COSMOS – A framework for containerised, distributed execution and analysis of hydraulic water distribution system models. In *Proceedings of the 2nd International Joint Conference on Water Distribution Systems Analysis (WDSA) & Computing and Control in the Water Industry (CCWI)*, Universitat Politècnica de València, Spain. <https://doi.org/10.4995/WDSA-CCWI2022.2022.14074>.
- [496] Jose, J., & Shenoy, G. S. (2024). An efficient framework for integrating DevOps practices in network configuration and monitoring. In 2024 3rd International Conference for Innovation in Technology (INOCON). <https://doi.org/10.1109/INOCON60754.2024.10512008>.
- [497] Ayas, H. M., Hebig, R., & Leitner, P. (2024). An empirical investigation on the competences and roles of practitioners in microservices-based architectures. *Journal of Systems and Software*, 213, 112055. <https://doi.org/10.1016/j.jss.2024.112055>.
- [498] Thooriqoh, H. A., Mulyo, B. M., & Rakhmadi, A. (2024). Advanced RESTful API testing: Leveraging Newman's command-line capabilities with Postman collections. In 2024 IEEE 10th Information Technology International Seminar (ITIS), Surabaya, Indonesia. <https://doi.org/10.1109/ITIS64716.2024.10845315>.
- [499] Watanabe, H., Ogura, N., & Hisazumi, K. (2024). Towards a DevOps modeling based on context-oriented programming. In *Companion Proceedings of the 8th International Conference on the Art, Science, and Engineering of Programming (‘Programming’ Companion ’24)*, Lund, Sweden, Mar. 11–15, 2024. ACM. <https://doi.org/10.1145/3660829.3660832>.
- [500] Xu, Q., Gao, Y., & Wei, J. (2024). An empirical study on Kubernetes operator bugs. In *Proceedings of the 33rd ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA ’24)*, Vienna, Austria. ACM. <https://doi.org/10.1145/3650212.3680396>.
- [501] Vuyyuru, H. K. (2024). Architecting robust information flows in advanced artificial intelligence systems. *International Journal of Computer Engineering and Technology (IJCET)*, 15(6), 1899–1908. https://doi.org/10.34218/IJCET_15_06_162.
- [502] Kaltenberger, F., Melodia, T., Ghauri, I., Polese, M., Knopp, R., Nguyen, T. T., Velumani, S., Villa, D., Bonati, L., Schmidt, R., Arora, S., Irazabal, M., & Nikaein, N. (2025). Driving innovation in 6G wireless technologies: The OpenAirInterface approach. *arXiv preprint, arXiv:2412.13295 [cs.NI]*. <https://arxiv.org/abs/2412.13295>
- [503] Rzig, D. E., Houerbi, A., Chavan, R. G., & Hassan, F. (2025). Empirical analysis on CI/CD pipeline evolution in machine learning projects. *arXiv preprint, arXiv:2403.12199v4 [cs.SE]*. <https://arxiv.org/abs/2403.12199>.
- [504] Sanjurjo, E., Pedreira, O., García, F., & Piattini, M. (2024). Assessing BizDevOps maturity using international standards: Case studies and lessons learned. *Journal of Software: Evolution and Process*, 36, e2646. Wiley. <https://doi.org/10.1002/smr.2646>.
- [505] Femminella, M., Palmucci, M., Reali, G., & Rengo, M. (2024). Attribute-based management of secure Kubernetes cloud bursting. *IEEE Open Journal of the Communications Society*, 5, 1276–1290. <https://doi.org/10.1109/OJCOMS.2024.3367461>.

- [506] Vangala, R. R. (2018). Adaptive resilience framework: A comprehensive study on dynamic orchestration and auto-scaling of microservices in cloud-native systems. *International Journal of Computer Engineering and Technology (IJCET)*, 9(6), 278–288. IAEME. ISSN: 0976-6367 (Print), 0976-6375 (Online). Available at: <https://iaeme.com/Home/issue/IJCET?Volume=9&Issue=6>.
- [507] Sokolowski, D., Spielmann, D., & Salvaneschi, G. (2024). Automated infrastructure as code program testing. *IEEE Transactions on Software Engineering*, 50(6), 1585–1601. <https://doi.org/10.1109/TSE.2024.3393070>.
- [508] Grönblom, M. (2024). Automated release management and testing for SHOT (Master's thesis, Faculty of Science and Engineering, Åbo Akademi University). https://www.doria.fi/bitstream/handle/10024/189669/2/gronblom_maxemilian.pdf.
- [509] Bykov, S. A., & Shiboldenkov, V. A. (2025). Infrastructure-as-code approach for IT-infrastructure. In *Proceedings of the 7th International Youth Conference on Radio Electronics, Electrical and Power Engineering (REEPE)*. IEEE. <https://doi.org/10.1109/REEPE63962.2025.10971160>.
- [510] Esther, D. (2024). Automated testing strategies for quality assurance in CI/CD pipelines. https://www.researchgate.net/publication/385286073_Automated_Testing_Strategies_for_Quality_Assurance_in_CICD_Pipelines.
- [511] Anumandla, S. K. R. (2024). Automating container orchestration: Innovations and challenges in Kubernetes implementation. *Robotics Xplore: USA Tech Digest*, 1(1), 29–43. Available at <https://hal.science/hal-04787298>.
- [512] Haverinen, H., Janhunen, T., Päivärinta, T., Kaartinen, S., & Lempinen, S., & Merilä, S. (2024). Automating cybersecurity compliance in DevSecOps with open information model for security as code. In *Proceedings of the 4th Eclipse Security, AI, Architecture and Modelling Conference on Data Space (eSAAM 2024)*, Mainz, Germany. ACM. <https://doi.org/10.1145/3685651.3686700>.
- [513] Yang, J.-R. (2024). Automating security operations in telecommunication networks with GitOps (Master's thesis, Aalto University, School of Science, Programme in Security and Cloud Computing). <https://aaltodoc.aalto.fi/items/1f3d9850-8129-477e-83b0-20ccd5fe1d81>.
- [514] Gopireddy, R. R. (2019). Automating cloud security with DevSecOps: Integrating AI for continuous threat monitoring and response. *International Journal of Core Engineering & Management*, 5(12). <https://doi.org/10.5281/zenodo.13929153>.
- [515] William, B., & Arunachalam, P. M. (2024). Automating security at scale: Best practices for DevSecOps in AWS environments. Article. Available at: <https://www.researchgate.net/publication/386566181>.
- [516] Kushwaha, M. K., David, P., & Suseela, G. (2024). Automation and DevSecOps: Streamlining security measures in financial system. In *2024 IEEE International Conference on Electronics, Computing and Communication Technologies (CONECCT)*. IEEE. <https://doi.org/10.1109/CONECCT62155.2024.10677271>.
- [517] Ros, R., Runeson, P., & Bjarnason, E. (2024). Evidence-based guidelines for advancing continuous experimentation. *IT Professional*. IEEE. <https://doi.org/10.1109/MITP.2024.3397541>.
- [518] Abbas, S. I., & Walia, R. (2024). Analysis of revolutionizing IoT solutions through cloud computing integration. In *2024 3rd International Conference on Sentiment Analysis and Deep Learning (ICSADL)*. <https://doi.org/10.1109/ICSADL61749.2024.00012>.
- [519] Saimler, M., İckin, S., Bernini, G., Toumi, N., Diamanti, M., Papavassiliou, S., Zivkovic, M., Akgul, O. U., & Khorsandi, B. M. (2025). The role of AI enablers in overcoming impairments in 6G networks. Preprint. <https://doi.org/10.1109/BigData62323.2024.10825499>.
- [520] Demchenko, Y. (2024). The importance of system engineering competences and knowledge for big data science and research infrastructure projects. In *2024 IEEE International Conference on Big Data (Big Data)*. IEEE. DOI: <https://doi.org/10.1109/BigData62323.2024.10825499>.

- [521] Craciun, P.-C., & Necula, R.-C. (2024). Why startups outpace multinationals in leveraging DevOps. In Proceedings of the 18th International Conference on Business Excellence 2024 (pp. 3421–3429). Sciendo. <https://doi.org/10.2478/picbe-2024-0277>.
- [522] Shevchenko, A., Malynovskyi, I., & Petrenko, M. (2024). Vulnerability detection in CI/CD pipelines with a case study on mobile applications. Artificial Intelligence. Available at <https://www.researchgate.net/publication/388852055>.
- [523] Rajasekar, V., Saračević, M., Karabašević, D., Stanujkić, D., Hasić, A., Azizović, M., & Thirumalai, S. (2024). Security-enhanced QoS-aware autoscaling of Kubernetes pods using horizontal pod autoscaler (HPA). Journal of Intelligent Management Decision, 3(3), 175–186. <https://doi.org/10.56578/jimd030304>.
- [524] Ccallo-Luque, M., & Quispe-Quispe, A. (2024). Adoption and adaptation of CI/CD practices in very small software development entities: A systematic literature review. Preprint. National University of the Altiplano of Puno. <https://arxiv.org/abs/2410.00623>.
- [525] Oyeniran, O., Adewusi, A., Komolafe, A. M., et al. (2024). The emergence and importance of DevSecOps: Integrating and reviewing security practices within the DevOps pipeline. World Journal of Advanced Engineering Technology and Sciences, 11(2). <https://doi.org/10.30574/wjaets.2024.11.2.0093>.
- [526] Thien, T. T. V. (2024). Analysing and designing CI/CD pipelines in an enterprise (Bachelor's thesis, Metropolia University of Applied Sciences). https://www.theseus.fi/bitstream/handle/10024/859468/Thanh_TranVietThien.pdf.
- [527] M. K. Singhal and C. Gunawat, "Mitigating Cloud Disruptions: An AI-Driven Approach to Proactively Assess and Resolve Impact on Customer Workflows," in Proc. 2024 Int. Conf. on Platform Technology and Service (PlatCon), IEEE, 2024. DOI: 10.1109/PlatCon63925.2024.10830739.
- [528] Schumann, M. (2024). Conceptual design of a container-based system landscape orchestrated by Kubernetes (Bachelor's thesis, University of Applied Sciences Zwickau). <https://libdoc.fh-zwickau.de/opus4/files/17421/BachelorThesisMaximilianSchumann.pdf>.
- [529] Cooper, Q. (2024). Expedited load tests for CI/CD microservice applications (Master's thesis, University of Calgary). Available at <https://prism.ucalgary.ca/handle/1880/119638>.
- [530] Moez, M., Mahmood, R., Asif, H., Iqbal, M. W., Hamid, K., Ali, U., & Khan, N. (2024). Comprehensive analysis of DevOps: Integration, automation, collaboration, and continuous delivery. Bulletin of Business and Economics, 13(1), 662–672. <https://doi.org/10.61506/01.00253>.
- [531] Saqib, M., Malhotra, S., Mehta, D., Jangid, J., Yashu, F., & Dixit, S. (2025). Optimizing spot instance reliability and security using cloud-native data and tools. Journal of Information Systems Engineering & Management, 10(14s). <https://doi.org/10.52783/jisem.v10i14s.2387>.
- [532] George, A. S. (2025). The critical role of micro-segmentation in modern cybersecurity architectures: A comprehensive review. Partners Universal Multidisciplinary Research Journal (PUMRJ), 2(2), 24–26. <https://doi.org/10.5281/zenodo.15063176>.
- [533] Cusick, J. J., & Basil, L. (2024). A global operational readiness review process: Improving cloud availability. In 2024 IEEE 35th International Symposium on Software Reliability Engineering Workshops (ISSREW). IEEE. <https://doi.org/10.1109/ISSREW63542.2024.00038>.
- [534] Terzi, S., & Stamelos, I. (2024). Architectural solutions for improving transparency, data quality, and security in eHealth systems by designing and adding blockchain modules, while maintaining interoperability: The eHDSI network case. Health and Technology, 14, 451–462. <https://doi.org/10.1007/s12553-024-00833-y>.
- [535] Gunathilake, K., & Ekanayake, I. (2024). K8s Pro Sentinel: Extend secret security in Kubernetes cluster. In 2024 International Conference on Advanced Computing and Emerging Technologies (ACET). <https://doi.org/10.1109/ACET61898.2024.10730456>.
- [536] James, C. (2024). Best practices for implementing TDD and BDD in CI/CD pipelines. Article. Available:

https://www.researchgate.net/publication/385275582_Best_Practices_for_Implementing_TDD_and_BDD_in_CICD_Pipelines.

[537] Muppala, P. K. (2025). Sustainable DevOps: Minimizing the carbon footprint of banking data centers. *International Journal of Science and Research Archive*, 14(1), 1780–1793. <https://doi.org/10.30574/ijrsra.2025.14.1.2372>.

[538] Jiang, L., An, J., Huang, H., Tang, Q., Nie, S., Wu, S., & Zhang, Y. (2024). BinaryAI: Binary software composition analysis via intelligent binary source code matching. In *Proceedings of the 46th International Conference on Software Engineering (ICSE '24)*, Lisbon, Portugal. ACM. <https://doi.org/10.1145/3597503.3639100>.

[539] Mahmoudi, B., Trabelsi, I., Tamzalit, D., Moha, N., & Guéhéneuc, Y.-G. (2025). BOAM: A business-oriented identification approach of microservices within legacy systems. In *ICSOC 2024: Service-Oriented Computing, Lecture Notes in Computer Science*, 15405 (pp. 123–137). Springer. https://doi.org/10.1007/978-981-96-0808-9_10.

[540] Ok, E. (2025). Bridging the gap: How DevOps can transform legacy systems for agile deployment. Article. Available:

https://www.researchgate.net/publication/389849833_Bridging_the_Gap_How_DevOps_Can_Transform_Legacy_Systems_for_Agile_Deployment.

[541] AbouZaid, A., Barclay, P. J., Chrysoulas, C., & Pitropakis, N. (2025). Building a modern data platform based on the data lakehouse architecture and cloud-native ecosystem. *Discover Applied Sciences*, 7, 166. Springer. <https://doi.org/10.1007/s42452-025-06545-w>.

[542] Bolish, S., Dodge, M., Mitalo, E., & Westing, R. (2024). Building cyber resilient systems from day 1. In *2024 IEEE Aerospace Conference*. IEEE. <https://doi.org/10.1109/AERO58975.2024.10521294>.

[543] Davis, M. (2020). Building resilient systems with DevOps methodologies. *International Journal of Advanced and Innovative Research*, 9(1), 729–735. Available at <https://www.researchgate.net/publication/383423678>.

[544] Kurz, A. F., Kampik, T., Pufahl, L., & Weber, I. (2025). Business process improvement with AB testing and reinforcement learning: Grounded theory-based industry perspectives. *Software and Systems Modeling*, 24, 87–109. <https://doi.org/10.1007/s10270-024-01229-2>.

[545] McIntosh, S. (2024). Mining our way back to incremental builds for DevOps pipelines. In *21st International Conference on Mining Software Repositories (MSR '24)*, Lisbon, Portugal. ACM. <https://doi.org/10.1145/3643991.3649106>.

[546] Shah, J. A., & Iyer, N. R. (2024). Building generative AI chatbot using Oracle Cloud Infrastructure. In *15th Annual Ubiquitous Computing, Electronics & Mobile Communication Conference (UEMCON)*. IEEE. <https://doi.org/10.1109/UEMCON62879.2024.10754774>.

[547] Sahu, A. (2024). Adoption of DevOps in the software development team: Challenges and recommendations (Master's thesis, Dublin Business School). <https://esource.dbs.ie/bitstreams/b0861c3e-f94a-403d-992e-6c5cc576746f/download>.

[548] Naresh, N. E., Murthy, S. V. N., Pareek, P. K., Reddy, K. T., S. D. S. L., & Pradeep Kumar, B. P. (2024). DevOps life cycle implementation on real life scenarios. In *2024 International Conference on Knowledge Engineering and Communication Systems (ICKECS)*. IEEE. <https://doi.org/10.1109/ICKECS2024.1269>.

[549] da Gão, H., Flores, A., Pereira, R., & Cunha, J. (2024). Chronicles of CI/CD: A deep dive into its usage over time. *arXiv preprint, arXiv:2402.17588 [cs.SE]*. <https://arxiv.org/abs/2402.17588>.

[550] Birchler, C.; Rohrbach, C.; Kim, H.; Gambi, A.; Liu, T.; Horneber, J.; Kehrer, T.; Panichella, S. (2023). Teaser: Simulation-based CAN bus regression testing for self-driving cars software. In *Proceedings of the 38th IEEE/ACM International Conference on Automated Software Engineering (ASE 2023)*, Kirchberg, Luxembourg, September 11–15, 2023, pp. 2058–2061. IEEE. <https://doi.org/10.1109/ASE56229.2023.00154>.

- [551] A. Mankotia, “Impact of AI and Language Models on DevOps and DevSecOps,” *International Journal of Management, IT & Engineering*, vol. 14, no. 07, Jul. 2024.
https://www.ijmra.us/project%20doc/2024/IJME_JULY2024/IJMIE7July24_11665.pdf.
- [552] Aina, A., Subramanian, A., Hsu, H.-W., Rama, S., Gowrishankar, V., & Cheng, Y.-C. (2024). Final report – Infrastructure and DevOps (CS5604). Virginia Tech, Blacksburg, VA, USA.
<https://hdl.handle.net/10919/118666>.
- [553] Alif, R. G., & Munggaran, L. C. (2024). Implementation of GitOps in containerized infrastructure. *RABIT: Jurnal Teknologi dan Sistem Informasi Univrab*, 9(1), 154–161.
<https://doi.org/10.36341/rabit.v9i1.3787>.
- [554] Jha, S., Arora, R., Watanabe, Y., Yanagawa, T., Chen, Y., Clark, J., Verma, M., et al. (2025). ITBench: Evaluating AI agents across diverse real-world IT automation tasks. *arXiv preprint*, arXiv:2502.05352.
- [555] Sivaraman, P., Prabakaran, G., Sarveshwaran, V., & Rajasekar, V. (2024). Efficient auto scaling of pods in Kubernetes: Accelerating continuous delivery with KEPTN. In *5th International Conference on Electronics and Sustainable Communication Systems (ICESC 2024)*. IEEE.
<https://doi.org/10.1109/ICESC60852.2024.10689854>.
- [556] Leonardi, N., & Profeta, D. (2024). DataOps, MLOps, and Secure CEDS Connectors (Deliverable D3.1, Version 1.0). CEDAR – Common European Data Spaces and Robust AI for Transparent Public Governance (Horizon Europe, Grant Agreement No. 101135577). Public deliverable submitted 28 June 2024. https://cedar-heu-project.eu/sites/default/files/materials/CEDAR_Deliverable_D3.1_final.pdf.
- [557] Ali, A. A. (2021). GitOps for configuration drift management in Kubernetes environments (Master’s thesis, Oslo Metropolitan University). <https://oda.oslomet.no/oda-xmlui/bitstream/handle/11250/3162980/no.oslomet%3Ainspera%3A232805187%3A125264241.pdf>.
- [558] Saraf, P., Mallick, S. S., Jagarlamudi, L., Chakraborty, A., & Simmhan, Y. (2025). CARL: Cost-optimized online container placement on VMs using adversarial reinforcement learning. *IEEE Transactions on Cloud Computing*, 13(1), 321–334. <https://doi.org/10.1109/TCC.2025.3528446>.
- [559] Geller, C., Haas, B., Kloecker, A., Hermens, J., Lampe, B., Beemelmans, T., & Eckstein, L. (2024). CARLOS: An open, modular, and scalable simulation framework for the development and testing of software for C-ITS. In *2024 IEEE Intelligent Vehicles Symposium (IV)*, Jeju Island, Korea. IEEE.
<https://doi.org/10.1109/IV55156.2024.1058850>.
- [560] James, C. (2024). Case studies on successful integration of TDD and BDD in CI/CD pipelines. <https://www.researchgate.net/publication/337076816>.
- [561] Sermpezis, E., Karapiperis, D., & Tjortjis, C. (2024). Integration of security in the DevOps methodology. In *2024 15th International Conference on Information, Intelligence, Systems & Applications (IISA)*. IEEE. <https://doi.org/10.1109/IISA62523.2024.10786669>.
- [562] Haengel, E., & Metzger, L. (2024). HawkEye 360’s CI/CD approach to automatic on-orbit firmware updates. In *38th Annual Small Satellite Conference (SSC24-S3-09)*. <https://doi.org/10.15142/v8g8-tz05>.
- [563] Yun, C.-Y., & Moon, S.-J. (2024). A fabricator design for metadata CI/CD in data fabric. *International Journal of Internet, Broadcasting and Communication*, 16(2), 193–202.
<https://doi.org/10.7236/IJIBC.2024.16.2.193>.
- [564] Ho, E., Chiu, Y.-H., & Liao, C.-F. (2024). Tactics for enhancing write ahead log for long-term system maintenance. In *2024 International Conference on Platform Technology and Service (PlatCon)*. IEEE. <https://doi.org/10.1109/PLATCON63925.2024.10830742>.
- [565] Vihovde, E. H., & Meng, Q. (2024). Test-driven development: Ensuring code quality in integration with CI/CD. In *2024 8th International Conference on Management Engineering, Software Engineering and Service Sciences (ICMSS)*. IEEE. <https://doi.org/10.1109/ICMSS61211.2024.00009>.
- [566] Diaz-de-Arcaya, J., López-de-Armentia, J., Miñón, R., Lasa Ojanguren, I., & Torre-Bastida, A. I., “Large Language Model Operations (LLMOps): Definition, Challenges, and Lifecycle Management,” in

- 2024 9th International Conference on Smart and Sustainable Technologies (SpliTech), IEEE, 2024. <https://doi.org/10.23919/SpliTech61897.2024.10612341>.
- [567] Awad, H., Alidra, A., Bruneliere, H., Ledoux, T., Leclercq, E., & Rivalan, J. (2024). VeriFog: A generic model-based approach for verifying fog systems at design time. In Proceedings of the 39th ACM/SIGAPP Symposium on Applied Computing (SAC '24) (pp. 1252–1261). ACM. <https://doi.org/10.1145/3605098.3635973>.
- [568] Khalaifat, W., & Abdulaziz, F. (2024). Challenges and opportunities of DevOps for IoT systems: DevOps-for-IoT. In 2024 International Conference on Innovation and Intelligence for Informatics, Computing, and Technologies (3ICT). IEEE. <https://doi.org/10.1109/3ICT64318.2024.10824302>.
- [569] Szandała, T. (2025). ChatGPT vs human expertise in the context of IT recruitment. *Expert Systems with Applications*, 264, 125868. Elsevier. <https://doi.org/10.1016/j.eswa.2024.125868>.
- [570] Wang, S.-K., Ma, S.-P., Lai, G.-H., & Chao, C.-H. (2024). ChatOps for microservice systems: A low-code approach using service composition and large language models. *Future Generation Computer Systems*, 161, 518–530. <https://doi.org/10.1016/j.future.2024.07.029>.
- [571] Ghaleb, T. A., Abduljalil, O., & Hassan, S. (2024). CI/CD configuration practices in open-source Android apps: An empirical study. *arXiv preprint*, arXiv:2411.06077.
- [572] Huang, M., & Lemieux, C. (2024). Directed or undirected: Investigating fuzzing strategies in a CI/CD setup (registered report). In Proceedings of the 3rd ACM International Fuzzing Workshop (FUZZING '24), Vienna, Austria. ACM. <https://doi.org/10.1145/3678722.3685532>.
- [573] Simhandl, G., & Zdun, U. (2024). Cloud programming languages and infrastructure from code: An empirical study. In Proceedings of the 17th ACM SIGPLAN International Conference on Software Language Engineering (SLE '24), Pasadena, CA, USA. ACM. <https://doi.org/10.1145/3687997.3695643>.
- [574] Owoade, S. J., Uzoka, A., Akerele, J. I., & Ojukwu, P. U. (2024). Cloud-based compliance and data security solutions in financial applications using CI/CD pipelines. *World Journal of Engineering and Technology Research*, 3(2), 011–021. <https://doi.org/10.53346/wjetr.2024.3.2.0059>.
- [575] Yadav, P. S. (2024). Migration to serverless architectures: A pathway to efficient cloud computing. *International Journal of Science and Research (IJSR)*, 13(10). <https://doi.org/10.21275/SR241002075058>.
- [576] Savant, R. V., Sunder, S. N., Seshadri, S., Panda, N., & Rajagopal, S. M. (2024). Cloud-native CDN monitoring using CI/CD. In 2024 15th International Conference on Computing Communication and Networking Technologies (ICCCNT), IIT Mandi, India. IEEE. <https://doi.org/10.1109/ICCCNT61001.2024.10724159>.
- [577] Vasa, R. (2025). Cloud-native middleware: AI as the driving force behind digital transformation. *International Journal of Computer Engineering and Technology (IJCET)*, 16(1), 3358–3374. https://doi.org/10.34218/IJCET_16_01_234.
- [578] Thota, R. C. (2024). Cloud-native DevSecOps: Integrating security automation into CI/CD pipelines. *International Journal of Innovative Research and Creative Technology (IJIRCT)*, 10(6). <https://doi.org/10.5281/zenodo.15036934>.
- [579] Afifah, A. S., Kabetta, H., Setiawan, H., & Buana, I. K. S. (2024). Code obfuscation in CI/CD pipelines for enhanced DevOps security. In 2024 International Conference on Artificial Intelligence, Blockchain, Cloud Computing, and Data Analytics (ICoABCD). IEEE. <https://doi.org/10.1109/ICoABCD63526.2024.10704536>.
- [580] Albin, S., Attebury, G., Bloom, K., Bockelman, K., Lundstedt, C., Shadura, O., & Thiltges, J. (2024). Coffea-Casa: Building composable analysis facilities for the HL-LHC. *EPJ Web of Conferences*, 295, 07009. <https://doi.org/10.1051/epjconf/202429507009>.
- [581] Patchamatla, P. S., & Owolabi, I. O. (2025). Comparative study of open-source CI/CD tools for machine learning deployment. *CogNexus: A Multidisciplinary, Multilingual, International, Peer-Reviewed, Open Access Journal*, 1(1), 239-250. <https://doi.org/10.63084/cognexus.v1i01.34>.

- [582] Cui, J. (2024). Comparative analysis of DevOps, DevSecOps, and AIOps in software development architectures: Impacts on software quality and development efficiency. Preprint. <https://doi.org/10.13140/RG.2.2.32206.68162>.
- [583] James, C. (2024). Comparing TDD and BDD approaches in enhancing CI/CD pipeline efficiency. https://www.researchgate.net/publication/385277944_Comparing_TDD_and_BDD_Approaches_in_Enhancing_CICD_Pipeline_Efficiency.
- [584] Talasila, P., Gomes, C., Vosteen, L. B., Iven, H., Leucker, M., Gil, S., Mikkelsen, P. H., Kamburjan, E., & Larsen, P. G. (2024). Composable digital twins on Digital Twin as a Service platform. SIMULATION: Transactions of the Society for Modeling and Simulation International. <https://doi.org/10.1177/00375497241298653>.
- [585] Horvath, K., Abid, M. R., Merino, T., Zimmerman, R., Peker, Y., & Khan, S. (2024). Cloud-based infrastructure and DevOps for energy fault detection in smart buildings. *Computers*, 13(1), 23. <https://doi.org/10.3390/computers13010023>.
- [586] Fredrikson, M. (2024). Configuring targeted dynamic parallel testing to a CI/CD pipeline (Bachelor's thesis, JAMK University of Applied Sciences). https://www.theseus.fi/bitstream/10024/858132/2/fredrikson_mikko.pdf.
- [587] George, A. S. (2024). Consequences of enterprise cloud migration on institutional information technology knowledge. *Partners Universal Innovative Research Publication (PUIRP)*, 2(2), 38–40. <https://doi.org/10.5281/zenodo.10938874>.
- [588] Deng, G., & Hu, H. (2024). Container acceleration method based on image block granularity optimization. In 2024 IEEE 4th International Conference on Power, Electronics and Computer Applications (ICPECA). IEEE. <https://doi.org/10.1109/ICPECA60615.2024.10471138>.
- [589] Manor, D., Rod, D., Fang, H., & Watkins, J. (2024). Containerization in cloud computing: A review. Preprint. https://www.researchgate.net/publication/389597107_Containerization_in_Cloud_Computing_A_Review.
- [590] Sharma, V. (2019). Continuous integration and continuous delivery (CI/CD): A comprehensive overview. *International Journal of Science and Research (IJSR)*, 8(10). <https://doi.org/10.21275/SR24115221653>.
- [591] Hornbeek, M. (2024). Continuous Testing, Quality, Security, and Feedback. Packt Publishing. ISBN 978-1-83546-224-9. Available: <https://www.packtpub.com/en-us/product/continuous-testing-quality-security-and-feedback-9781835462249>.
- [592] Ugwueze, V. U., & Chukwunweike, J. N. (2024). Continuous integration and deployment strategies for streamlined DevOps in software engineering and application delivery. *International Journal of Computer Applications Technology and Research*, 14(1), 1–24. <https://doi.org/10.7753/IJCATR1401.1001>
- [593] Tymoshenko, J., Savchenko, B., & Adeola, F. R. (2025). Continuous security testing for mobile apps in CI/CD pipelines: Best practices. *ITS Journal – Intelligent Transportation Systems Journal*. Available: https://www.researchgate.net/publication/388852091_Continuous_Security_Testing_for_Mobile_Apps_in_CICD_Pipelines_Best_Practices.
- [594] Nahar, G., & Jain, S. B. (2023, August). Continuous testing in Agile: A game-changer or a bottleneck? *Samdarshi*, 16(3), 1414–1418. ISSN 2581-3986. Available on https://www.researchgate.net/publication/389581409_Continuous_Testing_in_Agile_A_Game-Changer_or_a_Bottleneck.
- [595] Andrade, R., Torres, J., Flores, P., Cabezas, E., & Segovia, J. (2024). Convergence of AI for secure software development. In 2024 8th Cyber Security in Networking Conference (CSNet). IEEE. <https://doi.org/10.1109/CSNET64211.2024.10851473>.
- [596] Bandara, E., Bouk, S. H., Shetty, S., Roy, S., Mukkamala, R., Rahman, A., Foytik, P., Liang, X., Keong, N. W., & De Zoysa, K. (2025). Llama-Recipe—Fine-tuned Meta's Llama LLM, PBOM and NFT enabled 5G network-slice orchestration and end-to-end supply-chain verification platform. In 2025 IEEE

Consumer Communications & Networking Conference (CCNC) (pp. 1–8). IEEE.
<https://doi.org/10.1109/CCNC54725.2025.10976116>.

[597] Awad, H., Ledoux, T., Bruneliere, H., & Rivalan, J. (2025). VeriFogOps: Automated deployment tool selection and CI/CD pipeline generation for verifying fog systems at deployment time. In *Proceedings of the 40th ACM/SIGAPP Symposium on Applied Computing (SAC '25)* (pp. 1–10). ACM.
<https://doi.org/10.1145/3672608.3707854>.

[598] Tang, B., Zhang, X., Yang, Q., Qi, X., Alqahtani, F., & Tolba, A. (2025). Cost-optimized Internet of Things application deployment in edge computing environment. *International Journal of Communication Systems*, 38, e5618. <https://doi.org/10.1002/dac.5618>.

[599] Golis, T., & Dakić, P. (2024). Creating a self-service DevOps platform for black-box testing on Kubernetes. In X.-S. Yang et al. (Eds.), *Proceedings of the Ninth International Congress on Information and Communication Technology, Lecture Notes in Networks and Systems*, 1004 (pp. 345–350). Springer.
https://doi.org/10.1007/978-981-97-3305-7_28.

[600] Adamson, R., Bryant, P., Montoya, D., Neel, J., Palmer, E., Powell, R., Prout, R., & Upton, P. (2024). Creating continuous integration infrastructure for software development on U.S. Department of Energy high-performance computing systems. *Computing in Science & Engineering*, 26(1), 31–36. IEEE.
<https://doi.org/10.1109/MCSE.2024.336258>.

[601] Zheng, W., Yang, J., Chen, J., He, J., Li, P., Sun, D., Chen, C., & Meng, X. (2025). Cross-temporal knowledge injection with color distribution normalization for remote sensing change detection. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 18, 6249–6259.
<https://doi.org/10.1109/JSTARS.2025.3534583>.

[602] Orosz, M., Duffy, B., Charlton, C., Saunders, H., & Thomas, E. (2024). Unique challenges in mission engineering and technology integration. In D. Verma (Ed.), *Systems engineering for the digital age: Practitioner perspectives* (ch. 31). Hoboken, NJ: Wiley. <https://doi.org/10.1002/9781394203314.ch31>.

[603] Flores, A., Gião, H., Amaral, V., & Cunha, J. (2024). A two-level model-driven approach for reengineering CI/CD pipelines. In *Proceedings of the Conference on Model Driven Engineering Languages and Systems*. FCT, Portugal. <https://doi.org/10.54499/UIDB/50014/2020>.

[604] Cazacu, D.-L.; Oancea, C.-D. (2024). Monitoring the homogeneity of temperature fields in thermal chambers. In *2024 9th International Conference on Energy Efficiency and Agricultural Engineering (EE&AE)*, Ruse, Bulgaria. IEEE. <https://doi.org/10.1109/EEAE60309.2024.10600599>.

[605] Achuthan, B., & Alimohideen, M. A. (2024). Shifting gears: Integrating security audits into automotive DevSecOps. In *2024 IEEE International Conference on Vehicular Security and DevSecOps*. IEEE. (ISBN 979-8-3503-5551-2). DOI: <https://doi.org/10.1109/ICVTTS62812.2024.10763937>.

[606] Jeon, C. (2024). The evolution of Open RAN based on MLOps platform. In *2024 IEEE International Conference on Machine Learning and Communications*. IEEE, Red Hat & Seoul National University. DOI: <https://doi.org/10.1109/ICTC62082.2024.10827277>.

[607] Melén, J. (2025). Strategies for upgrading distributed edge cloud nodes (Bachelor's thesis, Metropolia University of Applied Sciences, Finland). <https://www.theseus.fi/handle/10024/878054>.

[608] Sehgal, J. (2025). Release engineering in the age of Kubernetes: Managing deployments in containerized environments. *International Journal on Science and Technology (IJSAT)*, 16(1). (ISSN 2229-7677). <http://www.ijstat.org/>.

[609] Balakrishna, B., Challa, N., Mooghala, S., & Tammana, P. K. (2024). Synergizing digital transformation. CARI Journals and Books Publishers. (ISBN 978-9914-746-17-4)

[610] Lazarus, J. I., Truett, L., & Fischer, B., & Kershner, C. (2024). DevSecOps process assessment collaboration tool: A novel method to inject R&M into agile development. In *2024 Annual Reliability and Maintainability Symposium (RAMS)*. IEEE. <https://doi.org/10.1109/RAMS51492.2024.10457824>.

- [611] Tahir, M. F. (2024). Role of AI/ML in decision making in software release management (Master's thesis, Lappeenranta–Lahti University of Technology LUT). https://lutpub.lut.fi/bitstream/handle/10024/168816/masterthesis_tahir_muhammadfaizan.pdf.
- [612] Bali, M. K., Mehdi, A., & Hariharan, H. (2024). AI-driven DevOps transformation: A paradigm shift in software development. In 2024 3rd International Conference on Sentiment Analysis and Deep Learning (ICSADL). <https://doi.org/10.1109/ICSADL61749.2024.00026>.
- [613] Sinha, P. K., Nand, P., Chauhan, S. S., & Tiwari, S. (2024). Study of failures in continuous integration environment. In 2024 International Conference on Electrical Electronics and Computing Technologies (ICEECT). IEEE. <https://doi.org/10.1109/ICEECT61758.2024.10739077>.
- [614] Vaish, P., Anand, N., & Sharma, G. (2024). Unleashing the power of DevOps and IoT: A framework for software delivery and system availability. *Multimedia Tools and Applications*. <https://doi.org/10.1007/s11042-024-19452-w>.
- [615] Nikolov, L., & Aleksieva-Petrova, A. (2024). Framework for integrating threat modeling into a DevOps pipeline for enhanced software development. Presented at IEEE Conference.
- [616] Khan, A. F., Khan, A. A., Mohamed, A., Ali, H., Moonlinti, S., Haroon, S., Tahir, U., Fazzini, M., Butt, A. R., & Anwar, A. (2025). LADs: Leveraging LLMs for AI-driven DevOps. *arXiv preprint, arXiv:2502.20825*.
- [617] Khonde, K. R., Shah, J., & Patel, P. (2024). EchoSense AI transcriber using DevOps. In 2024 Parul International Conference on Engineering and Technology (PICET). IEEE. <https://doi.org/10.1109/PICET60765.2024.10716195>.
- [618] Dincelli, E., & Yayla, A. (2024). Security and privacy implications of virtual reality applications in the Metaverse: A case of development, security, and operations (DevSecOps). *Journal of Information Systems Education*, 35(3), 261–270. <https://doi.org/10.62273/JMZA1065>.
- [619] Luca, C. (2025). Data privacy and ethical considerations in generative LSTM-based ad personalization for streaming media. https://www.researchgate.net/publication/388109531_Data_Privacy_and_Ethical_Considerations_in_Generative_LSTM-Based_Ad_Personalization_for_Streaming_Media.
- [620] Abbas, S. I., & Singh, M. (2024). Enhancing operational efficiency and user experience in virtual workspaces: A comprehensive analysis and action plan. In 2024 3rd International Conference on Sentiment Analysis and Deep Learning (ICSADL). IEEE. <https://doi.org/10.1109/ICSADL61749.2024.00014>.
- [621] Karlovs-Karlovis, U., & Nikiforova, O. (2024). Systematic literature review on model transformation for advanced generation of pipelines within DevOps solutions. In 2024 IEEE Conference Publication. IEEE. (ISBN 979-8-3315-3383-0). DOI: <https://doi.org/10.1109/ITMS64072.2024.10741933>.
- [622] Nouri, A., Cabrero-Daniel, B., Törner, F., Sivencrona, H., & Berger, C. (2024). Welcome your new AI teammate: On safety analysis by leashing large language models. In CAIN 2024: 3rd International Conference on AI Engineering – Software Engineering for AI. ACM, Lisbon, Portugal. <https://doi.org/10.1145/3644815.3644953>.
- [623] Mumtaz, M. H. (2024). Digital transformation in software development: Success factors, challenges, and impacts in transitioning from Waterfall to DevOps in industry (Master's thesis, Lappeenranta–Lahti University of Technology LUT). https://lutpub.lut.fi/bitstream/10024/167853/1/MasterThesis_Muhammad%20Hamza%20Mumtaz%20.pdf.
- [624] Lykousas, N., & Patsakis, C. (2024). Decoding developer password patterns: A comparative analysis of password extraction and selection practices. *Computers & Security*, 145, 103974. <https://doi.org/10.1016/j.cose.2024.103974>.
- [625] Rojpaisarnkit, R., Robles, G., Kula, R. G., Wang, D., Ragkhitwetsagul, C., Gonzalez-Barahona, J. M., & Matsumoto, K. (2024). Towards identifying code proficiency through the analysis of Python textbooks. In *Proceedings of the 2024 IEEE International Conference on Software Maintenance and Evolution (ICSME)* (pp. 288–300). IEEE. <https://doi.org/10.1109/ICSME58944.2024.00035>.

- [626] Qu, M., He, J., Tucaković, Z., Bartocci, E., Ničković, D., Isaković, H., & Grosu, R. (2024). DeepRIoT: Continuous integration and deployment of robotic-IoT applications. In DAC '24: Proceedings of the 2024 Design Automation Conference. ACM, San Francisco, CA, USA. <https://doi.org/10.1145/3649329.3658250>.
- [627] Valle, P., Arrieta, A., Han, L., Ali, S., & Yue, T. (2025). Defining and generating multi-level and uncertainty-wise test oracles for cyber-physical systems. *Software and Systems Modeling*. <https://doi.org/10.1007/s10270-025-01271-8>.
- [628] Thopalle, P. K. (2024). DevSecOps: Integrating security into the DevOps lifecycle with AI and automation. *International Journal of Advanced Research in Engineering and Technology (IJARET)*, 15(3), 452–466. Available: https://iaeme.com/MasterAdmin/Journal_uploads/IJARET/VOLUME_15_ISSUE_3/IJARET_15_03_038.pdf.
- [629] Rai, S. K. (2025). Demystifying cloud-native data engineering architectures. *International Journal of Information Technology and Management Information Systems (IJTMIS)*, 16(2), 959–993. https://doi.org/10.34218/IJTMIS_16_02_062.
- [630] Febrianto, R. F., Nugraheni, D. M. K., Suharto, E., Widodo, A. P., & Ariyanti, Y. D. P. (2024). Deployment strategy with integration testing implementation using DevOps method in development and production environment. In 2024 7th International Conference on Informatics and Computational Sciences (ICICoS). IEEE. <https://doi.org/10.1109/ICICoS60733.2024.00047>.
- [631] Kern, C. (2024). Developer ecosystems for software safety. *Communications of the ACM*, 67(6), 52–59. <https://doi.org/10.1145/3651621>.
- [632] Das, S., Deb, N., Chaki, N., & Cortesi, A. (2024). Minimising conflicts among run-time non-functional requirements within DevOps. *Systems Engineering*, 27(2), 177–198. Wiley. <https://doi.org/10.1002/sys.21715>.
- [633] Mekić, E. S., Jovanović, M. N., Kuk, K. V., Prlinčević, B. P., & Savić, A. M. (2024). Enhancing educational efficiency: Generative AI chatbots and DevOps in Education 4.0. *Computer Applications in Engineering Education*, 32(6), e22804. Wiley. <https://doi.org/10.1002/cae.22804>.
- [634] Ebert, C., Gallardo, G., Hernantes, J., & Serrano, N. (2025). DevOps 2.0. *IEEE Software*, 42(2), 24–31. <https://doi.org/10.1109/MS.2025.3525768>.
- [635] Chowdary, V. H. D., Kumar, B. S., Shanmukh, A., Khan, F., & Nikhil, T. P. (2024). DevOps 2.0: Embracing AI/ML, cloud-native development, and a culture of continuous transformation. In Proceedings of the 2024 4th International Conference on Pervasive Computing and Social Networking (ICPCSN). IEEE. <https://doi.org/10.1109/ICPCSN62568.2024.00112>.
- [636] Mekić, E. S., Jovanović, M. N., Kuk, K. V., Prlinčević, B. P., & Savić, A. M. (2024). Enhancing educational efficiency: Generative AI chatbots and DevOps in Education 4.0. *Computer Applications in Engineering Education*, 32, e22804. <https://doi.org/10.1002/cae.22804>.
- [637] Yousefnezhad, N., & Costin, A. (2024). Understanding SBOMs in real-world systems – A practical DevOps/SecOps perspective. In Proceedings of the 14th International Symposium on Business Modeling and Software Design (BMSD 2024), Lecture Notes in Business Information Processing (LNBIP), 523, 293–304. Springer. https://doi.org/10.1007/978-3-031-64073-5_20.
- [638] Dakkak, A., Daniele, P., Bosch, J., & Olsson, H. H. (2024). DevOps value flows in software-intensive system of systems. In Proceedings of the 2024 50th Euromicro Conference on Software Engineering and Advanced Applications (SEAA). IEEE. <https://doi.org/10.1109/SEAA64295.2024.00065>.
- [639] Langerman, J., & Leung, W. S. (2024). The effect of outsourcing and insourcing on Agile and DevOps. *Journal of Information Technology Teaching Cases*, 14(2), 192–199. <https://doi.org/10.1177/20438869231176841>.
- [640] Saxena, A., Singh, S., Prakash, S., Yang, T., & Rathore, R. S. (2024). DevOps automation pipeline deployment with IaC (Infrastructure as Code). In Proceedings of the 2024 IEEE Silchar Subsection Conference (SILCON 2024) (pp. 1–6). IEEE. <https://doi.org/10.1109/SILCON63976.2024.10910699>.

- [641] Akbar, M. A., Khan, A. A., Islam, N., & Mahmood, S. (2024). DevOps project management success factors: A decision-making framework. *Software: Practice and Experience*, 54, 257–280. Wiley. <https://doi.org/10.1002/spe.3269>.
- [642] Akbar, M. A., Rafi, S., Hyrynsalmi, S., & Khan, A. A. (2024). Towards people maturity for secure development and operations: A vision. In *Proceedings of the 28th International Conference on Evaluation and Assessment in Software Engineering (EASE 2024)*, Salerno, Italy, June 18–21. ACM. <https://doi.org/10.1145/3661167.3661238>.
- [643] Jayakody, J. A. V. M. K., & Wijayanayake, W. M. J. I. (2024). DevOps maturity: A systematic literature review. In *2024 International Research Conference on Smart Computing and Systems Engineering (SCSE)*. IEEE. <https://doi.org/10.1109/SCSE61872.2024.10550493>. ISBN: 979-8-3503-7568-8.
- [644] Jain, S., & Kumar, P. (2024). DevOps practices into machine learning. In *Proceedings of the 2024 IEEE International Conference on Intelligent Systems, Smart and Green Technologies (ICISSGT)*. IEEE. <https://doi.org/10.1109/ICISSGT58904.2024.00029>.
- [645] Goyal, A. (2024). Optimising cloud-based CI/CD pipelines: Techniques for rapid software deployment. *Technix International Journal for Engineering Research*, 11(11), 896–902. ISSN 2349-9249. https://www.researchgate.net/publication/386501005_Optimising_Cloud-Based_CICD_Pipelines_Techniques_for_Rapid_Software_Deployment.
- [646] Habibi, P., & Leon-Garcia, A. (2024). SliceSphere: Agile service orchestration and management framework for cloud-native application slices. *IEEE Access*, 12, 169024–169039. <https://doi.org/10.1109/ACCESS.2024.3492138>.
- [647] Arockiasamy, J. M. I. A. (2025). DevOps-driven real-time health analytics: A scalable framework for wearable IoT data. *International Journal for Multidisciplinary Research (IJFMR)*, 7(1). <https://doi.org/10.36948/ijfmr.2025.v07i01.37358>.
- [648] Bandara, E., Shetty, S., Mukkamala, R., Rahman, A., Foytik, P., Liang, X., Zoysa, K. D., & Keong, N. W. (2024). DevSec-GPT: Generative-AI (with custom-trained Meta’s Llama2 LLM), blockchain, NFT and PBOM enabled cloud native container vulnerability management and pipeline verification platform. In *2024 IEEE Cloud Summit*. IEEE. <https://doi.org/10.1109/Cloud-Summit61220.2024.00012>.
- [649] David, P., Kushwaha, M. K., & Suseela, G. (2024). DevSecOps in finance: Strengthening the security model of applications. In *Proceedings of the 2024 4th IEEE International Conference on Data Engineering and Communication Systems (ICDECS)*. IEEE. <https://doi.org/10.1109/ICDECS59733.2023.10502917>.
- [650] Chellappan, R. B. (2024). The future of DevOps: Intelligent, secure and scalable software delivery. Preprint. <https://doi.org/10.13140/RG.2.2.25699.26403>.
- [651] Omoike, O. (2024). DevSecOps in AWS: Embedding security into the heart of DevOps practices. *International Journal of Science and Research Archive*, 13(2), 1309–1313. <https://doi.org/10.30574/ijrsra.2024.13.2.2306>.
- [652] Benjamin, J., & Mathew, J. (2025). Enhancing continuous integration predictions: A hybrid LSTM–GRU deep learning framework with evolved DBSO algorithm. *Computing*, 107. Springer. <https://doi.org/10.1007/s00607-024-01370-2>.
- [653] Benjamin, M. (2025, March). DevSecOps and security-first development. Unpublished article, Obafemi Awolowo University. https://www.researchgate.net/publication/389816992_DevSecOps_and_Security-First_Development.
- [654] Abos, P. (2024, January). DevSecOps for secure software development in the cloud. Unpublished article, University of Southern Denmark. https://www.researchgate.net/publication/381459140_DevSecOps_for_secure_software_development_in_the_cloud.
- [655] Vangala, V. (2025, January). DevSecOps: Integrating security into the DevOps lifecycle. Unpublished article, ORRBA Systems LLC.

https://www.researchgate.net/publication/388555218_DevSecOps_Integrating_Security_into_the_DevOps_Lifecycle.

[656] Heikel, R. F. P., Guayuan, M. B., Cremona, M. L., Nemeth, H., & Mello-Román, J. C. (2024). Development of applications based on microservices: Case study of Ministry of Economy and Finance of Paraguay. In 2024 Latin American Computer Conference (CLEI). IEEE. <https://doi.org/10.1109/CLEI64178.2024.10700123>.

[657] Tarak, E., & Kılınç, H. H. (2024). DIA4M: A tool to streamline DevOps processes of distributed cloud-native systems. In 2024 9th International Conference on Computer Science and Engineering (UBMK). <https://doi.org/10.1109/UBMK63289.2024.10773447>.

[658] Arockiasamy, J. M. I. (2025). Digital healthcare evolution: The power of DevOps for better patient engagement. *International Journal of Intelligent Systems and Applications in Engineering*, 12(4), 5192–5198. Available at <https://ijisae.org/index.php/IJISAE/article/view/7314>.

[659] Flora, J., & Antunes, N. (2024). Doing more with less? A study on models for intrusion detection in microservices. In 2024 19th European Dependable Computing Conference (EDCC). IEEE. <https://doi.org/10.1109/EDCC61798.2024.00021>.

[660] Chiari, M., Xiang, B., Canzoneri, S., Nedeltcheva, G. N., Di Nitto, E., Blasi, L., Benedetto, D., Niculut, L., & Škof, I. (2024). DOML: A new modeling approach to infrastructure-as-code. *Information Systems*, 125, 102422. <https://doi.org/10.1016/j.is.2024.102422>.

[661] Devan, K. (2019). Enhancing Concourse CI/CD pipelines with real-time webhook triggers: A scalable solution for GitHub resource management. *International Journal of Creative Research in Computer Technology and Design*, 1(1), 1–9. Available at <https://jrctd.in/index.php/IJRCTD/article/view/78>.

[662] Oliveira, A. R. (2025). Dual-modeling approaches in CI/CD: An agnostic modeling framework to streamline DevOps (Master's thesis, NOVA University Lisbon). DOI: 10.5281/zenodo.13748531.

[663] Tiwari, U., Khurana, S., Rai, S., & Gaur, I. (2024). Dynamic container optimization for efficient microservices deployment. In 2024 International Conference on Electrical Electronics and Computing Technologies (ICEECT). IEEE. <https://doi.org/10.1109/ICEECT61758.2024.10739185>.

[664] Ndigande, A. P., Ari, I., & Ozer, S. (2024). Analysis and comparison of Dockerized and standalone Apache Spark configurations for efficient distributed data processing. In 2024 Innovations in Intelligent Systems and Applications Conference (ASYU). IEEE. <https://doi.org/10.1109/ASYU62119.2024.10757047>.

[665] Sharma, Y., Arya, V., Baghel, U., & Chawla, P. (2024). Innovative hosting with Hoster: Integrating AWS services, Docker, and real-time data for scalable deployment. In 2024 2nd International Conference on Advances in Computation, Communication and Information Technology (ICACCIT). IEEE. <https://doi.org/10.1109/ICACCIT65084.2024.00054>.

[666] Figueiredo, A. C., Pereira, R., & da Silva, M. Â. (2025). Exploring the integration of artificial intelligence and DevOps for agile product development. In *Digital Technologies and Transformation in Business, Industry and Organizations, Studies in Systems, Decision and Control* (Vol. 577). Springer. https://doi.org/10.1007/978-3-031-78412-5_2.

[667] Barbosa, D., et al. (2025). A cloud-native agile approach to cyber platform prototyping and integration for astronomy: The ENGAGE SKA case. *Journal of Instrumentation* (JINST). arXiv:2502.04039.

[668] Kumar, M. (2024). The design and implementation of automated deployment pipelines for Amazon Web Services: GitOps practices in the context of CI/CD pipelines using GitLab and infrastructure as code (Master's thesis, Aalto University & KTH). <https://aaltodoc.aalto.fi/items/c3119a69-595c-4fa0-b0b5-c6c1acda4d61>.

[669] Floroiu, M. S., Russo, S., Giamattei, L., Guerriero, A., Malavolta, I., & Pietrantuono, R. (2024). Anomaly detection and root cause analysis of microservices energy consumption. In 2024 IEEE International Conference on Web Services (ICWS). <https://doi.org/10.1109/ICWS62655.2024.00079>.

- [670] Gonzalez, J. L. H., Montebugnoli, S., Scotece, D., & Foschini, L. (2024). xSTART: xApp simulated evaluation environment for developers. In 2024 IEEE International Workshop on Metrology for Industry 4.0 and IoT (MetroInd4.0&IoT). IEEE. <https://doi.org/10.1109/MetroInd4.0IoT61288.2024.10584214>.
- [671] McElroy, D. R. (2024). Cloud, DevSecOps, and software supply chain security. In Effective vulnerability management (ch. 9). Wiley. <https://doi.org/10.1002/9781394277155.ch9>.
- [672] Saikiran, S. R. J., Padal, D. P., Mayan, A., C. A., & Jeslin Shanthamalar, J. (2024). Efficient application deployment: GitOps for faster and secure CI/CD cycles. In 2024 International Conference on Advances in Modern Age Technologies for Health and Engineering Science (AMATHE). IEEE. <https://doi.org/10.1109/AMATHE61652.2024.10582118>.
- [673] Baumgartner, N., Iyengar, P., Schoemaker, T., & Pulvermüller, E. (2024). AI-driven refactoring: A pipeline for identifying and correcting data clumps in Git repositories. *Electronics*, 13(9), 1644. <https://doi.org/10.3390/electronics13091644>.
- [674] Katevas, V., Fatouros, G., Kyriazis, D., & Kousiouris, G. (2024). Embedding automated function performance benchmarking, profiling and resource usage categorization in function-as-a-service DevOps pipelines. *Future Generation Computer Systems*, 160, 223–237. <https://doi.org/10.1016/j.future.2024.05.051>.
- [675] Ascensão, C., Teixeira, H., Gonçalves, J., & Almeida, F. (2024). Large-scale agile security practices in software engineering. *Information & Computer Security*, 32(4). <https://doi.org/10.1108/ICS-07-2023-0136>.
- [676] Sathe, C. A., & Panse, C. (2024). Modelling the enablers of productivity of enterprise-level agile software development process using modified TISM approach. *Journal of Modeling in Management*, 19(6), 1749–1772. <https://doi.org/10.1108/JM2-12-2023-0285>.
- [677] Akbar, M. A., & Alsanad, A. A. (2025). Empirical investigation of key enablers for secure DevOps practices. *IEEE Access*, 13, 43698–43725. <https://doi.org/10.1109/ACCESS.2025.3549183>.
- [678] Pangavhane, S., Shelar, K., Raktate, G., Wakchaure, R., Parjane, P., & Kale, J. N. (2024). AI-augmented software development: Boosting efficiency and quality. In 2024 International Conference on Decision Aid Sciences and Applications (DASA). IEEE. <https://doi.org/10.1109/DASA63652.2024.10836523>.
- [679] Vakhula, O., Kurii, Y., Opirskyy, I., & Susukailo, V. (2024). Security-as-Code Concept for Fulfilling ISO/IEC 27001:2022 Requirements. In Proceedings of CPITS-2024: Cybersecurity Providing in Information and Telecommunication Systems (pp. 59–72). CEUR Workshop Proceedings. Available: <https://ceur-ws.org/Vol-3654/paper6.pdf>.
- [680] Pillala, G. (2024). Empowering developers: Building a secure software factory with native platform integrations and fostering innersource culture for DevSecOps excellence. *International Journal of Information Security (IJIS)*, 3(1), 21–32. <https://doi.org/10.17605/OSF.IO/G7Z3J>.
- [681] Cuadra, J., Hurtado, E., Sarachaga, I., Estévez, E., Casquero, O., & Armentia, A. (2024). Enabling DevOps for fog applications in the smart manufacturing domain: A model-driven based platform engineering approach. *Future Generation Computer Systems*, 157, 360–375. Elsevier. <https://doi.org/10.1016/j.future.2024.03.053>.
- [682] Busch, J.-P., Reiher, L., & Eckstein, L. (2024). Enabling the deployment of any-scale robotic applications in microservice architectures through automated containerization. In 2024 IEEE International Conference on Robotics and Automation (ICRA), Yokohama, Japan. IEEE. <https://doi.org/10.1109/ICRA57147.2024.10611586>.
- [683] Baladari, V. (2020). End-to-end CI/CD deployment of RESTful microservices in the cloud. *International Journal of Core Engineering & Management*, 6(8). <https://doi.org/10.5281/zenodo.15020514>.
- [684] Runsewe, O., Osundare, O. S., Folorunsho, S. O., & Akwawa, L. A. (2024). End-to-end systems development in agile environments: Best practices and case studies from the financial sector. *International Journal of Engineering Research and Development*, 20(8), 522–529. Available: <https://www.researchgate.net/publication/384697507>.

- [685] Shriram, K. M. P. (2025). Engineering efficiency through CI/CD pipeline optimization. *International Journal of Science and Research Archive*, 14(1), 908–916. <https://doi.org/10.30574/ijrsra.2025.14.1.0107>.
- [686] Nouri, A., Cabrero-Daniel, B., Törner, F., Sivencrona, H., & Berger, C. (2024). Engineering safety requirements for autonomous driving with large language models. In *2024 IEEE 32nd International Requirements Engineering Conference (RE)*. IEEE. <https://doi.org/10.1109/RE59067.2024.00029>.
- [687] Shin, D., Kim, J., Pawana, I. W. A. J., & You, I. (2025). Enhancing cloud-native DevSecOps: A zero trust approach for the financial sector. *Computer Standards & Interfaces*, 93. Elsevier. <https://doi.org/10.1016/j.csi.2025.103975>.
- [688] Chittala, S. (2024). Enhancing developer productivity through automated CI/CD pipelines: A comprehensive analysis. *International Journal of Computer Engineering and Technology (IJCET)*, 15(5), 882–891. <https://doi.org/10.5281/zenodo.13929524>.
- [689] Kow, S.-L. (2025). Enhancing DevOps efficiency through cloud computing: A comprehensive analysis of automation, scalability, and continuous integration. *International Journal of DevOps (IJDO)*, 2(1), 1–10. Available: <https://www.researchgate.net/publication/389660973>.
- [690] Yulianto, S., & Ngo, G. N. C. (2024). Enhancing DevSecOps pipelines with AI-driven threat detection and response. In *2024 International Conference on ICT for Smart Society (ICISS)*. IEEE. <https://doi.org/10.1109/ICISS62896.2024.10751269>.
- [691] Bedoya, M., Palacios, S., Díaz-López, D., Laverde, E., & Nespoli, P. (2024). Enhancing DevSecOps practice with large language models and security chaos engineering. *International Journal of Information Security*, 23, 3765–3788. <https://doi.org/10.1007/s10207-024-00909-w>.
- [692] Martina, M. R., Bianchini, E., Sinceri, S., Francesconi, M., & Gemignani, V. (2024). Software medical device maintenance: DevOps based approach for problem and modification management. *Journal of Software: Evolution and Process*, 36, e2570. Wiley. <https://doi.org/10.1002/smr.2570>.
- [693] Ponce, F., Forti, S., Soldani, J., & Brogi, A. (2025). Enhancing failure resilience of cloud-edge microservices: The FREEDA approach. In *ESOCC 2025: European Conference on Service-Oriented and Cloud Computing (LNCS 15547)*, pp. 105–111. Springer. https://doi.org/10.1007/978-3-031-84617-5_9.
- [694] Dragomirescu, O.-A., Crăciun, P.-C., & Bologa, A. R. (2025). Enhancing invoice processing automation through the integration of DevOps methodologies and machine learning. *Systems*, 13(2), 87. <https://doi.org/10.3390/systems13020087>.
- [695] Shaikh, S., & Ghosh, S. (2024). Enhancing software reliability through machine learning: Prediction through evaluation metrics. In *Proceedings of the 2024 IEEE International Conference on Blockchain and Distributed Systems Security (ICBDS)*, COEP Technological University, Pune, India, IEEE. <https://doi.org/10.1109/ICBDS61829.2024.10837281>.
- [696] Boisrond, K., Tardif, P. M., & Jaafar, F. (2024). Ensuring the integrity, confidentiality, and availability of IoT data in Industry 5.0: A systematic mapping study. *IEEE Access*, 12, 107017–107045. DOI: <https://doi.org/10.1109/ACCESS.2024.3434618>.
- [697] Yarmolenko, D., Kononenko, A., & Adebayo, H. (2024). Enhancing DevSecOps with dynamic security testing tools. *Open Journal of Artificial Intelligence*, December 2024. Available: <https://www.researchgate.net/publication/388855430>.
- [698] Patchamatla, P. S. (2025). Enhancing software development efficiency: A comprehensive study on DevOps practices and automation. *HBRC Journal*, 8(2). <https://doi.org/10.5281/zenodo.14916413>.
- [699] Pattanayak, S., Murthy, P., & Mehra, A. (2024). Integrating AI into DevOps pipelines: Continuous integration, continuous delivery, and automation in infrastructural management: Projections for future. *International Journal of Science and Research Archive*, 13(1), 2244–2256. <https://doi.org/10.30574/ijrsra.2024.13.1.1838>.
- [700] Olsson, H. H., & Bosch, J. (2025). Strategic digital product management: Nine approaches. *Information and Software Technology*, 177, 107594. Elsevier. <https://doi.org/10.1016/j.infsof.2024.107594>.

- [701] Fuentes-Quijada, G., Ruiz-González, F., & Caro, A. (2024). Enterprise architecture and IT governance to support the BizDevOps approach: A systematic mapping study. *Information Systems Frontiers*. Springer. <https://doi.org/10.1007/s10796-024-10473-2>.
- [702] Trobo, I. P., Evangelou, K., Lossent, A., & Wagner, A. (2024). Version control and DevOps for accelerator and experiments: Experience and outlook. *EPJ Web of Conferences*, 295, 05024. <https://doi.org/10.1051/epjconf/202429505024>.
- [703] Schrittwieser, S., & Ianni, M. (2024). Software security. *ERCIM News*, 139, October. <https://ercim-news.ercim.eu/images/stories/EN139/EN139-web.pdf>.
- [704] Vogl, R., & Blank-Burian, M. (2025). An update on the Münster University Cloud: Technical architecture and user adoption. In *Proceedings of EUNIS 2024 Annual Congress in Athens (EPiC Series in Computing, Vol. 105, pp. 126-135)*. EasyChair. <https://doi.org/10.29007/dd24>.
- [705] Adhikari, N. B. (2024). Evaluating security tools in the context of DevSecOps. Master's thesis, Tampere University, Faculty of Information Technology and Communication Sciences. <https://trepo.tuni.fi/bitstream/handle/10024/156209/AdhikariNikesh.pdf>.
- [706] Dapshima, B., & Ahmad, S. K. (2024). Evaluation and assessment of software security risks and vulnerabilities within the realm of secure DevOps. Article, July. <https://www.researchgate.net/publication/382440594>.
- [707] Lange, F., & Kunz, I. (2024). Evolution of secure development lifecycles and maturity models in the context of hosted solutions. *Journal of Software: Evolution and Process*, 36, e2711. <https://doi.org/10.1002/smr.2711>.
- [708] Singh, C. P. (2024). Evolving CI/CD pipelines with AI: Intelligent error detection and performance optimization. *Journal of Artificial Intelligence, Machine Learning and Data Science*, 2(1), 2073–2076. <https://doi.org/10.51219/JAIMLD/Chandra-prakash-singh/455>.
- [709] Mainetti, G., Hernandez, F., Jammes, F., & Le Boulc'h, Q. (2024). Experience deploying an analysis facility for the Rubin Observatory's Legacy Survey of Space and Time (LSST) data. *EPJ Web of Conferences*, 295, 07017. <https://doi.org/10.1051/epjconf/202429507017>.
- [710] Luz, H., Peace, P., Luz, A., & Joseph, S. (2024). Explainable AI in CI/CD deployment orchestration. Article, December. <https://www.researchgate.net/publication/387556307>.
- [711] Bennett, L. (2019). Explore the application of GitOps principles to manage and deploy applications in distributed edge computing scenarios. *International Journal of Artificial Intelligence and Machine Learning in Engineering*, 4(1), 419–431. https://www.researchgate.net/publication/384695585_Explore_the_Application_of_GitOps_Principles_to_Manage_and_Deploy_Applications_in_Distributed_Edge_Computing_Scenarios.
- [712] Rodrigues-Filho, R., Sene Jr., I., Porter, B., Bittencourt, L. F., Kon, F., & Costa, F. M. (2025). Exploring emergent microservice evolution in elastic deployment environments. *Journal of Systems and Software*, 219, 112252. Elsevier. <https://doi.org/10.1016/j.jss.2024.112252>.
- [713] Ajitha, A. P. (2024). MedAI-DevOps imaging suite: Integrating CNN in diagnostic imaging with continuous deployment and real-time monitoring. In *2024 International Conference on Trends in Quantum Computing and Emerging Business Technologies (TQCEBT)*, CHRIST University, Pune Lavasa Campus, India. <https://doi.org/10.1109/TQCEBT59414.2024.10545123>.
- [714] Adnan, M., Rafi, S., Akbar, M. A., et al. (2024). MLOps-enabled security strategies for next-generation operational technologies. In *Proceedings of the 2024 ACM International Conference*. ACM. <https://doi.org/10.1145/3661167.3661283>.
- [715] Qureshi, J. N., Farooq, M. S., Ali, U., Khelifi, A., & Atal, Z. (2025). Exploring the integration of blockchain and distributed DevOps for secure, transparent, and traceable software development. *IEEE Access*, 13, 15489–15506. <https://doi.org/10.1109/ACCESS.2024.3509036>.

- [716] Bradbury, M. S. (2024). Exploring the synergy between Agile and DevOps methodologies for enhanced software development and delivery. *Frontiers in Engineering and Technology*, 5(2), 1-5. Available: <https://www.researchgate.net/publication/389823159>.
- [717] Ryan, I., Roedig, U., & Stol, K.-J. (2024). Training developers to code securely: Theory and practice. In *Proc. 2024 ACM/IEEE 4th Int. Workshop on Engineering and Cybersecurity of Critical Systems and 2nd Int. Workshop on Software Vulnerability (EnCyCriS/SVM '24)*, Lisbon, Portugal (pp. 1–8). ACM. <https://doi.org/10.1145/3643662.3643956>.
- [718] Kumar, A., Nadeem, M., & Shameem, M. (2024). Metaheuristic-based cost-effective predictive modeling for DevOps project success. *Applied Soft Computing Journal*, 163. Elsevier. <https://doi.org/10.1016/j.asoc.2024.111834>.
- [719] Kisaakye, J., Beyazit, M., & Demeyer, S. (2025). Extending a flakiness score for system-level tests. In *ICTSS 2024: Testing Software and Systems (LNCS 15383)*, pp. 292–312). Springer. https://doi.org/10.1007/978-3-031-80889-0_20.
- [720] Nagasundari, N. S., Manja, P., Mathur, P., & Honnavalli, P. B. (2025). Extensive review of threat models for DevSecOps. *IEEE Access*, 13, 45252–45278. <https://doi.org/10.1109/ACCESS.2025.3547932>.
- [721] Findra, M. (2024). Vulnerability assessment of container images. Master's thesis, Brno University of Technology, Faculty of Information Technology. https://theses.cz/id/sexirr/xfindr00_DIP_Vulntron_Archive.pdf.
- [722] Angi, A., Sacco, A., Esposito, F., Marchetto, G., & Clemm, A. (2024). NAIL: A network management architecture for deploying intent into programmable switches. *IEEE Communications Magazine*, 62(6), 28–34. <https://doi.org/10.1109/MCOM.001.2300313>.
- [723] Grujić, Ž., Milić, M., & Antović, I. (2024). Preliminary experiences of using the Azure DevOps platform in software development automation. In *2024 28th International Conference on Information Technology (IT)*. IEEE. <https://doi.org/10.1109/IT58873.2024.10451234>.
- [724] Bayuk, J. L. (2024). *Stepping through cybersecurity risk management: A systems thinking approach* (1st ed.). John Wiley & Sons. ISBN: 978-1-394-21397-9.
- [725] Romero-Álvarez, J., Alvarado-Valiente, J., Moguel, E., & Garcia-Alonso, J. M., Murillo, J. (2024). Quantum service-oriented computing: A proposal for quantum software as a service. *River Publishers Series in Rapids*. ISBN: 978-87-7004-199-7 (Paperback).
- [726] Zhang, X., Zhao, P., & Jaskolka, J. (2025). Navigating the DevOps landscape. *Journal of Systems and Software*, 223, 112331. Elsevier. <https://doi.org/10.1016/j.jss.2024.112331>.
- [727] Ghoddousi, A. (2024). Aspects of partially automated DevOps cycles and requirements for tracking in aerospace applications. *Elektrotechnik & Informationstechnik*, 141, 167–170. <https://doi.org/10.1007/s00502-024-01217-w>.
- [728] Bjørntvedt, N. K., & Sæther, K. (2024). Designing the CI/CD security maturity model. Master's thesis, University of Agder. <https://uia.brage.unit.no/uia-xmlui/handle/11250/3143420>.
- [729] Skias, D., Karkazis, P., Prusiel, I., et al. (2024). Advanced NEMO platform & laboratory testing results (Initial Version, Deliverable D4.2, Version 1.0). Horizon Europe Project “NEMO – Next Generation Meta Operating System,” Grant Agreement No. 101070118. December 2024. Available: https://meta-os.eu/wp-content/uploads/2025/01/NEMO_D4.2-Advanced-NEMO-platform-laboratory-testing-results.-Initial-version_-v1.0_compressed-1.pdf.
- [730] Tian, T., Peng, Y., Liu, Y., Shen, Y., Zhou, J., Yang, X., Zheng, J., Shao, H., & Xu, Y. (2025). Fault diagnosis of gear pump based on cyclic impulse characteristic decomposition. *IEEE Sensors Journal*, 25(4), 3777–3788. <https://doi.org/10.1109/JSEN.2024.3514675>.
- [731] Elgamal, Z. S., Elfangary, L., & Fahmy, H. (2025). The impact of using MLOps and DevOps on container-based applications: A survey. *Informatics Bulletin*, 7(1). Faculty of Computers and Artificial Intelligence, Helwan University. <https://doi.org/10.21608/fcihib.2025.261807.1105>.

- [732] Plant, O. H., Aldea, A., & van Hillegersberg, J. (2025). Improving DevOps team performance through context-capability coalignment: Towards a profile for public sector organizations. *Information and Software Technology*, 178, 107585. <https://doi.org/10.1016/j.infsof.2024.107585>.
- [733] Bachhav, A., Kharat, V., & Shelar, M. (2024). QOTUM: The query optimizer for distributed database in cloud environment. *Technical Journal*, 18(2), 172–177. <https://doi.org/10.31803/tg-20230501083155>.
- [734] Moon, J., Yang, S., & Lee, K. (2024). FedOps: A platform of federated learning operations with heterogeneity management. *IEEE Access*. <https://doi.org/10.1109/ACCESS.2024.3349691>.
- [735] Ribeiro, E., Restivo, A., Ferreira, H. S., & Dias, J. P. (2024). WASMICO: Micro-containers in microcontrollers with WebAssembly. *Journal of Systems and Software*, 214, 112081. Elsevier. <https://doi.org/10.1016/j.jss.2024.112081>.
- [736] Raghunathan, S. (2020). Strengthening Kubernetes: Strategies and tools for enhanced DevSecOps integration. *International Journal of Science and Research (IJSR)*, 9(2). <https://doi.org/10.21275/SR24401235010>.
- [737] Koopman, B. J., Bhimani, S., Galitzki, N., Hasselfield, M., Lashner, J., Nakata, H., Newburgh, L., Nguyen, D. V., Sakuma, T., & Yamada, K. (2024, June). The Simons Observatory: Deployment of the observatory control system and supporting infrastructure. Preprint: arXiv:2406.15703. <https://doi.org/10.48550/arXiv.2406.15703>.
- [738] Klinka, M. (2024). GitOps principles for deployment and management of the AFoLab platform. Master's thesis, Faculty of Informatics, Masaryk University, Brno. <https://is.muni.cz/th/kcqv/>.
- [739] Czekster, R. M. (2024). Continuous risk assessment in secure DevOps. arXiv preprint, arXiv:2409.03405 [cs.SE]. <https://arxiv.org/abs/2409.03405>.
- [740] Khanday, S. K. (2025). Optimizing performance and cost efficiency in AI-driven cloud infrastructures: A multi-objective approach. In *Proceedings of the 6th International Conference on Mobile Computing and Sustainable Informatics (ICMCSI)*. IEEE. <https://doi.org/10.1109/ICMCSI64620.2025.10883380>. ISBN: 979-8-3315-2266-7.
- [741] Lad, S. (2024). Enhancing CI/CD pipelines with AI: Advanced test automation, monitoring and alerts. *International Journal of Progressive Research in Engineering Management and Science*. <https://doi.org/10.5281/zenodo.13846422>.
- [742] Mishra, A., & Sharma, A. (2025). Improved DevOps lifecycle by integrating a novel tool V-Git Lab. *Recent Advances in Electrical & Electronic Engineering*, 18, 196–201. <https://doi.org/10.2174/2352096516666230517155221>.
- [743] Thomas, J., Rodriguez, L. M., Badger, A. R., Wortman, K., Wilson, D., & Heistand, C. (2024). Flight software for the Double Asteroid Redirection Test. In *2024 IEEE Aerospace Conference*. IEEE. <https://doi.org/10.1109/AERO58975.2024.10521169>.
- [744] Jarmusch, A., Cabarcas, F., Pophale, S., Kallai, A., Doerfert, J., Peyralans, L., Lee, S., Denny, J., & Chandrasekaran, S. (2024). CI/CD efforts for validation, verification and benchmarking OpenMP implementations. Preprint: arXiv:2408.11777 [cs.PL]. Available: <https://arxiv.org/abs/2408.11777>.
- [745] Singh, R., Yeboah-Ofori, A., Kumar, S., & Ganiyu, A. (2024). Fortifying cloud DevSecOps security using Terraform infrastructure as code analysis tools. In *Proc. Int. Conf. Electrical and Computer Engineering Researches (ICECER 2024)*. IEEE. <https://doi.org/10.1109/ICECER62944.2024.10920371>.
- [746] Pillala, G. (2024). Fostering agility DevSecOps practice: Dynamic secrets management and CMDB fusion reshaping enterprise development and deployment dynamics. *International Journal of Information Security (IJIS)*, 3(1), 14–20. <https://doi.org/10.17605/OSF.IO/R6KPY>.
- [747] Nagpal, A., Pothineni, B., Parthi, A. G., Maruthavanan, D., Banarse, A. R., Veerapaneni, P. K., Sankiti, S. R., & Jayaram, V. (2024). Framework for automating compliance verification in CI/CD pipelines. *International Journal of Computer Science and Information Technology Research (IJCSITR)*, 5(4), 17–27. <https://doi.org/10.5281/zenodo.14259679>.

- [748] Ferreira, A. M., Brito, M. A., & de Lima, J. (2024). Software quality in an automotive project: Continuous inspection. *International Journal of Automotive Technology*. Springer. <https://doi.org/10.1007/s12239-024-00132-5>.
- [749] Ali, Z., Khan, Z. A., Senapati, T., & Moslem, S. (2024). Frank-based TOPSIS methodology of development and operations challenges based on intuitionistic linguistic aggregation operators and their applications. *IEEE Access*, 12. <https://doi.org/10.1109/ACCESS.2024.3420711>.
- [750] Rüegger, J., Kropp, M., Graf, S., & Anslow, C. (2024). Fully automated DORA metrics measurement for continuous improvement. In *Proc. Int. Conf. Software and Systems Processes (ICSSP '24)*, München, Germany. ACM. <https://doi.org/10.1145/3666015.3666020>.
- [751] Gaeta, E., Haleem, M. S., Lopez-Perez, L., Manea, M., Cabrera Umpierrez, M. F., Arredondo Waldmeyer, M. T., Pecchia, L., & Fico, G. (2025). GATEKEEPER platform: Secure processing environment for European Health Data Space. *IEEE Access*, 13. <https://doi.org/10.1109/ACCESS.2025.3539559>.
- [752] Venkatesan, K., & Kumar, R. (2025). CI/CD pipelines for model training: Reducing turnaround time in offline model training with Hive and Spark. *Journal of Quantitative Science & Technology*, 2(1), Article 171. Available: <https://jst.org/index.php/j/article/view/171>.
- [753] Yang, B., Samba, A., Fraysse, G., & Cherrared, S. (2024). Generating commit messages for configuration files in 5G network deployment using LLMs. In *Proceedings of the 2024 20th International Conference on Network and Service Management (CNSM) (IFIP)*, Oct 2024. Available: <https://dl.ifip.org/db/conf/cnsm/cnsm2024/1571067267.pdf>.
- [754] Chinnam, S. K. (2024). GitOps demystified: A beginner's guide to streamlining Kubernetes deployments. *International Journal of Computer Engineering and Technology (IJCET)*, 15(4), 481–494. <https://doi.org/10.5281/zenodo.13309397>.
- [755] Kodakandla, N. (2022). GitOps: Why it's becoming the gold standard for infrastructure management. *TIJER – International Research Journal*, 9(10). <https://doi.org/10.1729/Journal.42726>.
- [756] Roy, A. (2024). *Hands-on Python for DevOps*. Packt Publishing Ltd. ISBN: 978-1-83508-116-7. Available: www.packtpub.com.
- [757] Díaz, J., Pérez, J., Alves, I., Kon, F., Leite, L., Meirelles, P., & Rocha, C. (2024). Harmonizing DevOps taxonomies — A grounded theory study. *The Journal of Systems and Software*, 208, 111908. <https://doi.org/10.1016/j.jss.2023.111908>.
- [758] Jose, A., & Poulose, J. (2024). Harnessing the power of AI: Transforming DevSecOps for enhanced cloud security. Conference paper. Available: <https://www.researchgate.net/publication/381003326>.
- [759] Gogineni, A. (2021). Helm for continuous delivery of serverless applications on Kubernetes. *International Journal of Innovative Research in Engineering & Multidisciplinary Physical Sciences (IJRMPS)*, 9(6). <https://doi.org/10.5281/zenodo.14880986>.
- [760] Garousi, V., Felderer, M., & Mäntylä, M. V. (2019). Guidelines for including grey literature and conducting multivocal literature reviews in software engineering. *Information and Software Technology*, 106, 101–121. <https://doi.org/10.1016/j.infsof.2018.09.006>.
- [761] Zhao, X., Clear, T., & Lal, R. (2024). Identifying the primary dimensions of DevSecOps: A multi-vocal literature review. *The Journal of Systems and Software*, 214, 112063. <https://doi.org/10.1016/j.jss.2024.112063>.
- [762] Sachdeva, M. S. (2024). Advanced cloud infrastructure security: Techniques, challenges, and impact. *International Journal of Computer Engineering and Technology (IJCET)*, 15(5), 892–900. <https://doi.org/10.5281/zenodo.13929652>.
- [763] Venkata, S. B. (2024). Architectural optimization of cloud-native data processing pipelines: A systematic analysis of serverless computing paradigms. *International Journal of Computer Engineering and Technology (IJCET)*, 15(6), 1492–1508. https://doi.org/10.34218/IJCET_15_06_124.

- [764] Kanakala, R. R. (2025). Enhancing CI/CD systems: A holistic approach to re-design, reliability, scalability, and performance. *International Journal of Computer Engineering and Technology (IJCET)*, 16(1), 3791–3802. https://doi.org/10.34218/IJCET_16_01_261.
- [765] Makani, S. T., & Jangampeta, S. (2021). DevOps security tools: Evaluating effectiveness in detecting and fixing security holes. *International Journal of DevOps (IJDO)*, 1(2), 1–12. Article ID: IJDO_01_02_001. Available: <https://iaeme.com/Home/issue/IJDO?Volume=1&Issue=2>.
- [766] Avuthu, Y. R. (2025). Cost-aware DevOps practices: Integrating financial metrics into CI/CD pipeline management. *International Journal of Innovative Research and Creative Technology (IJIRCT)*, 4(2). Available: <https://www.researchgate.net/publication/389713132>.
- [767] Sivaraman, H. (2020). Intelligent deployment orchestration using ML for multi-environment CI/CD pipelines. *International Journal of Innovative Research and Creative Technology (IJIRCT)*, 6(4). <https://doi.org/10.5281/zenodo.14250621>.
- [768] Parmar, T. (2025). Implementing CI/CD in data engineering: Streamlining data pipelines for reliable and scalable solutions. *International Journal of Innovative Research in Engineering & Multidisciplinary Physical Sciences (IJIRMP)*, 13(1). <https://doi.org/10.5281/zenodo.14762684>.
- [769] Vemuri, N., Thaneeru, N., & Tatikonda, V. M. (2024). AI-optimized DevOps for streamlined cloud CI/CD. *International Journal of Innovative Science and Research Technology (IJISRT)*, 9(2), 504–506. <https://doi.org/10.5281/zenodo.10673085>.
- [770] Pendyala, B. P. (2025). The convergence of platform engineering and service automation: Transforming enterprise cloud architecture. *International Journal of Information Technology and Management Information Systems (IJITMIS)*, 16(2), 225–239. https://doi.org/10.34218/IJITMIS_16_02_016.
- [771] Bommareddy, A. R. (2024). Automating DevOps in insurance: A case study of CI/CD implementation with micro-frontend architecture. *International Journal of Research in Computer Applications and Information Technology (IJRCAIT)*, 7(2), 2330–2339. https://doi.org/10.34218/IJRCAIT_07_02_171.
- [772] Aggarwal, A. (2025). Do software engineers need ML skills to be successful in the age of Gen AI. *International Journal of Research in Computer Applications and Information Technology (IJRCAIT)*, 8(1), 26–36. https://doi.org/10.34218/IJRCAIT_08_01_003.
- [773] Luz, H., Peace, P., Luz, A., & Joseph, S. (2024). Impact of emerging AI techniques on CI/CD deployment pipelines. https://www.researchgate.net/publication/387556393_Impact_of_Emerging_AI_Techniques_on_CICD_Deployment_Pipelines.
- [774] James, C. (2024). Impact of TDD and BDD on software quality and delivery speed in CI/CD pipelines. https://www.researchgate.net/publication/385278125_Impact_of_TDD_and_BDD_on_Software_Quality_and_Delivery_Speed_in_CICD_Pipelines.
- [775] Lumpatki, S. S., Patwardhan, S., & Kulkarni, M. (2024). Implementing “DevSecOps as a culture” — The concept, benefits, execution strategies, and challenges. In *Smart Trends in Computing and Communications (LNNS, Vol. 947)*. Springer. https://doi.org/10.1007/978-981-97-1326-4_16.
- [776] Haripriya, H. S. (2021). Advancing continuous security integration in DevOps pipelines: A strategic approach to fortifying network defense and enhancing resilience in site reliability engineering. *International Journal of Computer Science and Engineering Research and Development (IJCSERD)*, 11(1), 39–45. https://www.researchgate.net/publication/385904491_ADVANCING_CONTINUOUS_SECURITY_INTEGRATION_IN_DEVOPS_PIPELINES_A_STRATEGIC_APPROACH_TO_FORTIFYING_NETWORK_DEFENSE_AND_ENHANCING_RESILIENCE_IN_SITE_RELIABILITY_ENGINEERING.
- [777] Kohli, D., Kumar, J., Saini, J., Aggarwal, S., Singh, Y., & Garg, A. (2024). Implementing microservice architecture in e-commerce with DevOps practice. In *2024 International Conference on Intelligent Systems for Cybersecurity (ISCS)*. IEEE. <https://doi.org/10.1109/ISCS61804.2024.10581082>.

- [778] Ramaj, X., Sánchez-Gordón, M., Colomo-Palacios, R., & Gkioulos, V. (2024). Training and security awareness under the lens of practitioners: A DevSecOps perspective towards risk management. In *HCII 2024: Human-Computer Interaction International Conference (LNCS 14729)*, pp. 84–97. Springer. https://doi.org/10.1007/978-3-031-61382-1_6.
- [779] Akinola, O., Oyerinde, O., & Akinola, A. (2024). Implementing DevOps adoption within United States SMEs. *Journal of Artificial Intelligence General Science (JAIGS)*, 2(1), 217–219. Available at: <https://newjaigs.org/index.php/JAIGS/article/view/115>.
- [780] Pratana, V. J., Situmorang, R., & Kustandi, C. (2024). Digital talent of banking: Need Analyst of Development, Security and Operational (DevSecOps) by Bootcamp Approach. In *Proceedings of the 4th International Conference on Social Sciences and Education (ICSSE 2024)*. Universitas Negeri Jakarta. Available at: <https://proceeding.uns.ac.id/icsse/article/view/460>.
- [781] Yu, W., Jin, C., Qian, J., Fang, H., Xu, R., & Shi, X. (2024). Improving substation network security with DevSecOps and AIOps. In *2024 IEEE 10th Conference on Big Data Security on Cloud (BigDataSecurity)*. IEEE. <https://doi.org/10.1109/BigDataSecurity62737.2024.00027>.
- [782] Sobral, P., Teixeira, R., Marques, R., Figueiredo, N., & Antunes, M., Gomes, D. (2024). Improving automotive aftermarket forecasting with MLOps. In *2024 IEEE 11th International Conference on Future Internet of Things and Cloud (FiCloud)*. IEEE. <https://doi.org/10.1109/FiCloud62933.2024.00065>.
- [783] Moyón, F., Angermeir, F., & Mendez, D. (2024). Industrial challenges in secure continuous development. In *Proceedings of the 46th International Conference on Software Engineering: Software Engineering in Practice (ICSE-SEIP '24)*. ACM. <https://doi.org/10.1145/3639477.3639736>.
- [784] Akinbolaji, T. J., Nzeako, G., Akokodaripon, D., & Aderoju, A. V. (2024). Proactive monitoring and security in cloud infrastructure: Leveraging tools like Prometheus, Grafana, and HashiCorp Vault for robust DevOps practices. *World Journal of Advanced Engineering Technology and Sciences*, 13(2). <https://doi.org/10.30574/wjaets.2024.13.2.0543>.
- [785] QLeap Consortium. (2024). Containers as the quantum leap in software development. Project report, QLeap Project (2022–2024), Business Finland, University of Jyväskylä and partners. <https://arxiv.org/pdf/2501.07204>.
- [786] Haverinen, H., Päivärinta, T., Vänskä, J., & Joutsijoki, H. (2024). Information-centric adoption and use of standard compliant DevSecOps for operational technology: From experience to design principles. In S. Hyrynsalmi et al. (Eds.), *ICSOB 2023 (LNBIP, Vol. 500)*, pp. 400–415. Springer. https://doi.org/10.1007/978-3-031-53227-6_28.
- [787] Bannon, T. (2024). Infusing artificial intelligence into software engineering and the DevSecOps continuum. *IEEE Computer*, 57(9), 140–147. <https://doi.org/10.1109/MC.2024.3423108>.
- [788] Paulik, T. (2024). Security challenges of complex space applications: An empirical study. European Space Agency – Security for Space Systems (3S). arXiv preprint [arXiv:2408.08061](https://arxiv.org/abs/2408.08061).
- [789] Amgothu, S. (2024). Innovative CI/CD pipeline optimization through canary and blue-green deployment. *International Journal of Computer Applications*, 186(50). <https://doi.org/10.5120/ijca2024924141>.
- [790] Saleh, S. M., Sayem, I. M., Madhavji, N., & Steinbacher, J. (2024). Advancing software security and reliability in cloud platforms through AI-based anomaly detection. In *Proceedings of the 2024 Cloud Computing Security Workshop (CCSW '24)*. ACM. <https://doi.org/10.1145/3689938.3694779>.
- [791] Sedrakyan, G., Iacob, M.-E., & van Hillegersberg, J. (2024). Towards LowDevSecOps framework for low-code development: Integrating process-oriented recommendations for security risk management. In *ACM/IEEE 27th International Conference on Model Driven Engineering Languages and Systems (MODELS Companion '24)*. ACM. <https://doi.org/10.1145/3652620.3688335>.
- [792] Yu, L., Wang, Y., & Li, N. (2024). Integrated design method of development, operation and maintenance security in public computing environment. In *International Conference on Image Processing, Machine Learning and Pattern Recognition (IPMLP 2024)*. ACM. <https://doi.org/10.1145/3700906.3700982>.

- [793] Gugulotu, K. (2024). Integrating AI/ML into DevSecOps: Strengthening security and compliance in cloud-native applications. *International Journal of Computer Engineering and Technology (IJCET)*, 15(5), 1128–1148. <https://doi.org/10.5281/zenodo.14043775>.
- [794] Reis, L., Castro, F., Diniz, H., Mendes, B., & Silva, L., Viana, D. (2024). Integrating DevOps into software engineering education: A laboratory experience. In *XXIII Brazilian Symposium on Software Quality (SBQS 2024)*. ACM. <https://doi.org/10.1145/3701625.3701688>
- [795] Yadati, N. S. P. K. (2021). Integrating dynamic security testing tools into CI/CD pipelines: A continuous security testing case study. *International Journal of Science and Research (IJSR)*, 10(4). <https://doi.org/10.21275/SR24615152732>
- [796] Rida, A., & Ait Lahcen, A. (2024). Towards DevSecOps model for multi-tier web applications. *ITM Web of Conferences*, 69, 04018. <https://doi.org/10.1051/itmconf/20246904018>
- [797] Kånåhols, G., Hasan, S., & Strandberg, P. E. (2025). Integrating time series anomaly detection into DevOps workflows. *IEEE Access*, 13. <https://doi.org/10.1109/ACCESS.2025.3550665>
- [798] Ale, N. K. (2022). Integrating continuous integration / continuous deployment (CI/CD) with test automation: Enhancing software development efficiency. *International Journal of Science and Research (IJSR)*, 11(1). <https://doi.org/10.21275/SR24627045933>
- [799] Mahida, A. (2024). Integrating observability with DevOps practices in financial services technologies: A study on enhancing software development and operational resilience. *International Journal of Advanced Computer Science and Applications (IJACSA)*, 15(7). <https://doi.org/10.14569/IJACSA.2024.0150701>
- [800] Kim, J., Jo, U., Kim, T., Jung, S., & Kim, J. (2024). Sensing-to-sky intermittent connectivity realization for LTE-enabled drone platforms: Embedded design, measurement study, and positioning applications. *IEEE Access*, 12, 137360–137370. <https://doi.org/10.1109/ACCESS.2024.3462942>
- [801] Podelko, A. (2025). Performance engineering: New and conflicting trends. In *Companion of the 16th ACM/SPEC International Conference on Performance Engineering (ICPE Companion '25)*, May 5–9, 2025, Toronto, ON, Canada. ACM. <https://doi.org/10.1145/3680256.3721309>
- [802] Aktaş, K., & Kilinc, H. H. (2024). Interaction prediction and anomaly detection in a microservices-based telecommunication platform. In *Proceedings of the International Conference on Software and Systems Processes (ICSSP '24)*, September 04–06, 2024, München, Germany. ACM. <https://doi.org/10.1145/3666015.3666017>
- [803] Myllynen, T., Kamau, E., Mustapha, S. D., Babatunde, G. O., & Collins, A. (2024). Review of advances in AI-powered monitoring and diagnostics for CI/CD pipelines. *International Journal of Multidisciplinary Research and Growth Evaluation*, 5(1), 1119–1130. <https://doi.org/10.54660/IJMRGE.2024.5.1.1119-1130>
- [804] Somasundaram, P. (2024). Unified secret management across cloud platforms: A strategy for secure credential storage and access. *International Journal of Computer Engineering and Technology (IJCET)*, 15(2), 5–12. <https://doi.org/10.17605/OSF.IO/G96Z2>
- [805] Port, D., Taber, B., & Emkani, P. (2024). Investigating effectiveness and compliance to DevOps policies and practices for managing productivity and quality variability. *Journal of Systems and Software*, 213, 112030. <https://doi.org/10.1016/j.jss.2024.112030>
- [806] Oliveira, F. B., Di Felice, M., & Kamienski, C. (2024). IoTDeploy: Deployment of IoT smart applications over the computing continuum. *Internet of Things*, 28, 101348. <https://doi.org/10.1016/j.iot.2024.101348>
- [807] Grande, R., Vizcaíno, A., & García, F. O. (2024). Is it worth adopting DevOps practices in global software engineering? Possible challenges and benefits. *Computer Standards & Interfaces*, 87, 103767. <https://doi.org/10.1016/j.csi.2023.103767>

- [808] Chalvantzis, N., Vontzalidis, A., Kassela, E., Spyrou, A., Nikitas, N., Provatas, N., Konstantinou, I., & Koziris, N. (2024). IW-NET BDA: A big data infrastructure for predictive and geotemporal analytics of inland waterways. *IEEE Access*, 12, 52503–52517. <https://doi.org/10.1109/ACCESS.2024.3387315>.
- [809] Martina, M. R., Bianchini, E., Sinceri, S., Francesconi, M., & Gemignani, V. (2024). Software medical device maintenance: DevOps-based approach for problem and modification management. *Journal of Software: Evolution and Process*, 36, e2570. <https://doi.org/10.1002/smr.2570>.
- [810] Guduru, V. S. (2023). The future of DevOps in Salesforce: Implementing CI/CD with Copado. *Journal of Artificial Intelligence Machine Learning and Data Science*. <https://doi.org/10.51219/JAIMLD/venkat-sumanth-guduru/286>.
- [811] Joshi, P. K. (2021). CI/CD automation for payment gateways: Azure vs. AWS. *ESP Journal of Engineering & Technology Advancements*, 1(2), 163–175. <https://doi.org/10.56472/25832646/JETA-V1I2P118>.
- [812] Escott, E. (2024). Jidoka: Automation with a human touch. *Software and Systems Modeling*. Springer. <https://doi.org/10.1007/s10270-024-01256-z>.
- [813] Berti, L., Chabannes, V., Cladellas, J., Diallo, A., Elayam, M. M., Pinçon, P., & Prud'homme, C. (2024). Ktirio urban building: A computational framework for city energy simulations enhanced by CI/CD innovations on EuroHPC systems. *arXiv preprint arXiv:2407.14535*. <https://doi.org/10.48550/arXiv.2407.14535>.
- [814] Liang, P., Wu, Y., Xu, Z., Xiao, S., & Yuan, J. (2024). Enhancing security in DevOps by integrating artificial intelligence and machine learning. *Journal of Theory and Practice of Engineering Science*, 4(2). [https://doi.org/10.53469/jtpes.2024.04\(02\).05](https://doi.org/10.53469/jtpes.2024.04(02).05).
- [815] Aissat, S., Beaulieu, J., Bordeleau, F., Gascon-Samson, J., Motamedi, A., & Poirier, É. (2024). JuNo-OPS: A DevOps framework for the engineering of digital twins for built assets. In *ACM/IEEE 27th International Conference on Model Driven Engineering Languages and Systems (MODELS Companion '24)*, Linz, Austria. ACM. <https://doi.org/10.1145/3652620.3688266>.
- [816] Kose, B. O. (2024). Mobilizing DevOps: Exploration of DevOps adoption in mobile software development. *Kybernetes*. Emerald Publishing. <https://doi.org/10.1108/K-04-2024-0989>.
- [817] Schuster, B. (2024). Strategies to adopt DevOps model-driven engineering for automating enterprise networks (Doctoral dissertation). Walden University. <https://scholarworks.waldenu.edu/dissertations/16567>.
- [818] Zheng, Y., Yang, Y., Chen, M., & Quinn, A. (2024). Kgent: Kernel extensions large language model agent. In *Workshop on eBPF and Kernel Extensions (eBPF '24)*, Sydney, Australia. ACM. <https://doi.org/10.1145/3672197.3673434>.
- [819] James, C. (2024). Visualizing testing metrics to showcase ROI in CI/CD pipelines. https://www.researchgate.net/publication/385352767_Visualizing_Testing_Metrics_to_Showcase_ROI_in_CICD_Pipelines.
- [820] Oha, K. (2024). Advancements in microservice architectures: Tackling data communication, scalability, and CI/CD automation challenges (BSc thesis). Rhine-Waal University of Applied Sciences. <https://opus4.kobv.de/opus4-rhein-waal/files/2046/KO-ADVANCEMENTS-IN-MICROSERVICE-ARCHITECTURES.pdf>.
- [821] Yerramsetty, H. (2024). Kubernetes: The cornerstone of enterprise platform engineering. *International Journal of Research in Computer Applications and Information Technology (IJRCAIT)*, 7(2), 539–548. <https://doi.org/10.5281/zenodo.14009689>.
- [822] Kanerva, S. (2024). Lateral movement restriction in advanced DevOps environments (Master's thesis). Aalto University, Master's Programme in Computer, Communication and Information Sciences (CCIS). <https://aaltodoc.aalto.fi/items/4d4613f6-951a-482c-92cc-c6ddd2c473e6>.
- [823] Petty, J., Jones, D., & Hicks, J. (2024). *Learn PowerShell scripting in a month of lunches* (2nd ed.). Manning Publications. ISBN: 9781633438989.

- [824] Patan, L. (2024). Leveraging cloud-native architecture for scalable and resilient enterprise applications: A comprehensive analysis. *International Journal of Computer Engineering and Technology (IJCET)*, 15(5), 583–591. <https://doi.org/10.5281/zenodo.13861921>.
- [825] Akula, A. K. (2024). Leveraging AWS and Java microservices: An analysis of Amazon's scalable e-commerce architecture. *International Journal for Research in Applied Science and Engineering Technology (IJRASET)*, 12(9). <https://doi.org/10.22214/ijraset.2024.64275>.
- [826] Esther, D. (2024). Leveraging metrics to demonstrate the ROI of TDD and BDD in CI/CD pipelines. *International Journal for Research in Applied Science and Engineering Technology*. <https://doi.org/10.5281/zenodo.12345678>.
- [827] Shrestha, M., Johansen, C., & Johansen, J. (2024). LightSC: The making of a usable security classification tool for DevSecOps. *arXiv preprint arXiv:2410.01762*. <https://arxiv.org/abs/2410.01762>.
- [828] Fruth, M., & Scherzinger, S. (2024). Live patching for distributed in-memory key-value stores. *Proceedings of the ACM on Management of Data (SIGMOD)*, 2(6), Article 241. <https://doi.org/10.1145/3698816>.
- [829] Ashiwal, V., Finster, S., & Dawoud, A. (2024). LLM-based vulnerability sourcing from unstructured data. In *2024 IEEE European Symposium on Security and Privacy Workshops (EuroS&PW)*, pp. 579–584. IEEE. <https://doi.org/10.1109/EUROSPW61312.2024.00077>.
- [830] Méndez, Ó. A., Camargo, J., & Florez, H. (2025). Machine learning operations applied to development and model provisioning. In *Proceedings of ICAI 2024 (CCIS, Vol. 2236)*, pp. 73–88). Springer. https://doi.org/10.1007/978-3-031-75144-8_6.
- [831] Akbar, M. A., Khan, A. A., Mahmood, S., & Hyrynsalmi, S. (2025). Management of DevSecOps process: An empirical investigation. *Software: Practice and Experience*. Wiley. <https://doi.org/10.1002/spe.3419>.
- [832] Amaro, R., Pereira, R., & da Silva, M. M. (2025). Mapping DevOps capabilities to the software life cycle: A systematic literature review. *Information and Software Technology*, 177. Elsevier. <https://doi.org/10.1016/j.infsof.2024.107583>.
- [833] Kumar, P., & Madiseti, V. K. (2024). Sher: A secure broker for DevSecOps and CI/CD workflows. *Journal of Software Engineering and Applications*, 17, 321–339. <https://doi.org/10.4236/jsea.2024.175018>.
- [834] Immadisetty, A. (2024). Mastering data platform design: Industry-agnostic patterns for scale. *International Journal of Research in Computer Applications and Information Technology (IJRCAIT)*, 7(2), 2259–2270. <https://doi.org/10.5281/zenodo.14442699>.
- [835] Majumder, R. (2024). Maximizing efficiency: Automated software testing with CI/CD tools and Docker containerization for parallel execution (M.S. thesis). Russ College of Engineering and Technology, Ohio University. http://rave.ohiolink.edu/etdc/view?acc_num=ohiou1706815848305791.
- [836] Ok, E., & Eniola, J. (2024). Maximizing efficiency: How Jenkins transforms continuous integration and continuous delivery in business. *Open Access Journal Article*. Available at: https://www.researchgate.net/publication/387645291_Maximizing_Efficiency_How_Jenkins_Transforms_Continuous_Integration_and_Continuous_Delivery_in_Business.
- [837] Lapointe-Boisvert, A., Trudel, S., & Desharnais, J.-M. (2024). Measurement in DevOps standard: Proposed improvements. In *Proceedings of IWSM-Mensura 2024, CEUR Workshop Proceedings*, Montreal, Canada, Sep. 30 – Oct. 4, 2024. <https://ceur-ws.org/Vol-3407/paper1.pdf.org>. Available at: <http://ceur-ws.org/Vol-3852/paper8.pdf>.
- [838] Dakić, V., Redžepagić, J., Bašić, M., & Žgrablić, L. (2024). Methodology for automating and orchestrating performance evaluation of Kubernetes container network interfaces. *Computers*, 13(11). <https://doi.org/10.3390/computers13110283>.
- [839] Trabelsi, I., Popa, B., Péreyrol, J., & Beaulieu, P.-O., Moha, N. (2024). MicroMatic: Fully automated microservices identification approach from monolithic systems. In *Proceedings of the 2024 ACM/IEEE 6th*

- International Workshop on Software Engineering Research & Practices for the Internet of Things (SERP4IoT '24), Lisbon, Portugal, Apr. 20, 2024. ACM. <https://doi.org/10.1145/3643794.3648283>.
- [840] Gillespie, P. (2024). Security compliance in large private enterprise information systems utilizing DevOps: An exploratory study (Doctoral dissertation). University of the Cumberlands. <https://www.proquest.com/docview/3083269739>.
- [841] TaeiZadeh, A., Lotfi, Z., & Ramadhan, A. J. (2024). Microservices boundary determination migration in DevOps: A case study. BIO Web of Conferences, 97. EDP Sciences. <https://doi.org/10.1051/bioconf/20249700122>
- [842] Oberhauser, R. (2024). VR-DevOps: Visualizing and interacting with DevOps pipelines in virtual reality. In Proceedings of the 19th International Conference on Software Engineering Advances (ICSEA 2024), Venice, Italy, pp. 43-48. IARIA Press. <https://doi.org/10.1145/3725789.3725794>.
- [843] McIntosh-Smith, S., Alam, S. R., & Woods, C. (2024). Isambard-AI: A leadership class supercomputer optimised specifically for artificial intelligence. In Proceedings of the IEEE/ACM Conference on High Performance Computing, Data, and Intelligence (HPDC '24). IEEE/ACM. **DOI:** <https://doi.org/10.1145/3725789.3725794>.
- [844] Leshchenko, B., Snisar, B., Stupak, A., & Osadchyi, V. (2024). Integrating DevSecOps into the software development lifecycle: A comprehensive model for securing containerized and cloud-native environments. In Proceedings of CPITS-II 2024: Workshop on Cybersecurity Providing in Information and Telecommunication Systems II, October 26, 2024, Kyiv, Ukraine. CEUR Workshop Proceedings, Vol. 3826. <https://ceur-ws.org/Vol-3407/paper1.pdf.org>. Available at: <http://ceur-ws.org/Vol-3826/short1.pdf>.
- [845] Gourko, A. (2024). WebSocket communication between multiple users in scalable web-application environment (Master's thesis). Metropolia University of Applied Sciences. <https://www.theseus.fi/handle/10024/860920>.
- [846] Pousi, J. (2024). Migration from on-premise to public cloud: Case of Finnish governmental organization (Master's thesis). Aalto University, Nov. 10, 2024. <https://aaltoodoc.aalto.fi/items/9f12e94a-31ac-4b6d-a650-ce6adeefbb5d>.
- [847] Yu, J., Han, N., & Nguyen, C. T. (2024). NIST SP 800-204D: Software supply chain security in DevSecOps CI/CD pipelines. National Institute of Standards and Technology. <https://doi.org/10.6028/NIST.SP.800-204D>.
- [848] Mhlanga, S. T., & Chiyangwa, T. (2024). Modelling challenges in implementing DevOps in software development companies in South Africa: An empirical survey. In Proceedings of the 2024 4th International Multidisciplinary Information Technology and Engineering Conference (IMITEC). IEEE. <https://doi.org/10.1109/IMITEC60221.2024.10851190>.
- [849] Woźniak, A. P., Milczarek, M., & Woźniak, J. (2025). MLOps components, tools, process, and metrics: A systematic literature review. IEEE Access, 13, 22166–22191. <https://doi.org/10.1109/ACCESS.2025.3534990>.
- [850] Younes, F., Lahsen-Cherif, I., & El Ghazi, H. (2024). Toward a city digital twin: Design principles, and challenges. In Proceedings of the 2024 7th International Conference on Advanced Communication Technologies and Networking (CommNet). IEEE. <https://doi.org/10.1109/CommNet63022.2024.10793378>.
- [851] Thimmapuram, A. K. (2024). Modernizing middleware infrastructure: An American Airlines case study. International Journal of Research in Computer Applications and Information Technology (IJRCAIT), 7(2), 2239–2249. <https://doi.org/10.5281/zenodo.14399998>.
- [852] Giamattei, L., Guerriero, A., Pietrantuono, R., et al. (2024). Monitoring tools for DevOps and microservices: A systematic grey literature review. The Journal of Systems and Software, 208, 111906. <https://doi.org/10.1016/j.jss.2023.111906>.
- [853] Tripathi, A. A. (2024). Attacking and defending Kubernetes (Master's thesis). Dublin Business School. <hdl.handle.net/10788/4504>.

- [854] Asimiyu, Z. (2024). Strategic deployment of OmniScripts and cloud-native CI/CD pipelines: Bridging advanced VLSI design and socio-digital challenges. <https://www.researchgate.net/publication/337076816>. <https://doi.org/10.13140/RG.2.2.38699.27164>.
- [855] Pochu, S., Nersu, S. R. K., & Kathram, S. R. (2024). Multi-cloud DevOps strategies: A framework for agility and cost optimization. *Journal of Artificial Intelligence General Science (JAIGS)*, 7(1), 104-119. <https://doi.org/10.60087/jaigs.v7i01.300>.
- [856] Kumar, A., Nadeem, M., & Shameem, M. (2024). Multicriteria decision-making-based framework for implementing DevOps practices: A fuzzy best-worst approach. *Journal of Software: Evolution and Process*, 36(7). <https://doi.org/10.1002/smr.2631>.
- [857] Vysoká, D. (2024). Multi-domain change impact analysis for agile cyber-physical production systems engineering (Diploma thesis). TU Wien. [handle/20.500.12708/195707](https://hdl.handle.net/20.500.12708/195707).
- [858] Orosz, M., Duffy, B., Charlton, C., Saunders, H., & Shih, M. (2024). Scaling agile principles to an enterprise. In D. Verma (Ed.), *Systems engineering for the digital age: Practitioner perspectives* (ch. 10). Hoboken, NJ: Wiley. <https://doi.org/10.1002/9781394203314.ch10>.
- [859] Velásquez, A. P., Pérez-Salazar, C., Hernández-González, L. A., & Sánchez-García, Á. J. (2024). Systematic literature review of low-code and its future trends. In *Proceedings of the 2024 12th International Conference on Software Engineering Research and Innovation (CONISOFT)* (pp. 30–39). IEEE. <https://doi.org/10.1109/CONISOFT63288.2024.00015>
- [860] Huang, Z., Zhang, Q., Fan, H., et al. (2024). Multi-grained trace collection, analysis, and management of diverse container images. *IEEE Transactions on Computers*, 73(7), 1698–1712. <https://doi.org/10.1109/TC.2024.3383966>
- [861] Peter, H. (2024). Navigating the landscape of Kubernetes security threats and challenges. Preprint. University of Belgrade. https://www.researchgate.net/publication/389571313_Navigating_the_Landscape_of_Kubernetes_Security_Threats_and_Challenges.
- [862] Pochu, S., Nersu, S. R. K., & Kathram, S. R. (2024). Enhancing cloud security with automated service mesh implementations in DevOps pipelines. *Journal of Artificial Intelligence General Science (JAIGS)*, 7(1), 91–104. ISSN: 3006-4023 (Online). DOI: <https://doi.org/10.60087/jaigs.v7i01.300>.
- [863] Camacho, N. G. (2024). Unlocking the potential of AI/ML in DevSecOps: Effective strategies and optimal practices. *Journal of Artificial Intelligence General Science (JAIGS)*, 3(1), 106-121. <https://doi.org/10.60087/jaigs.v3i1.72>.
- [864] Horve, K.-H., Mathisen, M. E. B., Stenersen, F. L., & Strand, B. K. (2024). Creating a scalable log analytics pipeline with GitOps. Bachelor's thesis. Norwegian University of Science and Technology (NTNU). <https://ntnuopen.ntnu.no/ntnu-xmlui/bitstream/handle/11250/3139810/no.ntnu%3Ainspera%3A187443338%3A231873985.pdf>.
- [865] Ray, A. (2024). A complete CI/CD pipeline for automatizing application containerization and deployment in the Kubernetes cluster. Master's thesis. OsloMet – Oslo Metropolitan University. <https://oda.oslomet.no/oda-xmlui/handle/11250/3162981>.
- [866] Tamanampudi, V. M. (2024). End-to-end ML-driven feedback loops in DevOps pipelines. *World Journal of Advanced Engineering Technology and Sciences*, 13(1), 340–354. <https://doi.org/10.30574/wjaets.2024.13.1.0424>.
- [867] Nurhayati. (2024). Implementation of continuous integration and continuous deployment (CI/CD) to speed up the automation process of software delivery in the production process using Node.js, Docker, and React.js. *Jurnal Info Sains: Informatika dan Sains*, 14(2). <https://doi.org/10.54209/infosains.v14i02>.
- [868] Ramaj, X., Sánchez-Gordón, M., Gkioulos, V., & Colomo-Palacios, R. (2024). On DevSecOps and risk management in critical infrastructures: Practitioners' insights on needs and goals. In *Proceedings of the 4th International Workshop on Engineering and Cybersecurity of Critical Systems (EnCyCriS) and Second International Workshop on Software Vulnerability (SVM '24)*, Lisbon, Portugal. ACM. <https://doi.org/10.1145/3643662.3643954>.

- [869] Heck, P. (2024). What about the data? A mapping study on data engineering for AI systems. In Proceedings of CAIN 2024: Conference on AI Engineering – Software Engineering for AI, Lisbon, Portugal, Apr. 14–15, 2024. ACM. <https://doi.org/10.1145/3644815.3644954>.
- [870] Fluri, J., Fornari, F., & Pustulka, E. (2024). On the importance of CI/CD practices for database applications. *Journal of Software: Evolution and Process*, 36(7). <https://doi.org/10.1002/smr.2720>.
- [871] Sagdic, E., Bayram, A., & Islam, M. R. (2024). On the taxonomy of developers' discussion topics with ChatGPT. In Proceedings of the 2024 IEEE/ACM 21st International Conference on Mining Software Repositories (MSR '24), Lisbon, Portugal, Apr. 15–16, 2024. ACM. <https://doi.org/10.1145/3643991.3645080>.
- [872] Song, Y., Mahmood, T., & Rehman, U. U. (2024). Selection of software development methodology by employing a multi-criteria decision-making approach based on logarithmic bipolar complex fuzzy aggregation operators. *IEEE Access*, 12, 38163–38164. <https://doi.org/10.1109/ACCESS.2024.3373708>.
- [873] Panchal, D., Verma, P., Baran, I., Hsiung, T., Musgrove, D., & Lu, D. (2024). Reusable MLOps: Reusable deployment, reusable infrastructure and hot-swappable AI models and services. In Proceedings of the 2024 10th International Conference on Smart Computing and Communication (ICSCC). IEEE. <https://doi.org/10.1109/ICSCC62041.2024.10690392>.
- [874] Warnett, S. J., & Zdun, U. (2024). On the understandability of MLOps system architectures. *IEEE Transactions on Software Engineering*, 50(5), 1015–1031. <https://doi.org/10.1109/TSE.2024.3367488>.
- [875] Hashemi, S., & Mäntylä, M. (2024). OneLog: Towards end-to-end software log anomaly detection. *Automated Software Engineering*, 31(37). <https://doi.org/10.1007/s10515-024-00428-x>.
- [876] Gaur, I., Rai, S., Tiwari, U., & Khurana, S. (2024). Optimizing cloud applications with DevOps. In Proceedings of the 2024 International Conference on Computational Intelligence and Computing Applications (ICCICA). IEEE. <https://doi.org/10.1109/ICCICA60014.2024.10511890>.
- [877] Bokkena, B. (2024). Optimizing cloud infrastructure management using large language models: A DevOps perspective. In Proceedings of the 2nd International Conference on Self Sustainable Artificial Intelligence Systems (ICSSAS-2024). IEEE. ISBN: 979-8-3503-6841-3. <https://doi.org/10.1109/ICSSAS64001.2024.10760725>.
- [878] Dileepkumar, D. S. R., & Mathew, J. (2025). Optimizing continuous integration and continuous deployment pipelines with machine learning: Enhancing performance and predicting failures. *Advances in Science and Technology Research Journal*, 19(3), 108–120. <https://doi.org/10.12913/22998624/197406>.
- [879] Ragothaman, H., & Udayakumar, S. K. (2024). Optimizing service deployments with NLP-based infrastructure code generation – An automation framework. In Proceedings of the 2024 IEEE 2nd International Conference on Electrical Engineering, Computer and Information Technology (ICEECIT), University of Jember, Indonesia, Nov. 22–23, 2024. IEEE. ISBN: 979-8-3315-0437-3. <https://doi.org/10.1109/ICEECIT63698.2024.10859822>.
- [880] Soldani, J., Amadini, R., Brogi, A., Forti, S., Giallorenzo, S., Plebani, P., Vitali, M., & Zavattaro, G. (2024). Towards sustainable deployment of microservices over the cloud-edge continuum, with FREEDA. In Proceedings of the Workshop on Flexible Resource and Application Management on the Edge (FRAME '24), Pisa, Italy, June 3–7, 2024. ACM. <https://doi.org/10.1145/3659994.3660311>.
- [881] Gadani, N. (2024). Optimizing software development processes in cloud computing environments using agile methodologies and DevOps practices. *Asian Journal of Research in Computer Science*, 17. <https://doi.org/10.9734/ajrcos/2024/v17i7479>.
- [882] Patchamatla, P. S. (2024). Optimizing hyperparameter tuning in machine learning using open-source CI/CD tools. *International Journal for Multidisciplinary Research in Science, Engineering and Technology (IJMRSET)*, 7(6). <https://doi.org/10.15680/IJMRSET.2024.0706003>.
- [883] Gopireddy, S. R. (2022). Streamlining infrastructure as code in Azure DevOps: Automation strategies for scalability. *International Journal of Science and Research (IJSR)*, 11(8). <https://doi.org/10.21275/SR22810111317>.

- [884] Chittala, S. (2024). Orchestrating the cloud: AI-enhanced release automation in Kubernetes environments. *International Journal of Research in Computer Applications and Information Technology (IJRCAIT)*, 7(2), 864–878. <https://doi.org/10.5281/zenodo.14045753>.
- [885] McNally, S., & Curran, K. (2024). Web application vulnerabilities. In *Proceedings of the 2024 8th International Symposium on Computer Science and Intelligent Control (ISCSIC)* (pp. 359–367). IEEE. <https://doi.org/10.1109/ISCSIC64297.2024.00081>.
- [886] Subramaniam, K., Kumar, S., Mishra, A., Bhandari, A., Manja, J., & Chandrasekaran, G. (2024). PEaF-production environment analyzer framework: Assisting continuous deployment of 5G workloads using AI/ML. *IEEE Access*, 12. <https://doi.org/10.1109/ACCESS.2024.3472498>.
- [887] Dakić, V., Redžepagić, J., Bašić, M., & Žgrablić, L. (2024). Performance and latency efficiency evaluation of Kubernetes container network interfaces for built-in and custom tuned profiles. *Electronics*, 13(3972). <https://doi.org/10.3390/electronics13193972>.
- [888] Grebić, B., Ćirić Lalić, D., Savković, M., Gračanin, D., & Dakić, D. (2024). From fragile to agile: Building organisational resilience through agile transformation. In *Proceedings of the First International Conference FUTURE-BME 2024 (Forging the Future: Pioneering Approaches in Business, Management, and Economics Engineering to Overcome Emerging Global Challenges)*. Novi Sad, Serbia. <https://doi.org/10.24867/FUTURE-BME-2024-012>.
- [889] Kotliar, A., & Kotliar, V. (2024). Containerized computing with DevOps methodology use on the central Linux cluster at NRC ‘Kurchatov Institute’—IHEP. *Physics of Particles and Nuclei*, 55(3), 407–409. <https://doi.org/10.1134/S1063779624030547>.
- [890] Chaudhari, N. (2024). Pipelines have feelings too: A structured way to design CI/CD pipelines. In *Proceedings of the ACM/IEEE 27th International Conference on Model Driven Engineering Languages and Systems (MODELS Companion '24)*, Linz, Austria, Sep. 22–27, 2024. ACM. <https://doi.org/10.1145/3652620.3676876>.
- [891] Brabbs, J., & Jones, B. C. (2024). How the cloud is a mission enabler for embedded system development. In *Proceedings of the Ground Vehicle Systems Engineering and Technology Symposium (GVSETS)*, August 13–15, 2024, Novi, Michigan, USA. GI/NDIA. Available: <https://ndia-mich.org/images/events/gvsets/2024/papers/MSPV/4%2000PM%20How%20the%20Cloud%20is%20a%20Mission%20Enabler%20for%20Embedded%20System%20Development.pdf>.
- [892] Pérez-Sánchez, J., Rafi, S., Carrillo de Gea, J. M., Ros, J. N., García-Sánchez, R., & Fernández-Alemán, J. L. (2024). A taxonomy on human factors that affect DevOps adoption. Preprint. DOI: https://doi.org/10.1007/978-3-031-60221-4_31.
- [893] Alimam, M. N., & Kudsi, S. (2025). IDEAL-enhanced DevOps: A structured framework for continuous improvement in software engineering. Preprints.org. <https://doi.org/10.20944/preprints202503.1031.v1>
- [894] Alghawli, A. S. A., & Radivilova, T. (2024). Resilient cloud cluster with DevSecOps security model, automates a data analysis, vulnerability search and risk calculation. *Alexandria Engineering Journal*, 107, 136–149. Elsevier. <https://doi.org/10.1016/j.aej.2024.07.036>
- [895] García-Fernández, A., Parejo, J. A., Cavero, F. J., & Ruiz-Cortés, A. (2025). Racing the market: An industry support analysis for pricing-driven DevOps in SaaS. In *ICSOC 2024: International Conference on Service-Oriented Computing (LNCS 15405)*, pp. 260–275. Springer. https://doi.org/10.1007/978-981-96-0808-9_19.
- [896] Emma, L. (2025). MLOps on cloud platforms: End-to-end CI/CD pipelines for machine learning model deployment and monitoring. https://www.researchgate.net/publication/390700782_MLOps_on_Cloud_Platforms_End-to-End_CICD_Pipelines_for_Machine_Learning_Model_Deployment_and_Monitoring.
- [897] Johnson, E., Smith, J., Patel, R., Samuel, A. H., & Borra, P. (2023). Prioritizing security in CI/CD on cloud platforms: Strengthening DevSecOps practices. Preprint.

https://www.researchgate.net/publication/386245937_Prioritizing_Security_in_CICD_on_Cloud_Platforms_Strengthening_DevSecOps_Practices.

[898] Silva, C., Cunha, V. A., Barraca, J. P., & Salvador, P. (2024). Privacy-based deployments: The role of DevPrivOps in 6G mobile networks. *IEEE Communications Magazine*, 62(4), 66–73. <https://doi.org/10.1109/MCOM.004.2300405>.

[899] Tebib, M. E. A., Aktouf, O. E. K., André, P., & Graa, M. (2024). PRIVBENCH: A benchmark capturing privilege escalation vulnerabilities in Android. In *Proceedings of the 2024 1st Annual International Conference on Privacy, Security and Trust (PST)*. IEEE. <https://doi.org/10.1109/PST62714.2024.10788068>.

[900] Nogueira, A. F., & Zenha-Rela, M. (2024). Process mining software engineering practices: A case study for deployment pipelines. *Information and Software Technology*, 168, 107392. <https://doi.org/10.1016/j.infsof.2023.107392>.

[901] Nguyen, H. T., Usman, M., & Buyya, R. (2024). QFaaS: A serverless Function-as-a-Service framework for quantum computing. *Future Generation Computer Systems*, 154, 281–300. <https://doi.org/10.1016/j.future.2024.01.018>.

[902] Zhang, J. Y., & Zhang, Y. (2024). Quantitative DevSecOps metrics for cloud-based web microservices. *IEEE Access*, 12, 160317–160334. <https://doi.org/10.1109/ACCESS.2024.3486314>.

[903] Paulino, D., Netto, A. T., Brito, W. A. T., & Paredes, H. (2024). WebTraceSense—A framework for the visualization of user log interactions. *Eng*, 5(3), 2206–2222. <https://doi.org/10.3390/eng5030115>.

[904] Sriraman, G., & Shriram, S. R. (2024). Slide-block: End-to-end amplified security to improve DevOps resilience through pattern-based authentication. *Heliyon*, 10(2), e27196. <https://doi.org/10.1016/j.heliyon.2024.e27196>.

[905] Corona-Fraga, P., Hernandez-Suarez, A., Sanchez-Perez, G., Toscano-Medina, L. K., Perez-Meana, H., Portillo-Portillo, J., Olivares-Mercado, J., & García Villalba, L. J. (2025). Question–answer methodology for vulnerable source code review via prototype-based model-agnostic meta-learning. *Future Internet*, 17(1), 33. <https://doi.org/10.3390/fi17010033>.

[906] Desmond, O. C. (2024). The convergence of AI and DevOps: Exploring adaptive automation and proactive system reliability. *International Journal of Innovative Research in Computer and Communication Engineering (IJIRCCE)*, 12(9). <https://doi.org/10.15680/IJIRCCE.2024.1209045>.

[907] Tshakwanda, P. M., Kumar, H., Arzo, S. T., & Devetsikiotis, M. (2024). SE-DO: Navigating the 6G frontier with scalable and efficient DevOps for intelligent agents optimization. In *2024 14th Annual Computing and Communication Workshop and Conference (CCWC)*. IEEE. <https://doi.org/10.1109/CCWC60891.2024.10427717>.

[908] Kyler, T. (2024). AI-driven DevSecOps: Integrating security into continuous integration and deployment pipelines. University of Chicago.

[909] Sun, B., Zhou, H., Song, G., & Zhang, Q. (2024). Research on the technology architecture of full-stack cloud-native hosted cloud. *Proceedings of the 2024 11th International Conference on Machine Intelligence and Smart Systems (ICMISS)*. IEEE. <https://doi.org/10.1109/MiTA60795.2024.10751724>

[910] Niu, X., Yang, L., Liu, K., & Liu, Z. (2024). Research on the transformation path of DevOps in the digital era. In *26th International Conference on Advanced Communications Technology (ICACT)*, 2024, pp. 248–251. University of Chicago & China Academy of Information and Communications Technology. Available via CAICT & ResearchR: https://researchr.org/publication/NiuYLL24_Research_on_the_Transformation_Path_of_DevOps_in_the_Digital_Era

[911] Pastrana-Pardo, M. A., Ordoñez-Erazo, H. A., & Cobos-Lozada, C. A. (2024). Identification of characteristics in very small entities to implement DevOps: Determining strengths and opportunities for improvement. *Revista Científica*, 50(2), 1–17. <https://doi.org/10.14483/23448350.22071>

- [912] Linsbauer, S., Hark, R., Koziol, H., & Eskandani, N. (2024). Runtime orchestration of distributed control system services with TOSCA, Kubernetes, and GitOps. In 2024 IEEE 21st International Conference on Software Architecture Companion (ICSA-C). IEEE. <https://doi.org/10.1109/ICSA-C63560.2024.00013>
- [913] Rzig, D. E. (2024). Tackling DevOps and CI within ML projects (Doctoral dissertation). University of Michigan. https://deepblue.lib.umich.edu/bitstream/handle/2027.42/196334/Rzig_Dissertation_Tackling_DevOps.pdf.
- [914] Salášek, M. (2024). Design and implementation of the data-provenance subsystem for the AFoLab platform (Master's thesis). Masaryk University, Faculty of Informatics. https://is.muni.cz/th/w3kcx/posudek_vedouciho_Rebok.pdf.
- [915] Dalia, G., Visaggio, C. A., Di Sorbo, A., & Canfora, G. (2024). SBOM Ouverture: What we need and what we have. In ARES 2024: The 19th International Conference on Availability, Reliability and Security (pp. 1–9). ACM. <https://doi.org/10.1145/3664476.3669975>
- [916] Esther, D. (2024). Post-quantum cryptography: Securing networks against quantum threats. https://www.researchgate.net/publication/386381859_Post-Quantum_Cryptography_Securing_Networks_Against_Quantum_Threats.
- [917] Cui, J. (2024). The Enhancement of Software Delivery Performance through Enterprise DevSecOps and Generative Artificial Intelligence in Chinese Technology Firms. Preprint: arXiv:2411.02255. <https://doi.org/10.48550/arXiv.2411.02255>.
- [918] Balta, D., Balta, M., Kırık, H., & Esen, Y. E. (2024). Secure design and implementation of OPC standards-based log/process gathering framework for water management systems. In Proceedings of the 2024 International Conference on Artificial Intelligence, Metaverse and Cybersecurity (ICAMAC). IEEE. <https://doi.org/10.1109/ICAMAC62387.2024.10828837>.
- [919] Karanam, R. (2024). Securing CI/CD pipelines: Strategies for mitigating risks in modern software delivery. International Journal of Engineering and Technology Research (IJETR), 9(2), 1–9. <https://doi.org/10.5281/zenodo.13365012>.
- [920] Chittala, S. (2024). Securing DevOps pipelines: Automating security in DevSecOps frameworks. Journal of Recent Trends in Computer Science and Engineering (JRTCSE), 12(5), 31–44. <https://doi.org/10.70589/JRTCSE.2024.5.5>.
- [921] Redding, P., & Martin, S. (2025). Securing DevOps pipelines with AI-powered threat detection and mitigation. Preprint. https://www.researchgate.net/publication/388639873_Securing_DevOps_Pipelines_with_AI-Powered_Threat_Detection_and_Mitigation.
- [922] Abudalou, M. A. (2024). Security DevOps: Enhancing application delivery with speed and security. International Journal of Computer Science and Mobile Computing, 13(5), 100–104. <https://doi.org/10.47760/ijcsmc.2024.v13i05.009>
- [923] Luca, C. (2024). Security and compliance in multi-cloud Kubernetes orchestration. https://www.researchgate.net/publication/388527553_Security_and_Compliance_in_Multi-Cloud_Kubernetes_Orchestration.
- [924] Malik, G., & Prashasti, P. (2025). Shift left security. The Eastasouth Journal of Information System and Computer Science, 2(03), 219–245. <https://doi.org/10.58812/esiscs.v2i03>.
- [925] Koneru, N. M. K. (2025). Optimizing CI/CD pipelines for multi-cloud environments: Strategies for AWS and Azure integration. The Eastasouth Journal of Information System and Computer Science, 2(03), 288–310. <https://doi.org/10.58812/esiscs.v2i03>.
- [926] John, J., & Bello, S. (2025). Robust DevSecOps practices for AI model development and deployment. International Journal of Research and Analytical Reviews (IJRAR), 12(2), 25–30. Available at <https://www.researchgate.net/publication/391399108>.

- [927] Brahmandam, B. A. (2025). Beyond DevOps: The evolution toward intelligent IT operations with AIOps and MLOps. *The Review of Contemporary Scientific and Academic Studies*, 5(04). <https://doi.org/10.55454/rcsas.5.04.2025.002>.
- [928] Kolo, F. H. O. (2025). From framework to practice: Barriers and enablers to RMF adoption in mid-sized enterprises. *Asian Journal of Research in Computer Science*, 18(5), 459–479. <https://doi.org/10.9734/ajrcos/2025/v18i5667>.
- [929] Piridi, S., Asundi, S., & Hyatt, J. C. (2025). Cross-environment deployment strategies for Power Platform solutions: Investigating best practices for managing multi-environment deployments from development to production using managed environments and DevOps. *International Journal of Advanced Engineering Research and Science (IJAERS)*, 12(4), 66–75. <https://doi.org/10.22161/ijaers.124.8>.
- [930] Liang, P., Wu, Y., Xu, Z., Xiao, S., & Murphy, P. (2024). Optimizing DevOps security through AI and machine learning. *Journal of Theory and Practice of Engineering Science*, 4(2), 31–38. [https://doi.org/10.53469/jtpes.2024.04\(02\).05](https://doi.org/10.53469/jtpes.2024.04(02).05).
- [931] Cui, J. (2025). The impact of automated testing on DevOps efficiency in Chinese internet technology companies: The moderating role of human-AI interaction. Preprint. <https://www.researchgate.net/publication/337076816>. <https://doi.org/10.13140/RG.2.2.18018.64963>.
- [932] Noreña-Cardona, P., Durango, C., Suescún, E., & Pardo, C. (2025). Improving understanding of the DevOps framework using Essence: A visual representation. *Periodicals of Engineering and Natural Sciences*, 13(2), 305–326. <https://doi.org/10.21533/pen.v13i2.1232>.
- [933] Zhang, M. (2025). Optimization of medical device software lifecycle management based on DevOps. *Journal of Medicine and Life Sciences*, 1(3), 1–6. <https://doi.org/10.71222/etnrdv93>.
- [934] Yakkanti, P. R. (2025). AI-augmented DevOps for application modernization: Transforming software development and operations. *Journal of Computer Science and Technology Studies*, 7(2), 368–376. <https://doi.org/10.32996/jcsts.2025.7.2.38>.
- [935] Gudelli, V. R. (2024). Automating multi-cluster Kubernetes deployments with Terraform. *Hong Kong Journal of AI and Medicine*, 4(1), 382–397. <https://doi.org/10.5281/zenodo.15271228>.
- [936] Alfawzan, L., Alenezi, M., & Akour, M. (2025). Assessing DevOps adoption in Saudi software teams. *Journal of Posthumanism*, 5(4), 725–747. <https://doi.org/10.63332/joph.v5i4.1130>.
- [937] Singh, A. (2025). Configuration changes in Kubernetes configuration scripts (Master's thesis). Auburn University. <https://etd.auburn.edu/handle/10415/9682>.
- [938] Yang, S. (2025). The impact of continuous integration and continuous delivery on software development efficiency. *Journal of Computer, Signal, and System Research*, 2(3), 1–9. <https://doi.org/10.71222/pzvfqm21>.
- [939] Jeeri, K. (2025). Approaches to automating CI/CD processes in distributed teams. *The American Journal of Engineering and Technology*, 7(4), 75–82. <https://doi.org/10.37547/tajet/Volume07Issue04-13>.
- [940] Cui, J., & Shin, S. (2025). The effect of software automation testing efficiency on enterprise DevOps performance: The moderating role of GAI innovation. Preprint. <https://www.researchgate.net/publication/337076816>. <https://doi.org/10.13140/RG.2.2.30467.21283>.
- [941] Sundareswaran, A. P., & Kushwaha, A. S. (2025). Microservices and DevOps: Accelerating eCommerce delivery. *International Journal of Research in Humanities & Social Sciences*, 13(3). <https://doi.org/10.63345/ijrhs.net.v13.i3.7>.
- [942] Faseeha, U., Syed, H. J., Samad, F., Zehra, S., & Ahmed, H. (2025). Observability in microservices: An in-depth exploration of frameworks, challenges, and deployment paradigms. *IEEE Access*, 13, 72011–72026. <https://doi.org/10.1109/ACCESS.2025.3562125>.
- [943] Cui, J. (2025). Exploring the Git flow and Docker container standardization on enterprise DevOps development efficiency. Preprint. <https://www.researchgate.net/publication/337076816>. <https://doi.org/10.13140/RG.2.2.10183.82085>.

- [944] Cui, J. (2025). The effects of mobile and server-side automated testing on DevOps performance efficiency. *Journal of Computer Science and Digital Technology*, 1(1), 1000080. <https://doi.org/10.71204/c3f9r559>.
- [945] Patil, R. V., Kudande, V., Jagtap, S., Jadhav, S., & Jawalgekar, A. (2025). Self-healing infrastructure system. *International Journal of Electrical, Electronics and Computer Systems*, 14(1), 219–245. MRI India. <https://journals.mriindia.com/index.php/ijeecs/article/view/190>.
- [946] Kour, G. (2025). Challenges and solutions in managing the modernization of industrial software – Case study of Sentinel and FLMS (Bachelor’s thesis). Turku University of Applied Sciences, Finland. <https://www.theseus.fi/handle/10024/882243>.
- [947] Cellamare, F. P. (2024). AI-driven unit test generation (Master’s thesis). Politecnico di Torino, Master’s Degree in Communications and Computer Networks Engineering. <https://webthesis.biblio.polito.it/35229/1/tesi.pdf>.
- [948] Nimdia, A. (2025). Effectiveness of automated testing in container orchestration. *The American Journal of Engineering and Technology*, 7(4), 34–42. <https://doi.org/10.37547/tajet/Volume07Issue04-05>.
- [949] Kim, M. (2025). Guidelines for designing an IT undergraduate course on cloud-native technologies for microservice architectures. *International Journal of Internet, Broadcasting and Communication*, 17(1), 352–364. <https://doi.org/10.7236/IJIBC.2025.17.1.352>.
- [950] Awasthi, K. S. (2025). Automated software testing. In *Digital transformation in retail: Adapting to new consumer landscape* (pp. 91–105). Lucknow Public College of Professional Studies. ISBN: 978-93-6773-284-7
- [951] Ogety, S. S. (2025). Dependent tasks verification-aware task scheduling and resource allocation in cloud DevOps using Obl-Fuzzy and Smrnn. *IRJEMS: International Research Journal of Economics and Management Studies*, 4(3), 266–281. <https://doi.org/10.56472/25835238/IRJEMS-V4I3P130>.
- [952] Wik, M. (2025). Continuous improvement for RPA through process mining (Master’s thesis). Åbo Akademi University, Faculty of Science and Engineering. <https://www.doria.fi/handle/10024/192016>.
- [953] Tiwari, A., Agrawal, A., Kumar, A., & Raghav, P. K. (2025). DevOps implementation strategies: Evaluating different practices. SSRN Preprint. Adobe Systems, Noida, India. <https://doi.org/10.2139/ssrn.5205581>.
- [954] Dzeikalo, A. (2025). Architectural approaches to creating scalable and adaptable software solutions for oil and gas companies. Independent Research Publication. <https://doi.org/10.13140/RG.2.2.10667.25125>.
- [955] Zhao, G. Y., & Tu, C. Z. (2025). Adapt DevOps method to information systems capstone course projects. *The Journal of Computing Sciences in Colleges*, 40(6), 18–28. Proceedings of the 31st Annual CCSC Central Plains Conference (Drake University, Des Moines, IA). ISSN 1937-4771 (print), 1937-4763 (digital). <https://dl.acm.org/doi/10.5555/3729857.3729860>.
- [956] Chumwatana, T., & Hpone, A. K. K. (2025). Bridging the IT skill gap with industry demands: An AI-driven text mining approach to job market trends using large language model. *Journal of Theoretical and Applied Information Technology*, 103(6), 2270–2282. Little Lion Scientific. ISSN 1992-8645; E-ISSN 1817-3195. Available at: <http://www.jatit.org>.
- [957] Tanneru, B. (2025). Integrating security vulnerability management into agile DevOps pipelines. *International Journal on Science and Technology (IJSAT)*, 16(1), 1–5. E-ISSN: 2229-7677. <https://doi.org/10.71097/IJSAT.v16.i1.3153>.
- [958] Das, S., & Prasad, A. R. R. (2022). Using MLOps for deployment of opinion mining model as a service for smart city applications. *International Journal of Research in Engineering, Science and Management (IJRESM)*, 5(12), 183–189. <https://doi.org/10.37943/21CPQX5616>.
- [959] Alkan, E., & Ibazizene, K. (2025). Study of FATES properties in the MLOps field. Institut de Recherche Informatique de Toulouse; Université Toulouse 2 Jean Jaurès. HAL ID: hal-05014141. <https://hal.science/hal-05014141v1>.

- [960] Lase, S. D. J., & Suharyanto, C. E. (2025). Perancangan aplikasi Point of Sales (POS) pada Minimarket Berkat berbasis web. *Jurnal Teknologi Digital dan Sistem Informasi (JUTEKDISI)*, 2(1), 36–46. E-ISSN: 3046-9384. Available at: <https://ojsiibn1.indobarunasional.ac.id/index.php/JUTEKDISI/article/view/1063>.
- [961] Moreschini, S., Pour, S., Lanese, I., Balouek, D., Bogner, J., Li, X., Pecorelli, F., Soldani, J., Truyen, E., & Taibi, D. (2025). AI techniques in the microservices life-cycle: A systematic mapping study. *Computing*, 107, Article 100. <https://doi.org/10.1007/s00607-025-01432-z>.
- [962] Matt, M., Weah, J., & Christopher, G. (2023). Modern enterprise architecture in the AI era: Integrating cloud, DevOps, and DataOps. Preprint. <https://www.researchgate.net/publication/390175336>.
- [963] Max, M. O. (2025). Performance optimization techniques for continuous deployment in multi-cloud environments. Preprint. <https://www.researchgate.net/publication/390182354>.
- [964] Idowu, M., William, B., Owen, J., & Edwin, F. (2025). Driving efficiency with DevOps: Optimizing agile practices for continuous delivery. Preprint. <https://www.researchgate.net/publication/390204602>.
- [965] Lewis, B., & Willcot, G. (2025). AI-powered synergy: Integrating DataOps and DevOps for modern cloud enterprise optimization. Preprint. <https://www.researchgate.net/publication/337076816>.
- [966] Pereira, I., Carneiro, T., & Figueiredo, E. (2025). Manipulating a CI/CD pipeline in an IoT embedded project: A quasi-experiment. *Journal of Software: Evolution and Process*, 37, e70022. <https://doi.org/10.1002/smr.70022>
- [967] David, S., Jones, T., & Christopher, G. (2022). Intelligent infrastructure management: Merging AI and DevOps for optimized cloud-native applications. https://www.researchgate.net/publication/390172385_Intelligent_Infrastructure_Management_Merging_AI_and_DevOps_for_Optimized_Cloud-Native_Applications.
- [968] Matthew, M., Anderson, J., & Ogunrinde, V. (2022). AI and cloud computing synergy: Transforming enterprise architecture with DevOps and DataOps. https://www.researchgate.net/publication/390172452_AI_and_Cloud_Computing_Synergy_Transforming_Enterprise_Architecture_with_DevOps_and_DataOps.
- [969] Halilcevic, E. (2025). Software testing and test environment simulation in the cloud: A case study on open-source IaC and CI/CD technologies (Diploma thesis). Technische Universität Wien. <https://doi.org/10.34726/hss.2025.127221>.
- [970] Kondeti, N. (2025). Customization and extensibility in Guidewire Cloud: A developer's perspective. *International Journal on Science and Technology (IJSAT)*, 16(1), 1–13. IJSAT25012643. E-ISSN: 2229-7677. DOI: <https://doi.org/10.71097/IJSAT.v16.i1.2643>.
- [971] Jeyarajan, B., Murugan, A., Pandey, G., & Pugazhenth, V. J. (2025). AI for predictive monitoring and anomaly detection in DevOps environments. In *Proceedings of SoutheastCon 2025* (pp. 450–455). IEEE. <https://doi.org/10.1109/SOUTHEASTCON56624.2025.10971552>
- [972] Islavath, N. (2023). Unlocking the potential of Chef: Configuration management for scalable DevOps. *International Journal of Computer Techniques (IJCT)*, 10(2), 1–17. ISSN: 2394-2231. <https://ijctjournal.org/volume-10-issue-2-chef-devops-scalability>.
- [973] Ramaswamy, Y. (2025). AI enhanced bioinformatics pipelines in DevOps. *International Journal on Science and Technology (IJSAT)*, 16(1), 1-20. IJSAT25012136. E-ISSN: 2229-7677. <https://doi.org/10.71097/IJSAT.v16.i1.2136>.
- [974] Mehta, R. (2024). Ai-assisted code smell detection and refactoring: A continuous integration approach. Computer Science, University of Illinois Chicago (report). https://www.cs.uic.edu/~i440/Grad%20Papers/440%20Fall%202024/Papers/RiyaMehta_cs440.pdf.

- [975] Awad, H. (2025). Quality of service assurance before deployment of fog systems with model-based engineering and DevOps (Doctoral dissertation). Ecole nationale supérieure Mines-Télécom Atlantique, Nantes, France. HAL: <https://theses.hal.science/tel-05056618v1>.
- [976] Carignan, A., & Enoch, O. (2025). Enhancing DevOps efficiency: Best practices for cloud infrastructure management. Preprint. Preprints.org. <https://doi.org/10.20944/preprints202503.1143.v1>
- [977] Syed-Mohamad, S. M., Ngah, A., Ali, A.-F. M., & Keikhosrokiani, P. (2025). Measuring software maintainability: An analysis of evolving metrics across development paradigms. *Journal of Advanced Research in Applied Sciences and Engineering Technology*, 63(2), 181–195. <https://doi.org/10.37934/araset.63.2.181195>
- [978] William, E., & Laughter, J. (2025). Immutable infrastructure: Principles and implementations. Preprint. <https://www.researchgate.net/publication/337076816>.
- [979] Mabel, E. (2025). DevOps in the cloud: A guide to streamlining infrastructure for faster deployments. Preprint. <https://www.researchgate.net/publication/337076816>.
- [980] Artha, I. P. M. K., Wiguna, I. K. A. G., Wijaya, I. K. H. K., & Sudipa, I. G. I. (2025). Development and implementation of mobile applications standing instructions (SI) services for customers with the DevOps method. *ISAR Journal of Economics and Business Management*, 3(3), 28–31. <https://article.isarpublisher.com/viewArticle/Development-and-Implementation-of-Mobile-Applications-Standing-Instructions-SI-Services-for-Customers-with-the-DevOps-Method>.
- [981] Alexander, & Wella. (2025). Continuous integration pipeline with Jenkins. *Pilar*, 21(1). <https://doi.org/10.33480/pilar.v21i1.6062>
- [982] Gangu, K., & Mishra, R. (2025). DevOps and continuous delivery in cloud-based CDN architectures. *International Journal of Research in All Subjects in Multi Languages (IJRSMML)*, 13(1). Print ISSN: 2321-2853. <https://ijrsmml.org/devops-and-continuous-delivery-in-cloud-based-cdn-architectures>.
- [983] Wüstenberg, P. K. (2025). Agile development of satellite software and its automated testing and operation (Doctoral dissertation). Technische Universität Berlin. DOI: <https://doi.org/10.14279/depositonce-22394>.
- [984] Obloev, K. H. O. (2025). Machine learning model deployment using FastAPI and Docker: A modern approach to scalable AI services. *Pedagogik Tadqiqotlar Jurnalı*, 4, 69–73. ISSN: 3060-4923. Available at: <https://journalseeker.researchbib.com/view/issn/3060-4923>.
- [985] Velishala, S. (2025). Leveraging machine learning in DevOps pipelines to enhance patient data management systems. *International Journal of Computer Science and Engineering (ISCSITR)*, 6(1), 31–49. <https://doi.org/10.5281/zenodo.14792139>.
- [986] Benjamin, M. (2025). AI and machine learning in software development. Preprint. <https://www.researchgate.net/publication/337076816>. Available:
- [987] Vadisetty, R., Polamarasetti, A., & Nomula, V. K. (2025). AI-powered policy management: Implementing Open Policy Agent (OPA) with intelligent agents in Kubernetes. *Cuestiones de Fisioterapia*, 54(5), 19–27. <https://doi.org/10.48047/f5vc5s27>.
- [988] Vadisetty, R., Polamarasetti, A., & Nomula, V. K. (2025). AI-Driven Kubernetes orchestration: Utilizing intelligent agents for automated cluster management and optimization. *Cuestiones de Fisioterapia*, 54(5), 28–36. <https://doi.org/10.48047/brsw0z42>.
- [989] Ahmed, S. (2025). Integrating AI-driven automated code review in agile development: Benefits, challenges, and best practices. *International Journal of Advanced Engineering and Management Science (IJAEMS)*, 11(2), 1–10. <https://doi.org/10.22161/ijaems.112.1>.
- [990] Kothamali, P. R. (2025). AI-powered quality assurance: Revolutionizing automation frameworks for cloud applications. *Journal of Advanced Computer Systems (JACS)*, 5(3), 1–25. <https://doi.org/10.69987/JACS.2025.50301>.

- [991] Gudelli, V. R. (2025). Serverless computing in cloud environments: Advancing scalability in DevOps pipelines. *International Journal of Novel Research and Development (IJNRD)*, 10(2), 1–10. <https://doi.org/10.1729/Journal.43732>.
- [992] Misri, M. E. (2024). The impact of DevOps practices on software development lifecycle. *International Journal of Creative Research Thoughts (IJCRT)*, 12(12), 531–538. ISSN: 2320-2882. Available: <https://www.ijcrt.org/papers/IJCRT2412276.pdf>.
- [993] Johnson, S. (2021). DevOps in a box: An open-source starter kit for new startups. *International Journal of Applied Engineering and Technology*, 3(2), 143–160. ISSN: 2633-4828. Available at: https://www.researchgate.net/publication/389556453_DEVOPS_IN_A_BOX_AN_OPEN-SOURCE_STARTER_KIT_FOR_NEW_STARTUPS
- [994] Joshi, S. (2025). Introduction to generative AI and DevOps: Synergies, challenges and applications. *International Journal of Advanced Research in Science Communication and Technology (IJARSCT)*, 5(1), 205–223. <https://doi.org/10.48175/IJARSCT-23634>.
- [995] Cui, J. (2025). Enterprise DevOps efficiency on product delivery: The moderating role of automated testing capabilities. SolBridge International School of Business. SSRN paper. <https://doi.org/10.2139/ssrn.5178311>.
- [996] Pahl, C., Gunduz, N. G., Sezen, Ö. C., Ghamgosar, A., & El Ioini, N. (2025). Infrastructure as code – Technology review and research challenges. In *Proceedings of the 20th International Conference on Evaluation of Novel Approaches to Software Engineering (ENASE)*. <https://doi.org/10.5220/0013247700003950>.
- [997] Khan, Z., Ali, L., & Zubair, S. (2025). Challenges & opportunities of adopting DevOps in the PMI context. *Contemporary Journal of Social Science Review*, 3(1), 1357–1365. <https://doi.org/10.12345/k2wnbp74>
- [998] Riaz, S., Asif, A., Khan, Y., Ibrar, M., Afzal, S., Hamid, K., Gul, S., & Iqbal, M. W. (2025). Software development empowered and secured by integrating a DevSecOps design. *Journal of Computer Biomedical Informatics*, 8(2). <https://doi.org/10.56979/802/2025>
- [999] Benjamin, M. (2025). Emerging trends in sales automation and software development for global enterprises. Preprint. Available: <https://www.researchgate.net/publication/390704552>.
- [1000] Oladele, O. K. (2025). Cloud-native AI development: Building and deploying scalable machine learning models on AWS, Azure, and GCP. Preprint. <https://www.researchgate.net/publication/337076816>.
- [1001] Alić, A., Traljić, A., & Đonko, D. (2025). Methodology for evaluating the impact of DevOps principles. In *Proceedings of the 24th International Symposium INFOTEH-JAHORINA* (pp. 1–7). IEEE. <https://doi.org/10.1109/INFOTEH64129.2025.10959292>.
- [1002] Irabedra, J. I., & Solari, M. (2025). Active learning methodology for advanced agile software engineering and DevOps practices. In *Proceedings of the IEEE World Engineering Education Conference (EDUNINE)*. IEEE. <https://doi.org/10.1109/EDUNINE62377.2025.10981332>.
- [1003] Ashtagi, R., Bhalerao, Y., Belani, C., Bhalerao, P., & Chawle, A. (2025). Building resilient CI/CD pipelines: A DevOps security-first framework. In *Proceedings of the International Conference on Computing, Communication and Information Technology (ICCCIT)* (pp. 828–834). IEEE. <https://doi.org/10.1109/ICCCIT62592.2025.10927871>.
- [1004] Garg, A., Rajiv, A., Singh, T., Dhamodaran, D. S., & Sahoo, G. S. (2025). Optimizing cost efficiency in cloud-based automation with DevOps techniques. In *Proceedings of the International Conference on Automation and Computing (AUTOCOM)* (pp. 488–493). IEEE. <https://doi.org/10.1109/AUTOCOM64127.2025.10956958>.
- [1005] Chhatrria, B., Dharua, H., Tarai, S. K., & Panda, S. (2025). Emerging trends in software engineering: Implications for development and efficiency. In *Proceedings of the International Conference on Emerging*

Systems and Intelligent Computing (ESIC) (pp. 13–18). IEEE.
<https://doi.org/10.1109/ESIC64052.2025.10962758>.

[1006] Lukić, M. D., Ivković, D. S., & Poledica, A. M. (2025). MLOps tools for deployment: A case study on text classification. In Proceedings of the 29th International Conference on Information Technology (IT). IEEE. <https://doi.org/10.1109/IT64745.2025.10929797>.

[1007] Stanišić, S., Vesković, M., Ristić, O., & Đorđević, B. (2025). Security aspects of container orchestration in Kubernetes environments. In Proceedings of the 24th International Symposium INFOTEH-JAHORINA. IEEE. <https://doi.org/10.1109/INFOTEH64129.2025.10959185>.

[1008] Ndassimba, E., Zoh-Ponguele, N., & Kossingou, G. M. S.-J. (2025). The impact of TV white space (TVWS) and 5G networks on bimodal learning for master's programs in engineering science at the Higher Institute of Technology, University of Bangui. In Proceedings of the International Conference on Computing, Communication and Information Technology (ICCCIT) (pp. 880–885). IEEE.
<https://doi.org/10.1109/ICCCIT62592.2025.10927912>.

[1009] Buljic, I., Hadzajlic, N., Kadusic, E., Zivic, N., & Cvijanovic, T. (2025). Comparative performance analysis of leading backend frameworks for developers. In Proceedings of the 24th International Symposium INFOTEH-JAHORINA. IEEE. <https://doi.org/10.1109/INFOTEH64129.2025.10959250>.

[1010] Viradia, V., Donvir, A., Jain, A., & Ogety, S. S. (2025). Explainable AI (XAI) in predictive cloud optimization: Cost, workload and performance. In Proceedings of IEEE SoutheastCon 2025 (pp. 1424–1429). IEEE. <https://doi.org/10.1109/SOUTHEASTCON56624.2025.10971567>.

[1011] Krishnamurthy, D., & Neelanath, V. (2025). Establishing a robust LLMops framework for intelligent automation: Strategies and best practices. In Proceedings of the IEEE Conference on Emerging Technologies for Intelligent Systems (ETIS) (pp. 1–8). IEEE.
<https://doi.org/10.1109/ETIS64005.2025.10961869>.

[1012] Bidokhti, N. (2025). RelOps – Introduction to reliable operations. In Proceedings of the Annual Reliability and Maintainability Symposium (RAMS) (pp. 1–8). IEEE.
<https://doi.org/10.1109/RAMS48127.2025.10935243>.

[1013] Muthukrishnan, H., Viradia, V., & Yadav, D. (2025). Unified AI and ML framework in DevSecOps practices: Solving real-world problems. In Proceedings of IEEE SoutheastCon 2025 (pp. 1250–1257). IEEE. <https://doi.org/10.1109/SOUTHEASTCON56624.2025.10971458>.

[1014] (2025). Application deployment: The game-changing impact of containerization. In Proceedings of the 4th International Conference on Power, Control and Computing Technologies (ICPC2T). IEEE.
<https://doi.org/10.1109/ICPC2T63847.2025.10958660>.

[1015] Puvvadi, M., Arava, S. K., Santoria, A., Chennupati, S. S. P., & Puvvadi, H. V. (2025). Coding agents: A comprehensive survey of automated bug fixing systems and benchmarks. In Proceedings of the 14th IEEE International Conference on Communication Systems and Network Technologies (CSNT) (pp. 680–686). IEEE. <https://doi.org/10.1109/CSNT.2025.113>.

[1016] Dasanayake, S. D. L. V., Senanayake, J., & Wijayanayake, W. M. J. I. (2025). DevSecOps implementation for continuous security in financial trading software application development. In Proceedings of the 5th International Conference on Advanced Research in Computing (ICARC). IEEE.
<https://doi.org/10.1109/ICARC64760.2025.10963292>.

[1017] Patil, A., & Jadon, A. (2025). Next-generation bug reporting: Enhancing development with AI automation. In Proceedings of the 10th International Conference on Signal Processing and Communication (ICSC). IEEE. <https://doi.org/10.1109/ICSC64553.2025.10968932>.

[1018] Upeksha, S., Samarasinghe, D. T., Sanochana, M., Samarathunga, S. S., Rajamanthri, L., Samarakkody, T., & Aluthwala, C. (2025). The influence of generative AI on work-life balance among female software professionals in Sri Lanka. In Proceedings of the 5th International Conference on Advanced Research in Computing (ICARC). IEEE. <https://doi.org/10.1109/ICARC64760.2025.10963119>.

- [1019] Garousi, V., Felderer, M., & Mäntylä, M. V. (2019). Guidelines for including grey literature and conducting multivocal literature reviews in software engineering. *Information and Software Technology*, 106, 101–121. Elsevier. <https://doi.org/10.1016/j.infsof.2018.09.006>.
- [1020] Budakoglu, G., & Emekci, H. (2025). Unveiling the power of large language models: A comparative study of retrieval-augmented generation, fine-tuning, and their synergistic fusion for enhanced performance. *IEEE Access*. <https://doi.org/10.1109/ACCESS.2025>.